Build Once, Answer Many

Where disclosure demand converges for European seafood-processing SMEs

A diagnostic framework to transform compliance obligations into strategic advantage

Master's Thesis

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Abstract

European SMEs face a fragmented wall of sustainability-disclosure demands. Regulators, retail customers, certifiers, investors and NGOs each request seemingly different information through different channels, and for a firm with limited resources the rational response is reactive: answer the loudest demand, defer the rest, and build nothing that would serve more than the request in front of it. Yet beneath these superficially different requests lies a far smaller set of recurring *themes* and the *datapoints* that produce them. This thesis argues that a resource-constrained firm which identifies the few themes on which demand converges - and learns to generate their underlying datapoints once - can satisfy most of what it is asked from a single effort, shifting from reactive compliance to proactive strategy.

Taking European seafood processing as a case, the thesis demonstrates this in four steps. Step 1 maps the sector's disclosure *ecosystem*: 49 catalogued regulatory, customer, investor and industry instruments, of which 30 place a scorable demand on the firm. Step 2 establishes what mature firms actually report, using six large processors from among the sector's most advanced reporters. Even the deepest covers only 60 of 70 relevant ESRS disclosure requirements, and all six converge on a 13-requirement core regardless of how differently they source their fish.

Step 3 crosses the two, and is where the central finding appears. Scoring every topic against every instrument yields 1,023 combinations, of which just 18 per cent carry any demand at all. That sparse demand is heavily concentrated: a small set of topics already answers most of what the ecosystem asks. What presents itself as dozens of separate requests is, underneath, a handful of shared ones.

Step 4 turns this into a diagnostic. Topics are staged by how strongly and how widely they are demanded, producing a four-topic core every seafood processor should build first - energy, greenhouse-gas emissions, biodiversity and supply-chain traceability - extending to nine topics for a mixed-sourcing firm, eight for aquaculture and five for wild-capture. For the modelled firm, those build-first topics answer a demand raised by 26 of the 30 frameworks scored in the sector's landscape, from one dataset. The findings give seafood SMEs a sequenced prioritisation framework and provide empirical input to the EU sustainability-simplification debate.

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List of Abbreviations

ASC Aquaculture Stewardship Council

BAP Best Aquaculture Practices

CDP Carbon Disclosure Project

CSDDD Corporate Sustainability Due Diligence Directive

CSR Corporate Social Responsibility

CSRD Corporate Sustainability Reporting Directive

DR Disclosure Requirement (ESRS)

EC European Commission

EFRAG European Financial Reporting Advisory Group

ESG Environmental, Social and Governance

ESRS European Sustainability Reporting Standards

EU European Union

EUDR EU Deforestation Regulation

EU Taxonomy EU classification system for environmentally sustainable economic activities (Regulation (EU) 2020/852)

EPR Extended Producer Responsibility

F-gas Fluorinated Greenhouse Gas (EU F-gas Regulation)

GDST Global Dialogue on Seafood Traceability

GHG Greenhouse Gas

GLOBALG.A.P. Global Good Agricultural Practice

GRI Global Reporting Initiative

GSSI Global Sustainable Seafood Initiative

IFRS S1/S2 IFRS Sustainability Disclosure Standards 1 and 2 (ISSB)

ISSB International Sustainability Standards Board

ISO International Organization for Standardization

ISSF International Seafood Sustainability Foundation

IUU	Illegal, Unreported and Unregulated (fishing)
MSC	Marine Stewardship Council
NGO	Non-Governmental Organisation
OSH	Occupational Safety and Health
PPWR	Packaging and Packaging Waste Regulation
RSPO	Roundtable on Sustainable Palm Oil
SASB	Sustainability Accounting Standards Board
SBTi	Science Based Targets initiative
SFDR	Sustainable Finance Disclosure Regulation
SME	Small and Medium-sized Enterprise
SMETA	Sedex Members Ethical Trade Audit
STECF	Scientific, Technical and Economic Committee for Fisheries
TNFD	Taskforce on Nature-related Financial Disclosures
UN	United Nations
VSME	Voluntary Sustainability Reporting Standard for SMEs

1. Introduction

1.1 A crowded and fragmented field of demands

Companies today face a large and steadily growing volume of requests for sustainability information, and from a widening set of actors. Regulators legislate mandatory disclosures. Retail and business customers send sustainability questionnaires and make certification a condition of supply. Banks, investors and ratings agencies ask for climate and ESG data before extending capital or assigning a score. Industry coalitions and NGOs promote voluntary standards and commitments. Each actor asks through its own channel, on its own timetable, and in its own vocabulary.

These demands are less independent than they first appear, because many of them share a common origin. The European Union has made corporate sustainability reporting a central lever of its Green Deal: the Corporate Sustainability Reporting Directive (CSRD), implemented through the European Sustainability Reporting Standards (ESRS), defines mandatory disclosures across environmental, social and governance topics on a double-materiality basis (EFRAG, 2021). These obligations bind only large undertakings directly, but they do not stop there. A company in scope must gather data from its suppliers, and passes the requirement down its value chain as contractual questionnaires. Banks and investors, themselves regulated under the SFDR, ask the same of the companies they finance. A single regulatory requirement therefore resurfaces, in different forms, through several actors at once. What a small supplier experiences as 'a customer demand' is frequently a regulatory demand that has cascaded one or two steps down the chain (a pattern this thesis establishes directly in Chapter 4's source-coding test, Section 4.1). This is a dynamic the EU has itself acknowledged by legislating, in its 2026 simplification package, a 'value-chain cap' to limit how far such requests may be pushed onto small suppliers (European Commission, 2026).

Seen one request at a time, the combined effect is a proliferating, uncoordinated wall of separate demands: the 'alphabet soup' of standards and frameworks that has prompted sustained calls for harmonisation (Afolabi, Ram & Rimmel, 2022). Goals are not the sticking point so much as alignment between the actors pursuing them: the STAGE Index, which measures how far government and enterprise sustainability agendas coincide, registers only about 51 per cent convergence in 2026 (The European House – Ambrosetti, 2026). For the firm on the receiving end, the practical experience is simply that the volume and variety of what is asked keeps growing, and that keeping up takes ever more effort.

1.2 Different questions, shared answers

The observation motivating this thesis is that, beneath their different wording, these demands draw on a much smaller body of shared data. A customer's carbon-footprint question, an investor's climate metric and a regulatory emissions disclosure all rest on the same greenhouse-gas accounting. A retailer's traceability requirement, a certification's chain-of-custody audit, a due-diligence law and a value-chain disclosure standard all rest on the same lot-level tracking data. Behind a large number of superficially different and fragmented questions sits a much smaller number of shared datapoints, and those datapoints cluster into a handful of recurring themes. Establishing that this overlap is real, and measuring how large it is, is precisely the work of the analysis that follows.

That overlap is not evenly spread. Some datapoints are demanded by nearly every actor, others by only one. A few central themes are where the most actors converge, and these underpin a firm's licence to operate and to sell; a long tail of remaining topics is asked less often and by fewer actors. The practical implication is direct. A firm that identifies the key themes and learns to produce their datapoints *once* - as a structured, reusable process rather than a one-off answer - can satisfy most of what it is asked from a single effort, instead of rebuilding an answer for every request.

1.3 Capacity, not intent: why small firms stay reactive

The barrier to capturing this prize - answering many demands from one well-built set of data – is not a matter of intent but of capacity. Small and medium-sized enterprises (SMEs) operate with limited financial and managerial resources, little dedicated reporting expertise and weak data infrastructure. SMEs do not operate simply as miniature large firms, and their performance should be assessed against benchmarks that fit the circumstances of small firms rather than standards built for large-firm capacity (Battisti & Perry, 2011). Under those constraints the up-front investment in structured data processes, the very investment that would let a firm answer many frameworks from one dataset, is rarely made. Instead the firm responds *reactively*: each new questionnaire, audit or regulation is handled as a separate, one-off exercise, its data assembled by hand and then set aside (the pattern the scoping conversations described in Section 1.4 pointed to). Fragmented governance thus pushes resource-constrained firms toward minimum-viable compliance rather than strategic integration (Chan, Lai & Kim, 2022).

The consequence follows directly: because little or no reusable capability was built, each new regulation or customer request has to be met from a standing start. Data is assembled again for each request, and the firm stays a step behind - mostly catching up to the last demand, rarely prepared for the next - the reactive posture developed fully in Chapter 2 (Section 2.2). Two licences are at stake at once, the *legal* licence to operate, which direct regulation makes non-negotiable, and the *commercial* licence to sell, which large customers and their financiers increasingly attach to the same sustainability data. The stakes are concrete. Buyers increasingly require suppliers to complete ESG assessments before contracting. A supplier that cannot answer risks being scored down, dropped from the supply chain, or screened out at the rating stage before a relationship starts. Both are documented dynamics of buyer-side screening platforms such as EcoVadis (Tanase, 2025), whose reach across supplier networks makes the assessment a de facto condition of trade rather than an optional exercise. A firm that builds the underlying data once avoids repeating that risk with every buyer; one that answers each demand in isolation carries it every time.

1.4 Seafood processing in Europe: a revealing case

To make the problem tangible and empirically grounded, this thesis studies one sector in depth: European seafood processing. Three features make it a revealing case.

First, it is overwhelmingly a sector of small firms. Around two-thirds of the roughly 3,245 EU enterprises that process fish as their main activity are micro-enterprises of ten employees or fewer, and only about two per cent employ 250 people or more (STECF, 2025). The great majority of the sector therefore sits in the resource-constrained position described above.

Second, it carries a dense and fragmented governance load. Seafood processors sit beneath the full stack of horizontal EU sustainability law (the CSRD, CSDDD, the Packaging and Packaging Waste Regulation, the Empowering Consumers Directive and more), a layer of sector-specific regulation (the Common Fisheries Policy, the IUU and Control Regulations, the EU Deforestation Regulation), *and* a voluntary layer that stacks certification (MSC and RFM in wild capture; ASC, GLOBALG.A.P. and BAP in aquaculture), buyer rating platforms (EcoVadis, Sedex) and traceability standards (GDST) on top of one another. A fourth strand runs across all three: ethical and human-rights demand, reaching the firm through the CSDDD's due-diligence duty, the Forced Labour Regulation and buyer codes of conduct. Few of these are individually onerous - and several are interoperable, with MSC and ASC both working with GDST. The load comes from their number as a *stack*: each is answered separately, on its own timetable and in its own format.

Third, this compounding pressure is not merely inferred from the regulatory map. Informal scoping conversations with small seafood processors in Belgium and Germany, held at the outset of this research, pointed the same way. Those firms described simultaneous, overlapping requests from regulators, retail customers and finance providers, which they answered one at a time because they had no shared process for answering them together.[1]

Seafood is a case, however, not the point. The same structure, many actors, overlapping underlying data, resource-constrained firms, recurs across manufacturing and agri-food SME sectors facing the same widening set of demands.

[1] These conversations were exploratory and unstructured, conducted to orient the research rather than as a data-collection method. They are not coded, quoted or treated as evidence anywhere in this thesis; the empirical claims rest entirely on the document analysis of Chapters 4 to 7.

Seafood processing is a sharp instance that makes the general problem visible and testable; the diagnostic approach developed here is intended to be transferable.

1.5 Aim, approach and contribution

The EU has not been blind to this burden. Its main simplification response - the Voluntary Sustainability Reporting Standard for SMEs (VSME), which EFRAG developed and the Commission endorsed by recommendation in 2025, alongside the wider Omnibus revision now cutting ESRS datapoints by more than seventy per cent (European Commission, 2026) - lightens the *volume* of what a small firm must report. But it standardises *what* a proportionate disclosure contains, not *which* themes an SME should build first amid its own particular mix of regulators, customers and financiers. It answers what a lighter report looks like; it does not answer where, among those demands, a firm should begin.

This thesis addresses that second question: where, amid the overlapping demands, a resource-constrained European seafood-processing SME should begin. Its aim is to identify the key disclosure themes - and the underlying datapoints - that, if built first, satisfy the largest share of the demands the firm faces across regulators, customers, investors and industry initiatives, and thereby to support a shift from reactive compliance to proactive strategy while helping secure both the licence to operate and the licence to sell. Neither licence is new: buyers have long imposed financial, quality and food-safety conditions beyond what the law requires, and in some areas - microbiological limits, or human-rights due diligence - the legal floor is the stricter of the two. What is new is the volume and formalisation of the sustainability-specific demand now attached to both.

It approaches the problem in four steps, each forming one empirical chapter. Step 1 (Chapter 4) scans the *ecosystem*: the full landscape of instruments acting on the sector, who demands what, organised by how obligatory each demand is and which stakeholder drives it, using the mandatory ESRS as a common reference frame. Step 2 (Chapter 5) scans the *peers*: benchmarked against that ESRS structure, what some of the sector's most advanced reporters actually disclose, establishing the realistic frontier of seafood disclosure and a universal core of topics every mature firm reports. Step 3 (Chapter 6) *crosses the two*: mapping which instruments demand which topics reveals where demand converges, the hotspots at which a single theme, once its datapoints are in place, answers many actors at once. Step 4, carried out in the same chapter because staging and tool-building draw on the same underlying matrix, *builds the tool*: it stages the surviving topics by convergence and stakeholder breadth and operationalises the result as a diagnostic with which an individual SME can locate its own position and identify what to build first. Figure 1 summarises the four steps. Chapter 3 states the research questions and intended contributions; Chapter 2 reviews the literature and develops the conceptual framework (the reactive-proactive distinction, stakeholder pressure and legitimacy); Chapter 3 sets out the shared research design. Chapter 7 draws out the implications for seafood SMEs and for EU disclosure policy, and concludes.

The contribution is threefold. For seafood SMEs, the thesis offers a diagnostic framework and a prioritised, sequenced set of topics tailored to a firm's situation, intended as a basis for strategic discussion within the firm rather than a mere compliance checklist. For the academic literature, it provides a systematic mapping of cross-stakeholder disclosure demand at sector level and an empirically grounded application of the reactive/proactive distinction. For EU policy, it supplies evidence to the ongoing simplification debate, the Omnibus package, the VSME, the postponement of sector standards, by identifying where the disclosure ecosystem converges, and therefore where coordinating or capping demand would deliver the largest reductions in SME burden. The analysis that follows shows precisely which themes those are and which datapoints they require.

2. Literature review and conceptual framework

This chapter situates the thesis in existing work and then builds the conceptual frame the rest of the analysis uses. Section 2.1 reviews the relevant literature and identifies the gap it addresses. Sections 2.2–2.4 then assemble the frame in three moves, each introducing a theoretical lens only where that lens does explanatory work for the argument: the reactive–proactive spectrum the thesis seeks to move firms along (3.2); why resource-constrained firms become stuck on its reactive side (3.3); and why building a few shared capabilities is the way off it (3.4). The necessity gradient that orders the demands themselves is developed where it is first used, in Chapter 4.

2.1 Literature review

This section situates the thesis within five strands of existing work: the growth and fragmentation of corporate sustainability reporting, the particular position of SMEs within it, the reactive–proactive posture and maturity literature, the theoretical lenses used to read multi-stakeholder demand, and the governance of seafood sustainability specifically. It closes by identifying the gap the thesis addresses.

The growth and fragmentation of sustainability reporting. Corporate sustainability reporting has moved over two decades from a voluntary, largely narrative practice to an increasingly regulated and, in the European Union, mandatory field (Hummel & Jobst, 2024; Dinh, Husmann & Melloni, 2023). Its centre of gravity has shifted toward standardised, decision-useful metrics and the double-materiality principle (Nielsen, 2023; Nahrstedt, 2026). Yet the same period produced a proliferation of overlapping standards, frameworks and ratings that observers have repeatedly criticised as fragmented and insufficiently comparable, prompting sustained calls for harmonisation (Afolabi, Ram & Rimmel, 2022). A growing body of work now examines the European Sustainability Reporting Standards (ESRS) directly - their information architecture (Fragidis & Papafloratos, 2025), their operationalisation into assessment frameworks (Ahmad et al., 2025), and their implications for manufacturing firms (Bataleblu et al., 2024) – but this literature is overwhelmingly oriented toward the large undertakings the standards were written for.

SMEs and the proportionality problem. Another strand documents the gap between this expanding reporting apparatus and the capacity of small and medium-sized enterprises. SMEs operate with limited financial and managerial resources and little dedicated reporting expertise, which constrains their environmental and sustainability engagement (Battisti & Perry, 2011). The proportionality response - reduced-scope voluntary standards such as the VSME - has begun to attract study, including of its applicability to agri-food SMEs specifically (Tugliani, Guareschi & Arfini, 2026). What this strand establishes is that lighter standards reduce the *volume* of what an SME must report, but do not resolve the prior question of *where*, within a firm's own mix of demands, a proportionate effort should be focused.

Reactive and proactive postures, and reporting maturity. A long line of strategy research distinguishes reactive from proactive environmental and sustainability postures and links the proactive posture to the development of valuable organisational capabilities and to long-run performance (Sharma & Vredenburg, 1998; Torugsa, O'Donohue & Hecker, 2013; Chan, Lai & Kim, 2022; Fazli et al., 2023). A parallel maturity literature characterises the stages through which firms progress and the difficulty of moving between them (Baumgartner & Ebner, 2010; Cöster et al., 2020; Lozano, 2013). Together these provide the posture-and-maturity vocabulary the thesis adopts, but they are largely pitched at the level of the individual firm's strategy and say little about how the *external structure of demand* shapes whether a small firm can make the reactive-to-proactive transition at all.

Theoretical lenses on multi-stakeholder demand. Three established lenses help read that external structure. Stakeholder-salience theory (Mitchell, Agle & Wood, 1997) explains how firms triage competing claims by power, legitimacy and urgency. Institutional theory (DiMaggio & Powell, 1983) explains why disclosure conventions converge across firms through coercive, mimetic and normative isomorphism even where formally voluntary. Legitimacy theory (Suchman, 1995) distinguishes pragmatic, moral and cognitive legitimacy, a gradient that maps onto the progression from legally enforced to normatively expected obligation. The thesis draws these together in its conceptual framework (Sections 2.2–2.4).

The governance of seafood sustainability. Finally, a sector-specific literature characterises seafood sustainability governance as a crowded, multi-actor field in which regulators, retailers, certifiers, NGOs and investors exert parallel and overlapping demands (Bush & Oosterveer, 2019; Barclay & Miller, 2018), coordinated through certification schemes, improvement projects and voluntary initiatives. This work describes the governance landscape richly, but treats it largely from the standpoint of the movement and its institutions rather than from the standpoint of a single resource-constrained firm trying to answer all of them at once.

The gap. Across these strands, the reporting literature maps frameworks, the strategy literature studies posture, the theoretical literature explains multi-stakeholder pressure, and the seafood literature describes sector governance - but they remain largely separate. No existing work systematically maps cross-stakeholder disclosure *demand* at the level of a single sector and firm, in a way that tells a resource-constrained SME which disclosure topics, if built first, would satisfy the largest share of the demands it actually faces. Supplying that mapping - and the diagnostic that follows from it - is the contribution this thesis makes. The remainder of the chapter builds the conceptual framework through which it is done.

2.2 The reactive–proactive sustainability spectrum

The reactive–proactive distinction (Sharma & Vredenburg, 1998; Torugsa et al., 2013) provides the thesis's central organising frame. It is the spectrum along which a firm's sustainability posture sits, and the shift from its reactive to its proactive end is the change the thesis ultimately seeks to enable. The two postures differ along several dimensions:

Dimension	Reactive posture	Proactive posture
Trigger for action	External demand (questionnaire, audit, regulatory deadline)	Strategic anticipation; voluntary commitments
Disclosure approach	Transactional, question-by-question; minimum viable	Capability-driven, integrated; goes beyond legal minimum
Data infrastructure	Ad-hoc, manual, rebuilt per request	Systematic, automated, reusable across demands
Strategic positioning	Compliance cost center	Competitive differentiation; risk-adjusted financial benefit
Stakeholder relationship	Defensive; minimum communication	Proactive engagement; trust-building

Empirical CSR research also associates the proactive posture with positive long-term financial performance (Torugsa et al.). The barrier to adoption among resource-constrained SMEs is not rejection of the proactive posture in principle but the absence of a tractable path from reactive practice to proactive capability, given the fragmented and unpredictable nature of external demand. Why such firms remain on the reactive side of the spectrum despite its costs - and why that is a rational response rather than a failure of intent - is the question the next section takes up.

2.3 Why resource-constrained firms stay reactive

The reactive posture is best read not as a failure of intent but as the predictable response to how sustainability demand is structured. Stakeholder-salience theory (Mitchell, Agle & Wood, 1997) explains how a firm triages competing claims by their power, legitimacy and urgency. Where one stakeholder dominates, the firm can order its response cleanly. In the seafood sustainability field, however, regulators, retail customers, certifiers, investors and NGOs each press legitimate but uncoordinated demands through their own channels and on their own timetables. With no single claim dominant and no coordination among them, the firm is left continually re-triaging whichever demand is most urgent at the moment - and continual re-triage, meeting each demand as it lands, *is* the reactive posture. Salience theory thus supplies the mechanism behind the reactivity that Section 2.2 describes: it is the structure of fragmented demand, not indifference, that keeps a small firm firefighting.

The fragmented governance of the seafood sustainability field, regulators, brands, retailers, certifiers, NGOs and investors all exercising parallel and overlapping demands (Bush & Oosterveer, 2019; Barclay & Miller, 2018), is a sharp instance of this multi-stakeholder structure. Each demand is individually legitimate, yet their cumulative weight outstrips what a resource-constrained firm can systematically plan for, so meeting them one at a time becomes the rational adaptation this thesis takes as its starting premise.

2.4 Why building shared capabilities is the way out

If fragmentation is what creates the reactive trap, the way out lies in its mirror image: beneath the many differently worded demands sits a much smaller body of shared data, and the demands converge on it. Two ideas turn that observation into a strategy.

The first explains why the convergence is real rather than coincidental. Institutional theory (DiMaggio & Powell, 1983) shows why disclosure conventions converge across firms and instruments even where formally voluntary: coercive isomorphism (regulatory pressure), mimetic isomorphism (copying perceived leaders) and normative isomorphism (the professionalisation of the sustainability function) all push independent actors toward the same disclosure choices. When a customer certification, an investor questionnaire, an industry framework and a regulation all ask for the same topic, that agreement reflects a shared, institutionally reinforced expectation rather than chance. This is what makes it meaningful to map where demand converges, and to read a topic demanded through many channels as a genuine sector-wide priority - the interpretation the cohort's universal core is given in Chapter 5. The mimetic mechanism bears particularly on SMEs, whose limited managerial resources and dedicated reporting expertise (Battisti & Perry, 2011) leave little capacity to form independent positions on what to disclose; following what peers and buyers already treat as standard is the lower-cost route.

The second explains why acting on that convergence pays. Where the same topic is demanded by a customer, an investor, an industry framework and a regulator at once, a firm that builds the capability to report it earns compound returns: one investment answers many demands. By *capability* this thesis means something concrete - the ability to produce a disclosure theme's underlying datapoints reliably and repeatably, as a standing process rather than a one-off assembly. This is the resource-based logic of the firm (Barney, 1991) and the dynamic-capabilities literature (Teece, 2007): a capability, once built, is fungible across every demand that draws on it. Because what one framework asks is so often also asked by another, a capability built to satisfy a mandatory requirement already supplies much of what the voluntary and value-chain requirements on the same topic need. The marginal cost of proactive disclosure is therefore far lower than it appears when each demand is costed in isolation: the proactive firm recognises the overlap and builds once, whereas the reactive firm rebuilds requirement by requirement and perpetually pays the full isolated cost of each.

This is the conceptual bridge from reactive to proactive, and it sets the agenda for the four empirical steps. Acting on convergence requires knowing three things, then acting on them. First, which instruments demand what, and how bindingly: the ecosystem and its necessity gradient, scanned in Chapter 4 (Step 1), where the progression from legally enforced to normatively expected obligation is read through legitimacy theory (Suchman, 1995). Second, what some of the sector's most advanced firms actually report - the realistic frontier, scanned in Chapter 5 (Step 2). Third, where demand converges across the two into the topics an individual firm should build first (Step 3). Acting on that finding is Step 4, which turns it into a diagnostic tool. Steps 3 and 4 are both mapped in Chapter 6.

3. Research questions and research design

3.1 Research questions

The research is organised around three interconnected questions, one for each of the first three of the four empirical steps that follow (Chapters 4–6; the third question is answered in two steps carried out in the same chapter - locating convergence, then building the diagnostic on it):

1. **RQ1 - Descriptive (the ecosystem):** What is the landscape of sustainability-disclosure instruments acting on a European seafood-processing SME, and how does it order when each instrument is classified by how binding it is (its *necessity*) and which stakeholder drives it (its *source*)?

2. **RQ2 - Empirical (the frontier):** Benchmarked against the ESRS structure, what do some of the sector's most advanced reporters actually disclose - and what does this reveal about the realistic upper bound of seafood disclosure depth and the universal core of topics every mature firm reports?

3. **RQ3 - Prescriptive (the convergence):** Across that landscape and that topic set, where does cross-stakeholder demand *converge*, and which disclosure topics - if built first - let a resource-constrained processor satisfy the largest share of the demands it faces, when this prioritisation is operationalised as a diagnostic tool the firm can apply to its own situation?

Each question is answered by one step: RQ1 by the ecosystem scan (Step 1, Chapter 4), RQ2 by the cohort scan (Step 2, Chapter 5), and RQ3 by the cross-framework demand mapping and the diagnostic tool built on it (Steps 3 and 4, both in Chapter 6).

The study approaches its question in four steps rather than as a single linear pipeline: an ecosystem scan, a cohort scan, the overlap between them, and the tool built on that overlap. Figure 1 sets them out.

Figure 1: The thesis's four-step methodology, answering RQ1–RQ3 above: Step 1 scans the disclosure ecosystem (Chapter 4, RQ1), Step 2 scans the cohort frontier (Chapter 5, RQ2), and Steps 3 and 4 - crossing the two to find where demand converges, then building the diagnostic tool on that finding - are carried out together in Chapter 6 (RQ3). Neither scan is actionable on its own: the ecosystem scan returns 49 catalogued instruments and the cohort scan 70 disclosure requirements, and it is their intersection rather than either list that yields a workable set of priorities. Chapters 5–7 each return to this figure as their starting point.

3.2 Intended contributions

The thesis contributes to three audiences:

- **For seafood SMEs:** the thesis delivers a diagnostic framework and a prioritised, sequenced set of build-first disclosure topics - and the datapoints beneath them - tailored to the firm's own mix of demands, intended as a basis for strategic discussion within the firm rather than a compliance checklist.
- **For EU policy:** it offers empirical input to the ongoing simplification debate (Omnibus, VSME extension, sector standards postponement). The cross-stakeholder hotspot analysis identifies where the regulatory and voluntary disclosure ecosystem is converging and therefore where coordination or capping of demand would deliver disproportionate burden reduction for SMEs. This aligns with the Competitiveness Compass target of cutting administrative burden by $\geq$35% for SMEs (European Commission, 2025).
- **For academia:** it provides a first systematic mapping of cross-stakeholder disclosure demand at the sector level using a defined coverage scale, extends the sustainability-reporting maturity-model literature by operationalising it as a diagnostic tool, and applies the reactive/proactive CSR distinction to empirical material.

3.3 Research design

Organisation of the thesis, and its reasoning. The thesis integrates its methodology and its results rather than presenting them in separate chapters, because the four steps use different data and different methods. *Each of the three empirical chapters is therefore self-contained*: Chapters 4, 5 and 6 each open with the reasoning behind their step, set out their own data and coding procedure, report their own findings, and close by interpreting what those findings do and do not establish. Chapter 6 carries two steps rather than one, because locating convergence and building the diagnostic on it are drawn from the same matrix.

This structure is deliberate. A pooled methodology chapter would have to describe three unrelated procedures in the abstract, before the reader knows what any of them is for, and a pooled results chapter would then separate each finding from the method that produced it. Keeping them together makes each step independently auditable.

What remains genuinely shared is presented in the remainder of this section: the common measurement framework, the cohort, the data and coding procedures, and the validation strategy. Chapter 7 then draws the three sets of findings together, reads them against the conceptual framework of Chapter 2, and states the limitations that apply across all of them.

A common reference frame. To compare and aggregate demand across a fragmented field of instruments, the analysis adopts the European Sustainability

Reporting Standards (ESRS) as a common coordinate system - the most comprehensive and legally anchored topic structure available, its ten topical standards (E1–E5, S1–S4, G1) resolving into a defined set of disclosure requirements. Instruments and reports written to other frameworks are read onto this structure, so that every demand and every disclosure is expressed in one vocabulary. The choice, and its limits, are developed where the frame is first used (Chapter 4 for the instrument landscape, Chapter 5 for the cohort).

Cohort and case selection. The empirical work centres on six large European seafood processors, selected purposively as among the sector's most advanced sustainability reporters - not as a representative sample but as an upper-bound proxy for what the sector can currently disclose. The selection spans the main sourcing models so that the disclosure pattern is not an artefact of a single production system; the full rationale and the cohort table are given in Chapter 5.

Data and coding. The analysis rests on two hand-built, hand-coded datasets: the cohort firms' most recent sustainability reports, coded at disclosure-requirement level against the ESRS structure (Chapter 5); and a catalogue of the regulatory, customer, investor and industry instruments active in the sector, each coded for the demand it places on each disclosure topic (Chapters 4 and 6). Coding was performed manually from primary documents and cross-checked against the underlying datapoints; the per-cell coding protocol is set out in Chapter 6.

Validation. The framework rests on three legs that do not depend on outside confirmation. Its construction is grounded in the theory of Chapter 2 and in primary regulatory and standard-setting documents. Its coding is checked for consistency against defined datapoints and logged cell by cell. And it is applied to a worked case (Chapter 6) to show that it yields a usable prioritisation.

A fourth leg is external and, at the time of writing, incomplete. The instrument classification of Chapter 4 was discussed with industry contacts at Seafood Europe during its construction, and the completed classification and cohort coding have been submitted to the sustainability leads at the six cohort firms for review. This is member checking (Lincoln & Guba, 1985): a corroborating step, not a formal reliability test. Any corrections it returns will be incorporated before final submission, and the analysis does not depend on it. Practitioner validation of the worked case with a firm itself is left to future work.

4. The disclosure ecosystem

The first step scans the *ecosystem*: it establishes which sustainability instruments act on a European seafood-processing SME, and classifies each by how obligatory it is and which stakeholder drives it. In doing so it constructs the first component of the study's central instrument - the demand matrix - by fixing and ordering its columns. The question it answers is prior and structural, and it is the thesis's first: which actors make demands of a seafood processor, and how binding those demands are. Establishing that is what makes the topic-by-topic scoring in Chapter 6 interpretable, because a demand's weight depends on who makes it and on whether it can be declined. This is Step 1 of the methodology set out in Figure 1.

4.1 Constructing the necessity–domain map

Identifying the instruments. The instrument set was assembled in three steps. First, candidate instruments were identified from two complementary sources: the sustainability reports of the six large seafood processors that form this study's cohort (Chapter 5), which show the regulations, certifications and frameworks that leading firms in the sector actually respond to; and a structured scan of EU sustainability law and the principal voluntary standards, ratings and initiatives active in the sector. Second, each candidate was screened against two inclusion gates: that it bears on an environmental, social or governance matter (excluding purely fiscal, customs and data-protection law), and that it places a binding obligation on the *organisation itself* - a disclosure, due-diligence, certification or market-access requirement - rather than merely binding Member States or granting an individual employee an entitlement. Third, the classification was built against how the demands are experienced in practice rather than against formal legal status alone. Industry contacts at Seafood Europe were consulted on the necessity ranking while it was being drawn up, and the completed classification has been submitted to the cohort firms for review (Chapter 3, 3.3 Research design).

A status for each instrument. The catalogue deliberately keeps the *whole* field in view: every instrument that acts on the sector is placed in the map, whether or not it ultimately enters the scored analysis. Each therefore carries a status alongside its tier, domain, source and locus - the four classifying attributes introduced below. An instrument is *scored*; held out as the CSRD/ESRS *baseline* reference; set aside as *context*, binding Member States rather than the processor, as the Common Fisheries Policy does; or *excluded* for failing a screening gate, as a purely fiscal law does, or a benchmark that rates other schemes rather than demanding anything of the processor.

The necessity axis: four tiers. The framework's spine ranks each instrument by how imperative it is from the SME's own perspective, on a gradient of obligation that runs from *coercive*, legally enforced requirements at the base to *normative*, voluntary commitments at the top - the institutional gradient developed in Chapter 2 (DiMaggio & Powell, 1983; Suchman, 1995). The four tiers make it operational, each a point on that gradient:

- **Tier 1 - legal licence to operate.** Binding law, without which the firm cannot lawfully trade: the IUU and Fisheries Control Regulations, packaging law, occupational-safety law - coercive obligation in its purest form. The tier is deliberately restricted to legal obligation. Customer requirements can be just as non-negotiable commercially, but they are separated into Tier 2 below, because the consequence of failing them differs in kind: a firm that breaches Tier 1 cannot lawfully trade at all, whereas one that fails Tier 2 loses a particular customer.

- **Tier 2 - market licence to operate.** Legally optional but commercially non-negotiable: a recognised certification, social audit or buyer assessment - MSC and ASC certification, SMETA audits and EcoVadis ratings are the most common examples in this sector, not an exhaustive list - without which a major retail or foodservice customer will not list a supplier. US Foods, for instance, requires any GSSI-benchmarked certification (MSC or BAP are given only as examples) for its top supplier tier and reserves the right to end a supplier relationship over unresolved non-compliance (US Foods Holding Corp., 2025); several major retailers make MSC or ASC specifically a sourcing-policy condition (Blank, 2025). The obligation is enforced by the market rather than the state, and which specific scheme satisfies it varies by buyer and species.

- **Tier 3 - competitive expectation.** Increasingly normalised reporting frameworks whose absence is a competitive disadvantage - the VSME, GRI, and the value-chain disclosures cascaded from customers under CSRD. This is the mimetic-normative middle, where firms adopt what peers and buyers expect.

- **Tier 4 - strategic differentiation.** Genuinely voluntary commitments that signal leadership (SBTi, the UN Global Compact, TNFD). Adoption is elective and reputational.

The tier assignment classifies the *type* of obligation an instrument carries, not the standing the market accords it, and the two can diverge. The ISSF is the clearest instance: it meets the Tier 4 definition, being voluntary and reputational, but sector practitioners read it functionally as a transitional step rather than a leadership signal. On this view a tuna processor sourcing against a full MSC certification has no remaining need for ISSF conformance, so sustained ISSF membership in place of MSC can indicate the absence of certification rather than differentiation beyond it. The instrument is retained in Tier 4 because the classification tracks obligation type, but the reading is recorded here because it bears on how the tier should be interpreted.

The tiers rank the *instruments* by how binding each is; they do not by themselves say which *topics* a firm should build first. A firm that answers only the binding tiers as each demand lands is reactive; the proactive move (Sharma & Vredenburg, 1998; Torugsa et al., 2013; the distinction developed in Chapter 2) is to build, ahead of demand, the shared datapoints that recur across many instruments regardless of tier, determined instead by how far demand *converges* across the ecosystem, how many stakeholder channels ask for a topic, not by tier position.

The domain axis. Cutting across the tiers, the domain axis separates *seafood-sector-specific* instruments (the fisheries and aquaculture regulations and certifications, and the GDST traceability standard) from *general corporate-sustainability* instruments that apply to the processor as an enterprise like any other (climate, packaging, labour and governance law and the cross-sector reporting frameworks). The general domain includes the ethical and human-rights strand: the CSDDD's due-diligence duty, the Forced Labour Regulation, and the social audits and buyer assessments (SMETA, EcoVadis) through which buyers verify labour conditions. It is worth naming separately because it is the clearest case in which the legal floor can sit *above* the customer requirement rather than below it, and because it is where the ESRS own-workforce and value-chain worker requirements (S1 and S2) carry most of their weight.

Two attributes: source and obligation locus. Each in-scope instrument carries two further tags. Its *source* is the stakeholder whose demand the processor actually answers, assigned by a single test - does the instrument legally bind the processor? Where it does, the source is the *regulator* (the Fisheries Control Regulation binds the processor directly). Where it does not, the source is the stakeholder that transmits the demand: a retailer-required certification such as MSC codes to *customer*; a capital-market standard such as SFDR to *investor*; an NGO-led framework such as GDST to *industry/NGO*. This test deliberately reassigns instruments whose nominal issuer is a regulator to the channel through which a sub-threshold SME actually meets them: CSRD and CSDDD, written for large undertakings, reach the small processor not as law it must obey but as data its large customers must extract from it, and so code to *customer*. The same reasoning codes the VSME to the customer channel: although a voluntary standard, in practice a small seafood processor adopts it to answer a large customer's value-chain request rather than to address capital markets. This does partly double-count the ESRS content that also reaches the firm through CSDDD - a limitation accepted here because, for seafood processors specifically, that demand genuinely arrives through the customer channel.

The *obligation locus* records, separately from source, whether the processor is itself the bound party: *direct* where the processor is the obligated, certified or reporting entity (a processor holding MSC chain-of-custody certification, which it maintains and is audited against); *indirect* where the obligation sits elsewhere and reaches the processor through a buyer's demand or its own disclosure duty (CSDDD, which binds the large customer and arrives as a value-chain questionnaire); and *context* where the instrument binds a Member State or another actor with no mechanism that transmits an obligation to the processor at all (the Common Fisheries Policy). Holding source and locus apart prevents the common error of reading the legally bound party off the identity of the actor who actually applies the pressure.

Two caveats on the map. First, the instrument set is calibrated to leading firms: it was screened from the reports of six large processors - selected on criteria set out in Section , which also addresses what "leading" can and cannot be taken to mean - so it captures what mature, larger corporates respond to. It may therefore both include instruments a small SME does not yet face and understate informal or local demands it does. Second, necessity is profile-dependent. The tiering assumes the modelled SME - a business-to-business supplier selling to large EU retailers and exporting - for which retailer-demanded certifications are a genuine market licence; a firm selling locally or direct to consumers would re-tier several instruments.

4.2 The instrument landscape

The catalogue comprises **49 instruments** (Figure 2) - the full field acting on a European seafood processor. Read as counts, the landscape has three features.

Of the 49, **30** are carried forward into the demand scoring of Chapter 6. The division is not one of importance but of what can be scored. An instrument enters the matrix only if it places a demand on the processor *itself* that can be traced to a specific disclosure topic. The remaining 19 fail that test for one of three reasons: they bind a Member State or another actor rather than the firm (the Common Fisheries Policy, the Marine Strategy Framework Directive); they set no organisation-level reporting obligation at all (the UN SDGs, Fishery Improvement Projects); or they rate the firm from outside without asking it for anything (the Dow Jones Best-in-Class Indices, Coller FAIRR). All 49 are catalogued here because the chapter's question is what the demand environment *contains*; only the 30 can answer Chapter 6's question, which is what that environment *asks for*.

By *necessity tier*, it is weighted toward the binding base: 19 instruments sit at Tier 1 (legal licence to operate), 10 at Tier 2 (market licence), 11 at Tier 3 (competitive expectation) and 9 at Tier 4 (strategic differentiation). The mandatory legal floor is the single largest block.

By *stakeholder source*, regulators account for 19 instruments and customers for 14 - together two-thirds of the field - while investors (6) and industry or NGO initiatives (10) make up the remainder.[2] The demand environment is thus overwhelmingly a regulator-and-customer one. This records where demands come from, not a judgement on why they are made. Both channels exist to move the sector toward outcomes it has largely endorsed itself, and retailer requirements in particular often codify what a well-run processor would do anyway. The argument here is not that the demands are unwelcome. It is that answering them one at a time is what makes them expensive.

By *domain*, 19 instruments are seafood-sector-specific - the fisheries and aquaculture regulations and certifications together with the GDST traceability standard - and 30 are general corporate-sustainability instruments. Most of what acts on a seafood processor is not peculiar to seafood at all.

[2] These counts apply the VSME reassignment set out above: the VSME is issued for capital-market and value-chain use, but a sub-threshold seafood processor adopts it to answer a large customer, so it is coded to the customer channel rather than the investor one. The source workbook records its issuer-based classification, which is why it shows 13 customer and 7 investor instruments.

Of these 49, thirty enter the scored demand matrix in Chapter 6; the remainder are held out as the CSRD/ESRS baseline, set aside as context because they bind Member States rather than the processor, or excluded for failing a screening gate. Every instrument named here is defined in full - the organisation behind it, its legal reference, current status and official source - in the Appendix A - Ecosystem Glossary.

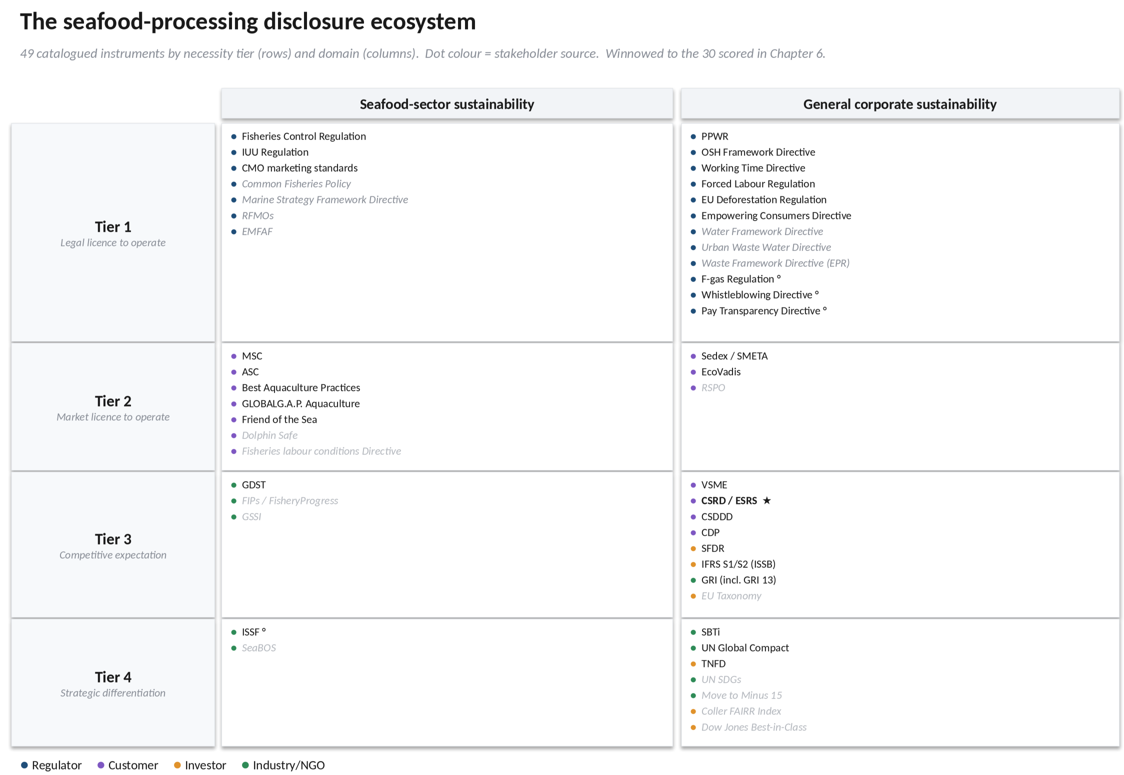

Figure 2: The seafood-processing disclosure ecosystem: all 49 catalogued instruments arranged by necessity tier (rows) and domain (columns). Dot colour marks the stakeholder source. Instruments set in *grey italic* are catalogued but not scored, either because they bind Member States rather than the processor or because they were excluded at screening; the reason in each case is recorded in the Appendix A - Ecosystem Glossary. A degree symbol (“ ° ”) marks a firm-profile opt-in and ⋆ the CSRD/ESRS baseline, which is carried as a reference point and excluded from the sums. The winnowing to the 30 scored instruments is set out in Chapter 6. A full reference entry for each instrument - the body behind it, its legal reference, status and official source - is given in the Appendix A - Ecosystem Glossary.

This map fixes the columns of the demand matrix: the actors making the demand, and the degree to which that demand is binding. What each instrument actually demands, topic by topic, and where those demands converge into build-first priorities, is taken up in Chapter 6. First, however, Chapter 5 turns to the rows - establishing, from the disclosures of the sector's leading reporters, what a mature seafood firm actually reports.

5. The seafood disclosure frontier: evidence from six large reporters

This chapter answers the thesis's second research question: *what do mature European seafood processors actually disclose?* It establishes that practice from primary documents, setting the assumed *upper bound* of sector disclosure depth. Both the demand mapping that follows and the expectations placed on smaller firms should be read against it. The cohort is not a picture of typical practice; it is a picture of the best-in-class frontier. This is Step 2 of the methodology set out in Figure 1.

Scope note: this is a coverage analysis, not a judgement of reporting quality. It measures whether a topic is disclosed at all, coded from each firm's own published sustainability report or ESRS-index appendix, using only public information - no non-public data, site visits or firm-specific verification. A firm scoring lower may simply have a different production system, framework, or reporting year, not weaker sustainability performance; no ranking or evaluative claim about the six companies is intended.

5.1 The frontier as a disclosure-maturity question

Corporate sustainability disclosure is not adopted all at once but develops through stages of organisational maturity: firms move from ad-hoc, reactive reporting toward systematic, integrated disclosure as capability accumulates (Baumgartner & Ebner, 2010), and the completeness and quality of a firm's reporting can be read as a position on that maturity gradient (Cöster et al., 2020). The most advanced reporters in a sector therefore mark its practical *frontier*: the upper bound of what is currently feasible - and relevant to seafood processing - to disclose given real data systems and resources. This research therefore treats the frontier as a ceiling that resource-constrained SMEs can grow into, not one they can replicate today.

Two features make that frontier analytically useful here. First, when processors as different as a vertically integrated salmon operation and a pure whitefish processor nonetheless disclose the same handful of topics, that agreement is unlikely to be coincidence. It reflects a shared expectation - what the sector has collectively come to treat as standard practice for a credible reporter - rather than each firm's independent choice; institutional theory calls this the mimetic and normative pressure that makes practices converge across an industry (DiMaggio

& Powell, 1983). This is an interpretive reading, and a competing one is possible: firms may converge because the same topics carry the greatest reputational risk, or simply because they are the easiest to report. That each firm's disclosures are anchored in its own double-materiality assessment is a partial check against pure convenience, though not a decisive one. Read either way, a universal core is the sector's *revealed* consensus on what a serious reporter covers - what counts as complete disclosure in practice, not just a coincidental overlap between six firms' independent choices. Second, the frontier calibrates the realistic ambition for a resource-constrained firm: if even the best-resourced reporters leave requirements uncovered, completeness is the wrong yardstick for an SME, and the question becomes which subset of disclosures matters most - the question the demand mapping in the following chapter takes up.

A qualification is needed on what that diversity establishes. The six firms differ in sourcing model, ownership and regulatory regime. They are alike, however, in the respect that most plausibly drives disclosure: all are seafood processors selling into a common European buyer base, meeting overlapping certification demands, and advised by an overlapping professional community. A practitioner reviewer put the objection directly - beneath the sourcing-model differences, the cohort is substantially homogeneous. The observation is correct, and it shapes how the convergence result should be read. What the cohort establishes is that firms facing a common demand environment converge on a common disclosure core. What it cannot establish is that the same core would emerge among firms facing different demand environments. The first is what this thesis claims. It is also what the institutional account above predicts: convergence under shared pressure is the expected outcome rather than a surprising one. The value of the core is therefore not that it is unexpected, but that it is identifiable. It marks where the sector's demand environment has already settled, which is what a firm entering that environment needs to know.

Existing work characterises sustainability-reporting maturity in general terms (Baumgartner & Ebner, 2010) and describes seafood sustainability governance at the level of the movement and its institutions (Bush & Oosterveer, 2019; Barclay & Miller, 2018), but does not map, at disclosure-requirement level, what the sector's leading firms actually report. This chapter supplies that map and reads it for three things: how *much* the leading firms disclose (the height of the frontier), which topics *every* firm reports (the universal core), and which relevant topics *none* of them report (the sector's collective gaps).

5.2 The cohort and case selection

Six large seafood processors were selected purposively as among the most advanced sustainability reporters in the sector rather than as a representative sample. The selection deliberately spans the sector's main sourcing models, so that the disclosure pattern that emerges cannot be an artefact of a single production system: wild-capture tuna (Thai Union, Bolton), aquaculture salmon (Mowi), an integrated wild-plus-aquaculture operation (Profand), and pure processing/branded firms whose variation comes from the species and origins they source rather than from a distinct production system of their own (Nomad Foods, Espersen). All six are seafood *processors*, but they differ in how far they integrate upstream - from Mowi, which farms its own salmon and mills its own feed, to Espersen, a near-pure whitefish processor. This matters for reading the reports: Mowi discloses feed and farming as its *own* operations, while Espersen meets the same topics only as *sourcing* questions about its suppliers - the two are not directly comparable without accounting for this difference.

The six were chosen on three criteria: they are among the largest seafood processors by turnover; each publishes a full, structured sustainability report under the ESRS or the GRI Standards; and together they span its main sourcing models.

The second criterion carries most of the weight, and it is worth being precise about what it does and does not establish. "Most advanced" here is a sampling frame rather than a measured rank. Of the roughly 3,245 EU enterprises that process fish as their main activity, about two-thirds are micro-enterprises and only some two per cent employ 250 people or more (STECF, 2025). The overwhelming majority therefore sit below the CSRD reporting thresholds and face no obligation to publish a structured sustainability report at all. Publishing one regardless is itself the distinguishing act, and the six are drawn from the small population that does. No claim is made that they are the six *best* reporters in the sector, only that they sit inside the set that reports at all and, within it, span the sourcing models and the size range.

The frame admits more firms than the six coded here, and one is worth naming. Parlevliet & van der Plas, a large European fishing and processing group, publishes a sustainability report with a GRI content index and so satisfies every criterion above. It was not coded. The cohort was capped at six to keep requirement-level manual coding tractable within the scope of this study, and the six were chosen to span the sourcing models and both reporting frameworks; the cap, not the frame, excluded the seventh. The omission is stated rather than argued away, and its consequence can be bounded rather than assumed. The leave-one-out and leave-two-out testing in Section shows that the universal core is retained under every removal of one or two firms from this cohort, and that removing a firm can only widen it. A core defined on more firms is therefore no less likely to contain these 13 requirements, though a seventh firm could of course have narrowed the wider near-universal set, which is one reason no claim in this chapter rests on that set.

Whether firms selected this way are advanced in absolute terms is a separate question, and one this chapter's own results answer rather than assume. They are not complete: the deepest reporter covers 60 of the 70 topical requirements and the shallowest 33, and seven requirements are reported by no firm in the cohort (Sections 5.4 and 5.6). The frontier they mark is therefore a realistic ceiling rather than a standard of completeness - which is precisely why it is a usable benchmark for a smaller firm.

The aim of the selection is therefore not representativeness but reach: to assemble the firms most likely to have pushed disclosure to its current limit, so that the pattern common to *them* marks the sector's realistic ceiling. By 2024 turnover the six span more than a tenfold range: Mowi (EUR 5.6bn) and Thai Union (THB 138.4bn, approx. US$4.0bn / EUR 3.6bn) are the largest, followed by Bolton Group (EUR 3.5bn group-wide, of which approx. EUR 2.4bn is its food division) and Nomad Foods (EUR 3.1bn), with Profand (approx. EUR 1.1bn) and Espersen (DKK 3.3bn, approx. EUR 440m) smaller but still among the sector's largest specialist processors.[3] The selection also gives geographical spread across the sector's main European processing regions - from southern Europe (Bolton in Italy, Profand in Spain) to the Nordics (Espersen in Denmark, Mowi in Norway) and the UK (Nomad Foods) - with Thai Union added as one of the largest seafood processors supplying the EU market from outside it, extending the cohort's reach beyond Europe without abandoning the EU-market focus. Table 1 summarises the cohort.

Table 1: Cohort overview and reporting coverage. Coverage is reported disclosure requirements (DRs) as a share of the 70 topical ESRS disclosure requirements (E1–E5, S1–S4, G1); see the Section 5.3 section.

Company	**HQ**	**Sourcing model**	**Reporting basis**	**DRs**	**Coverage**
Thai Union	Thailand	Wild-capture tuna	GRI, in accordance	60	86%
Nomad Foods	UK	Multi-species	GRI, with reference	52	74%
Bolton	Italy	Wild-capture tuna	ESRS	45	64%
Profand	Spain	Wild + aquaculture	CSRD double materiality; GRI supporting	43	61%
Espersen	Denmark	Multi-species	ESRS	34	49%
Mowi	Norway	Salmon aquaculture	ESRS	33	47%

Turnover and coverage do not track each other within the cohort: Mowi, the largest firm by 2024 turnover, reports the fewest requirements of the six (47%), while the smaller Profand and Nomad Foods both report more. Firm size is therefore not read here as a driver of reporting depth; what varies across the six is each firm's own reporting maturity and sourcing model, not the resources it commands.

[3] 2024 group turnover, most recent published results, converted to EUR at approximate contemporaneous rates where the firm did not report in EUR: Thai Union (*Thai Union FY2024 press release*, thaiunion.com); Mowi (*Mowi Q4 2024 Report*, mowi.com); Bolton Group (Undercurrent News, "Tuna sales boost Bolton's turnover, profit in 2024," Aug. 2025); Nomad Foods (Q4/FY2024 earnings release, nomadfoods.com); Profand (Undercurrent News, "Spain's Profand surpasses EUR1bn revenue in 2024," Apr. 2025); Espersen (Undercurrent News / SeafoodSource coverage of FY2024 results). Four of the six - Bolton, Mowi, Nomad Foods and Thai Union - also appear among Mordor Intelligence's top-ranked players in the European seafood market (mordorintelligence.com/industry-reports/europe-seafood-market/companies); that source does not cover Profand or Espersen, so cohort-wide ranking evidence is partial rather than complete.

The six also differ sharply in structure, and it is worth stating why they remain comparable. Mowi is a vertically integrated aquaculture producer; Thai Union depends heavily on externally sourced raw material; Profand combines fishing, aquaculture, processing and commercialisation in one group. Those differences shape both which topics each firm's double-materiality assessment returns as material and how deeply it is expected to report on them. Three features of the design absorb that. First, the denominator is external and identical for all six: the 70 topical ESRS disclosure requirements, not a cohort-derived list, so no firm is scored against another's materiality assessment. Second, the measure is breadth at requirement level - whether a topic is addressed at all, a question that can be put to any business model - rather than depth or quality. Third, and most important, the load-bearing result of this chapter is not the per-firm figure but the set of requirements reported by all six. Structural difference makes that intersection harder to reach, so agreement across six unlike firms is stronger evidence of a genuine sector core than agreement across six similar ones would be. Where the differences do bite is on the individual coverage figures, which is one reason the chapter draws no ranking from them.

One feature of the cohort must be stated plainly at the outset, because it bounds every claim that follows: the cohort is a purposive *upper-bound* sample. It is taken as a proxy for what is feasible and important to disclose in this sector - an assumption the robustness checks below probe - not for what a typical firm does, and no inference about average practice is drawn from it. How the six firms' disclosures, reported under two different frameworks, were placed on a single comparable measure is set out next.

5.3 Data and method

The ESRS as the common ruler. The measure is built on the European Sustainability Reporting Standards (ESRS), the disclosure taxonomy that operationalises the EU's Corporate Sustainability Reporting Directive. The ESRS are used here not because every cohort firm reports under them - three do not - but because they are the most comprehensive and legally anchored topic structure available. Ten topical standards (E1–E5, S1–S4, G1)[4] resolve into a defined, published set of disclosure requirements. That fixed structure gives one stable ruler onto which disclosures written under any framework can be placed and compared. It is what makes a cross-firm comparison - and, in later chapters, a cross-instrument one - possible at all. Coverage in this chapter is therefore always *ESRS-aligned coverage in practice*, never a judgement of legal ESRS compliance. As the standard mandatory for CSRD-scoped firms and the most complete public sustainability taxonomy in force, the ESRS is also the natural benchmark: even under the Omnibus simplification it remains the reference against which other frameworks - including the VSME and, through the interoperability mapping

[4] E1 Climate change; E2 Pollution; E3 Water and marine resources; E4 Biodiversity and ecosystems; E5 Resource use and circular economy; S1 Own workforce; S2 Workers in the value chain; S3 Affected communities; S4 Consumers and end-users; G1 Business conduct.

used below, the GRI - are aligned (Hummel & Jobst, 2024).

Data and scope. The primary sources are the firms' own published sustainability reports: the most recent report for each firm, from financial year 2024 and, where already published, 2025 (the exploratory year-over-year comparison later in the chapter uses the three firms for which both are available), supplemented where available by each firm's CSRD-aligned ESRS-index appendix. Coding proceeds from each report's own framework index - the ESRS index appendix where the firm publishes one, and the GRI content index where it does not - rather than from a full reading of the report body. This is a deliberate choice rather than a shortcut. Reading six reports of very different length and structure in full, one analyst and without a fixed protocol, would introduce exactly the inconsistency the measure is meant to avoid: a long report read attentively would score above a short one read quickly, and the resulting differences would record the reading rather than the reporting. The index is the firm's own declaration of what it reports and where, it is available for every firm in the cohort, and it is applied identically to all six. Each indexed disclosure was coded against the *topical* ESRS standards - the ten environmental, social and governance standards (E1–E5, S1–S4, G1) and their disclosure requirements. The two cross-cutting standards, ESRS 1 and ESRS 2 (general principles and general disclosures), are deliberately out of scope: they govern *how* a firm reports rather than *which* sustainability topics it addresses, and it is topical coverage that this chapter measures.

Unit of analysis: material topics, not IROs. Coding is at the level of the ESRS topical *disclosure requirements* (DRs) - the material sustainability topics a report addresses - and deliberately not at the level of the firm-specific impacts, risks and opportunities (IROs) that sit beneath a double-materiality assessment. Three reasons make the DR the right grain. First, IROs are entity-specific by design: under ESRS 1 they are the output of each undertaking's own double-materiality assessment, so one firm's impacts, risks and opportunities are not defined on the same basis as another's and cannot be aggregated into a cross-firm coverage measure. Second, the question here is one of *breadth* - which topics a mature reporter addresses at all - for which presence at DR level is the natural unit; IROs speak to a firm's internal prioritisation, not to the sector's disclosure frontier. Third, only three of the six firms report under the ESRS, and the GRI reporters produce no ESRS-style IROs at all, so the topical DR is the only common denominator on which all six firms can be placed.

Translating GRI to ESRS. Only three firms report directly under the ESRS (Mowi, Bolton, Espersen); the other three were coded through their GRI content indexes, and those three sit at three different places on a spectrum. Thai Union reports *in accordance with* the GRI Standards, the fuller of the two GRI application levels. Nomad Foods reports *with reference to* them, citing selected disclosures without claiming full application. Profand is a third case again: its 2025 report is structured around a CSRD double-materiality assessment, and GRI appears only as a supporting correspondence table appended to a document whose primary architecture is the ESRS. Its GRI index is therefore a secondary artefact rather than the report's organising spine - a distinction that matters for what the coding can see, and one returned to below. A bridge was built to score all six on one ruler. The bridge is the GRI-EFRAG interoperability index (GRI & EFRAG, 2024) applied in full: every GRI disclosure listed in that index was carried across with the complete set of ESRS disclosure requirements EFRAG assigns to it, including the cases where a single management-approach disclosure maps onto requirements in more than one topical standard. Where the index states that a GRI disclosure is addressed through the cross-cutting minimum disclosure requirements (MDR-P, MDR-A, MDR-T) or as an entity-specific metric rather than through a named topical requirement, no topical coverage was recorded. Sector-specific disclosures drawn from GRI 13, for which EFRAG publishes no correspondence, were mapped by the author and are identified as such in the bridge workbook. Three rules governed that adjudication. Where the interoperability index records a full correspondence, the mapping follows it directly. Where it records a partial one, the GRI disclosure was accepted as evidence for the ESRS requirement only if it addressed the requirement's substantive subject, not merely its topic area - so a GRI water-withdrawal metric maps to E3-4 but not to the E3-1 water policy requirement. Where one GRI disclosure spans several ESRS requirements it was credited to each; where several GRI disclosures were needed to satisfy one, the requirement was credited once. Every mapping decision is recorded with its source clause in the bridge workbook, so the chain from a firm's index entry to a coverage score can be reconstructed row by row. The with-reference firms supply fewer mappable disclosures by construction, which is one reason their coverage sits mid-range rather than at the top; it is a feature of reporting choice, not of sustainability performance. Every firm's disclosures therefore sit on the same ESRS structure regardless of native framework.

What index-based coding measures, and what it misses. Working from the index rather than the report body is a deliberate choice with a known cost. The benefit is reproducibility: the index is the firm's own declaration of what it reports and where, it is applied identically to all six firms, and it is how a customer's due-diligence team reads a report in practice. The cost is that a firm whose index is selective scores below what its report body contains. That gap is systematic rather than random. GRI 3-3, management of material topics, is the disclosure carrying a firm's policies, actions and targets for each topic, and it is the principal bridge to seven ESRS policy-and-action requirements - E1-2, E1-3, E3-1, E4-2, E4-3, E5-1 and E5-3. It is optional for firms reporting *with reference to* the standards, and it is omitted entirely where the GRI table is a supporting crosswalk rather than the report's primary index. Thai Union's index carries 27 GRI 3-3 entries and Nomad Foods' 16; Profand's carries none, in either reporting year. The distinction is narrow but decisive: Profand does index GRI 3-1 and 3-2, the process for determining material topics and the resulting list, in both 2024 and 2025. Those two disclosures map to ESRS 2, the general disclosures excluded from scope above. Only GRI 3-3 carries the management approach -

the policies, actions and targets - and only GRI 3-3 bridges to the topical ESRS policy requirements. Profand is therefore coded as reporting none of those seven requirements, although its report body describes a water-management policy aligned to ISO 46001, a biodiversity policy applying the mitigation hierarchy, and 2030 circular-packaging targets. The coding did not miss these; the index it was built from does not carry them. Profand's figure of 39 requirements should be read as a floor.

The ceiling the bridge imposes. A second and distinct effect sits alongside the first. Seven of the 70 topical disclosure requirements have no counterpart anywhere in the GRI-ESRS interoperability index: E1-1, E1-8, E2-6, E3-5, E4-6, E5-6 and G1-6. No GRI disclosure maps to any of them, so no firm coded through a GRI index can score them however fully it reports. E1-1, the climate transition plan, is the clearest case: Profand's report sets out a Net Zero roadmap with 2035 and 2050 milestones, and the coding still records E1-1 as absent - as it does for Thai Union and Nomad Foods, for the same structural reason. Bolton and Mowi, both ESRS reporters, are the only firms in the cohort recorded as reporting it. Where the first effect concerns how completely a firm indexes its own disclosures, this one concerns what the bridge between two frameworks can carry at all, and it applies uniformly to every GRI-coded firm.

The seven form a coherent set: the climate transition plan (E1-1), internal carbon pricing (E1-8), payment practices (G1-6), and all four anticipated-financial-effects requirements (E2-6, E3-5, E4-6, E5-6). These are the forward-looking and financially quantified disclosures for which the GRI Standards have no counterpart, so the gap reflects a genuine difference in what the two frameworks ask rather than an artefact of the crosswalk.

Because the bridge cannot reach these seven, they were screened separately. For each of the three GRI-indexed firms, and for both reporting years, the report body was read directly for evidence of each of the seven requirements, with any finding recorded against a page reference. The rule was fixed before the screening: where the interoperability index provides no route to a requirement, coverage is determined by direct reading of the report body. Only E1-1 was found. All three firms disclose a climate transition plan - Thai Union's Path to Net Zero, Nomad Foods' SBTi-validated targets and 2025 transition plan, and Profand's Net Zero 2050 roadmap - and E1-1 is coded accordingly for each, flagged in the workbook as body-sourced rather than index-sourced. The remaining six were not found in any of the six reports. Thai Union describes assessing a shadow carbon price but does not operate one, which does not meet E1-8.

After that screening the ceiling no longer binds. Six of the seven are reported by no firm in the cohort, on either instrument, so nothing is hidden behind them. The seventh, E1-1, is now recorded for five of the six firms and enters the near-universal set. The universal core of 13 is therefore exact rather than a floor.

The opposite effect: requirements the bridge over-credits. A third effect runs in the opposite direction to the first two. Because GRI 3-3 and several topical GRI disclosures each map to more than one ESRS requirement, a firm can be credited with a requirement its report does not in fact address. Two such cases surfaced when Nomad Foods' sustainability lead and reporting manager reviewed the coding of their own firm. E1-7, GHG removals and mitigation projects, is reached through GRI 305-5, the *reduction* of GHG emissions; the interoperability index records that route, but a reduction is not a removal, and the firm reports no removals in either year. E4-1, the resilience of the business model to biodiversity risk, is reached through the GRI 304 management approach, and again the firm publishes no biodiversity transition plan. The mapping is EFRAG's in both cases rather than this author's, and overriding it case by case would substitute private judgement for a published crosswalk; both are therefore left as the index dictates and recorded here instead.

The first two effects point the same way: under-counting can only shrink the set of requirements common to all six firms, so the convergence claim is not at risk from them. The third runs the other way and needs a different answer, since over-crediting could in principle manufacture a shared requirement. It does not do so here: E1-7 is recorded for two of the six firms and E4-1 for four, so neither enters the universal core of 13 nor the near-universal set, and no convergence claim in this chapter rests on either. What all three biases distort is the firm-by-firm comparison in Table 1, which should be read as the coverage the interoperability index attributes to each firm rather than as a verified account of what each firm discloses, and which is a further reason no ranking of the six is intended.

The mapping and the DR-level coding both rest on analyst judgement. As a check on that judgement, the coding has been submitted to the sustainability leads at the cohort firms for review (Chapter 3, 3.3 Research design). One review has returned. Profand's sustainability lead identified eight disclosure requirements substantively addressed in the report body but not reachable through its GRI index; that exchange is the source of both limitations set out above, and the coding was left unchanged so that the measure remains consistent across the cohort. This corroborates the coding rather than testing its reliability formally, and the analysis does not depend on it.

Coding and consolidation. Coverage is coded as binary presence or absence at DR level: a firm scores 1 on a requirement where its reporting contains a disclosure mapping to that requirement, and 0 otherwise. This measures *breadth*, not depth, quality or assurance - a requirement touched in a single sentence and one reported exhaustively both score 1. The per-firm codings were consolidated and analysed in a Jupyter notebook, which builds the firm-by-requirement coverage matrix and the frequency distribution from which the results in the following sections are drawn.

The universe and the funnel. The denominator is fixed and externally defined, not cohort-derived: it is the full set of 70 topical ESRS disclosure requirements - every DR across the ten topical standards (E1–E5, S1–S4, G1) as set out in Commission Delegated Regulation (EU) 2023/2772 (European

Commission, 2023), the ESRS Set 1 delegated act EFRAG's draft standards were adopted into. Beginning from this complete set, rather than from what the cohort happens to report, is what lets the coverage figures be read as a share of the whole relevant field. Applying the coding produces a natural narrowing: of the 70 DRs, 63 are reported by at least one firm (the remaining seven, reported by none, are examined in Section 5.6), 28 by at least five of the six, and 13 by all six. This funnel is the structure the results then read for the height of the frontier, its universal core, and its gaps. The per-standard version of this same frontier - what is screened against what is actually covered - is shown directly below (Figure 3).

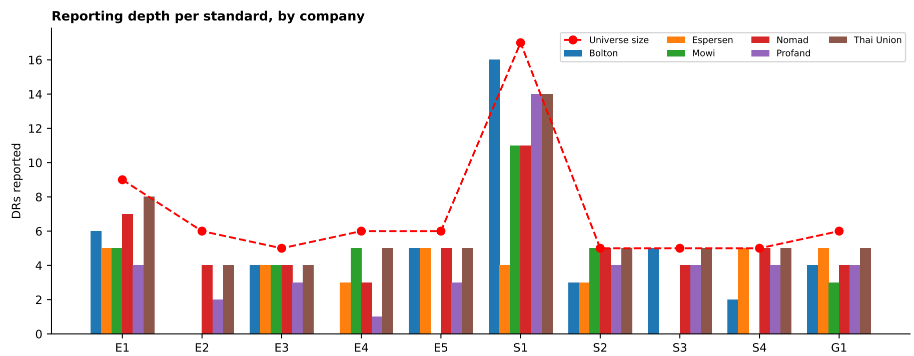

Figure 3: Reporting depth per topical standard, by company (bars), against the number of screened DRs in each standard (red line). The gap between the bars and the red line is widest for E2 (pollution) and E4 (biodiversity), the sector's thinnest topics.

What frequency does and does not mean. One caveat governs the reading of this funnel: it measures *reporting convention* among large, mostly CSRD-scoped reporters, not materiality for a small processor. A frequently reported requirement is one an SME is likely to be *asked about* as that convention cascades down the value chain - a demand-relevance proxy - rather than a topic asserted to be material for any individual firm. Whether these revealed priorities fit a particular firm's situation is a separate question, taken up in the cross-framework mapping and the worked case that follow.

5.4 The height of the frontier: a coverage gradient

Coverage ranges from 33 to 60 of the 70 screened DRs (Figure 4). Thai Union is the deepest reporter at 60 DRs (86%), followed by Nomad Foods (52; 74%) and Bolton (45; 64%); the median large reporter covers roughly 44 DRs, around 63% of the 70-DR universe. The immediate and consequential observation is that *even the frontier is incomplete*: the most advanced reporter in the sector leaves around one relevant requirement in seven uncovered (10 of 70, roughly 14 per cent), and the median large reporter closer to two in five (26 of 70, roughly

37 per cent). If the sector's most resourced firms do not disclose the full set of relevant requirements, the realistic ceiling for a resource-constrained SME sits well below completeness - a point that reframes the SME question from 'how do we report everything?' to 'which subset matters most?', which the demand mapping in the following chapters answers.

The company-by-standard heatmap also shows that firms specialise: Bolton reports most fully on own-workforce disclosure (S1, 16 of 17) and affected communities (S3, 5 of 5); Espersen reports fewer S1 items (4 of 17) and no S3 items in this coding, but is complete on consumers/end-users (S4, 5 of 5); Mowi's disclosure does not extend to affected communities (S3) or consumers (S4) in this coding. The frontier is therefore not only incomplete in height but uneven in shape from firm to firm.

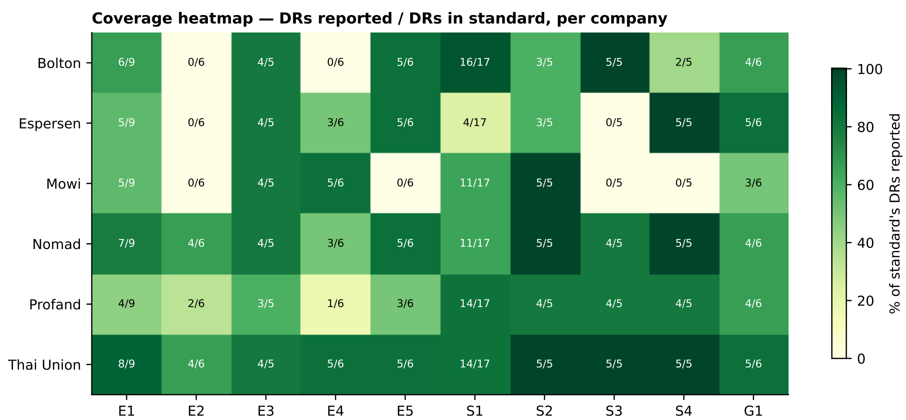

Figure 4: Coverage heatmap: each cohort firm's reported DRs as a fraction of the screened DRs in each of the ten topical standards (cell label "reported / in standard"; shading is the percentage). Rows sum to each firm's total in Table 1; the standard totals sum to the 70 screened DRs.

5.5 The universal core: what every firm reports

Beneath the gradient lies a stable core. Reading the universe by the number of firms reporting each requirement produces a clean funnel: of the 63 DRs reported by at least one firm, 28 are reported by at least five, and 13 by *all six* (Figure 5). These 13 requirements are the sector's empirically revealed *baseline* - the disclosures every mature reporter makes irrespective of sourcing model, and so, on the reading above, the minimum a credible seafood reporter is now taken to be expected to cover. They fall into four blocks:

- **Climate (3):** energy consumption and mix (E1-5), gross Scope 1–3 and total GHG emissions (E1-6), and climate targets (E1-4).

- **Water (3):** actions and resources (E3-2), targets (E3-3), and water consumption (E3-4).
- **Governance (2):** business-conduct policies and corporate culture (G1-1), and prevention and detection of corruption and bribery (G1-3).
- **Social (5):** own-workforce policies (S1-1), workforce characteristics (S1-6), health-and-safety metrics (S1-14), and, for value-chain workers, policies (S2-1) and remediation channels (S2-3).

This narrowing is the core descriptive finding of the chapter: the sector does not disclose evenly across a broad front, but converges onto a compact, shared backbone.

One qualification bounds these two counts. Seven of the 70 requirements cannot be reported by more than three firms, because the GRI-ESRS bridge offers no route to them and half the cohort is coded through GRI indexes (Section 5.3). Six of those seven are reported by no firm at all, so the qualification binds on a single requirement: E1-1, the climate transition plan, which could in principle reach five of six were all three GRI-indexed firms to disclose one. The core of 13 is unaffected either way, since none of the seven could reach six of six.

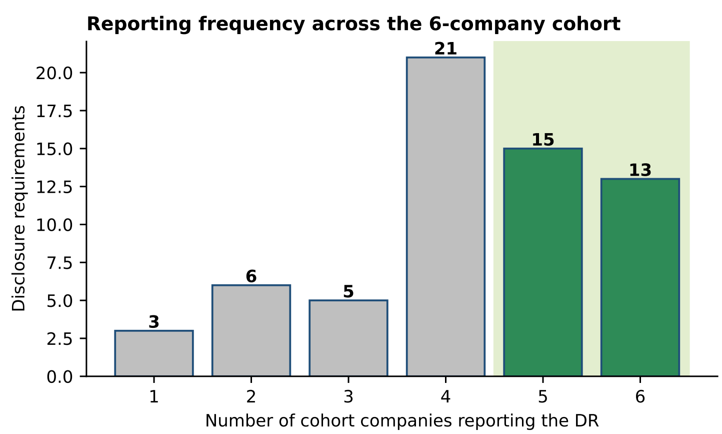

Figure 5: Number of cohort firms reporting each DR, over the 63 requirements reported by at least one firm. The highlighted band (reported by five or six firms) contains 28 requirements, of which 13 are reported by all six - the universal core.

5.6 The shape of the frontier: volume, consistency, and blind spots

Among the 63 reported requirements, disclosure is markedly social-heavy by volume: 32 are Social, 26 Environmental, and 5 Governance, with the own-workforce standard (S1) alone accounting for 17. This social weighting holds within firms as well as across the universe: for every cohort member, Social is the largest block of reported DRs and Governance the smallest (Figure 6, panels A–B). This is a volume comparison, however, and the ESRS universe itself is uneven across pillars - 32 DRs each for Environmental and Social, but only 6 for Governance. Normalized by each pillar's own size (Figure 6, panel C), Governance is in fact the *most* thoroughly covered pillar per firm on average (mean 69% of its 6 DRs, range 50–83%, against a mean 56% for Environmental and 70% for Social): the sector reports on a smaller absolute footprint of governance topics not because it neglects them, but because the ESRS asks fewer governance questions to begin with.

Volume, however, is not the same as consensus. Measuring each topic by the mean number of firms reporting its requirements reveals where the sector genuinely agrees (Figure 7): consistency is highest for water (E3, mean 5.75 of 6), value-chain workers (S2, 5.00), governance (G1, 5.00) and resource use and waste (E5, 4.60), and lowest for pollution (E2, 2.50), biodiversity (E4, 3.40) and affected communities (S3, 3.60). The frontier is thus concentrated on water, workforce, governance and resource-use topics and thins markedly across pollution, biodiversity and community disclosure.

The seven requirements reported by no firm sharpen this picture, though they need careful reading. Four are the "anticipated financial effects" disclosures - for pollution (E2-6), water (E3-5), biodiversity (E4-6) and circular economy (E5-6) - the forward-looking, financially quantified requirements that are the most demanding and are phased in under the standards. The remaining three are substances of concern (E2-5), internal carbon pricing (E1-8) and payment practices (G1-6). The sector's clearest collective gap is therefore the financial quantification of environmental impact, followed by pollution disclosure - a pattern consistent with the low consistency of E2 across the cohort.

One qualification applies before this is read as a sector-wide finding. Six of the seven have no route through the GRI-ESRS bridge, so for the three GRI-coded firms they were screened directly in the report bodies rather than through the index (Section 5.3); none was found. The absence is therefore observed for all six firms rather than inferred, but by two different instruments. A second point is substantive: the anticipated-financial-effect requirements are phased in under the ESRS on a delayed timeline, so their absence reflects current transitional relief as much as a genuine reporting gap - they are not yet required, rather than being avoided. A parallel movement appears in materiality itself: across 241 large listed EU companies, Nahrstedt (2026) finds double-materiality assessments declaring fewer topics material between FY2024 and FY2025, the decline concentrated in the environmental standards other than climate. Firms are not only disclosing less here, they are increasingly judging these topics immaterial - consistent with a landscape still calibrating rather than one that has settled.

This bears on how absence should be read throughout the chapter. A requirement recorded as unreported may have been assessed immaterial by the firm rather than overlooked: under both the ESRS and the GRI, a firm that judges a topic immaterial is not expected to report against it, and a zero in the coding is therefore consistent with a deliberate and defensible reporting decision. Nomad Foods, reviewing its own entry, notes that affected communities (S3) did not emerge as material in its double-materiality assessment. The coding registers what a report covers, not why a topic is absent, and no inference about a firm's intent or capability is drawn from a zero.

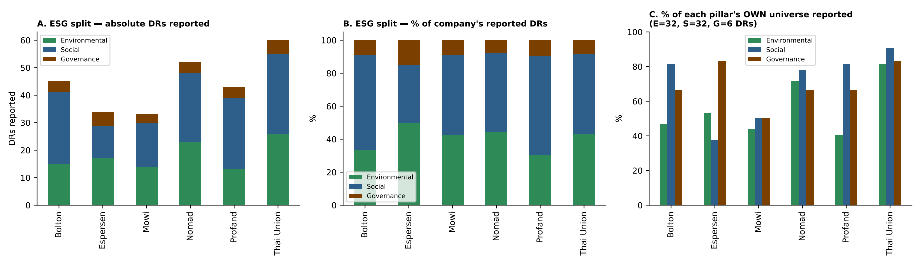

Figure 6: ESG composition of each firm's reported DRs: in absolute terms (panel A), as a share of the firm's total reported DRs (panel B), and as a share of each pillar's own ESRS universe - 32 DRs each for Environmental and Social, 6 for Governance (panel C). By volume, Social is the largest block and Governance the smallest for every firm (A–B); normalized by pillar size, Governance is the most thoroughly covered pillar (C).

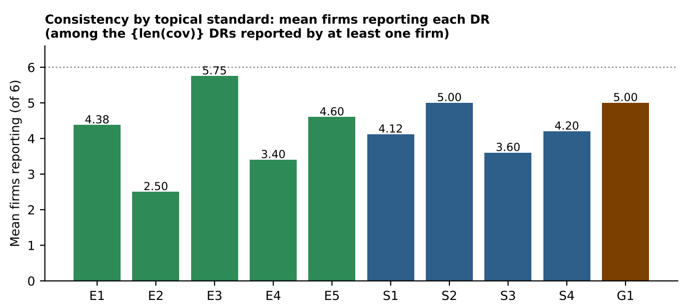

Figure 7: Consistency by topical standard: mean number of cohort firms reporting each DR, among the 63 DRs reported by at least one firm. Consistency is highest for water (E3) and governance (G1); lowest for pollution (E2) and affected communities (S3).

5.7 Robustness across sourcing models and reporting year

Does the core depend on sourcing model? Because the cohort deliberately spans four sourcing models, the universal core can be tested for whether it is an artefact of one type of firm. It is not: the 13 core requirements are reported across all four sourcing models - wild-capture, aquaculture, integrated wild-plus-aquaculture, and diversified multi-species sourcing alike. All six firms remain seafood *processors* throughout; what varies between the four groups is only what and how each sources - not whether processing is what they do. Sourcing model changes only the *edges* of the picture, not the centre: model-specific topics such as feed and animal health (aquaculture) or wild-stock status (wild-capture) appear only in the long tail, while the climate, water, governance and workforce core is common to all. This is confirmed rather than merely illustrated: because these 13 requirements are by definition reported by *all six* cohort firms, every sourcing-model subgroup necessarily reports them too - there is no possible split by which they could fail to be cross-model.

Two further checks confirm the core is not an artefact of the sample. First, *threshold sensitivity*: the near-universal set contracts smoothly with the cut - 60 requirements are reported by at least two firms, 54 by at least three, 49 by at least four, 28 by at least five, and 13 by all six - so the $\geq$5-of-6 line sits on a gradient rather than a cliff edge: the shortlist shrinks smoothly as the threshold tightens, with no sharp jump at any single cut, so the choice of $\geq$5-of-6 is not an arbitrary point that the result hinges on. Second, *cohort composition*. Re-deriving the near-universal set on the five European firms alone, dropping the non-European Thai Union at the equivalent $\geq$4-of-5 cut, returns the identical 28-requirement set. That result is real but it is not general, and systematic leave-one-out testing shows why it should not be read as evidence that the backbone is independent of the cohort. Dropping Thai Union leaves the set unchanged; dropping any other single firm changes it, in some cases substantially - to 31 requirements without Nomad Foods, 35 without Bolton, 37 without Profand, 38 without Espersen and 41 without Mowi (Figure 8). The reason is mechanical. At a proportional threshold, removing the broadest reporter withdraws mostly affirmative observations and nothing new clears the cut, whereas removing a narrower reporter withdraws mostly negative ones, promoting requirements that sat just below the line. Thai Union, the broadest reporter at 60 of 70, is precisely the firm whose removal is least likely to move the set. The near-universal set is therefore a property of these particular six firms and is treated as such here.

The universal core behaves differently, and it is the core rather than the near-universal set that carries the argument of this chapter. Because the core is the intersection of what all six firms report, removing a firm can only relax the condition that defines it: a requirement in the core cannot leave the core when the cohort shrinks. The 13 are accordingly retained under every one of the six leave-one-out subcohorts and under all fifteen leave-two-out subcohorts, growing rather than shrinking when the narrower reporters are removed - to 19 requirements without Espersen, and 19 without Mowi. They are a floor beneath any subset of this cohort rather than an artefact of its composition, and that is the sense in which the core, but not the near-universal set, is robust to who is included.

Thai Union itself, the single non-EU firm and the broadest reporter, reports *more* widely on biodiversity, affected communities and pollution than the European average. This thesis does not investigate why that pattern holds, and no assumption about its cause, whether GRI's broader reporting scope, company-specific policy, or any other factor, is made here; the observation is reported descriptively and left at that. What matters is only that one requirement is reported uniquely by Thai Union, so the core survives the EU and non-EU divide as common ground rather than an EU-regulatory artefact. The same 13-requirement core therefore emerges whichever way the cohort is cut - by the reporting threshold, by which firms are included, and by EU versus non-EU - which is what lets it be read as a feature of the sector rather than a quirk of this particular sample. The near-universal set does not carry that property, and no claim in this thesis depends on it doing so.

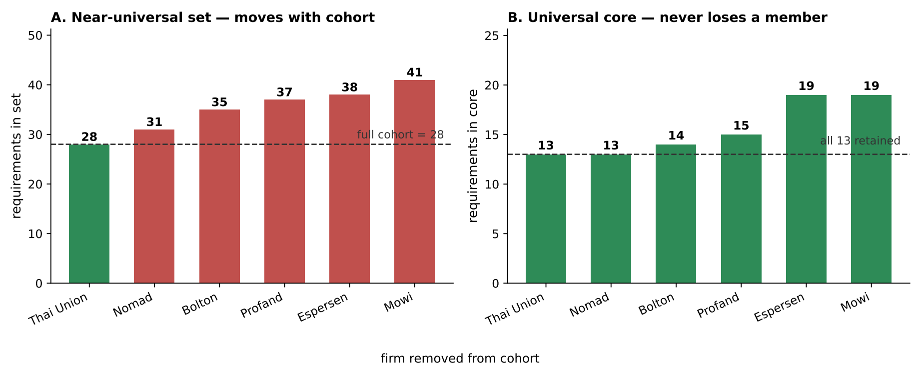

Figure 8: Sensitivity of the two sets to cohort composition, under leave-one-out. Panel A: the near-universal set changes size when any firm other than Thai Union is removed, rising as the narrower reporters drop out. Panel B: all 13 universal-core requirements are retained in every leave-one-out subcohort, the core growing rather than shrinking. The core is a floor beneath any subset of the cohort; the near-universal set is not.

Finally, the cohort mixes 2024 and 2025 reporting years, and for the three firms with two comparable reports coverage was flat or fell very slightly. That comparison is exploratory rather than a statistical result, and it rests on a reading of the indexes rather than of the disclosures themselves, so it is set out separately in Appendix I - Year-over-year reporting in the cohort and no claim here depends on it. Two cautions belong with it. Only the three firms publishing in both years can be examined at all, so the year-over-year view scrutinises half the cohort and is silent on the rest. And the direction it shows runs against the assumption of ever-expanding disclosure, which makes it the kind of finding that invites over-reading. The main analysis therefore uses each firm's most recent report throughout (the figures in Table 1).

5.8 What the cohort establishes - and what it does not

The cohort establishes three things. It fixes the height of the sector's disclosure frontier at 60 of the 70 screened requirements for the deepest reporter, and around 63% of that set for the median large firm; it identifies a 13-requirement universal core spanning climate, water, governance and workforce that constitutes the sector's de facto baseline expectation; and it locates the sector's collective gaps in the anticipated-financial- effect and pollution requirements. Together these calibrate the realistic disclosure ambition for a smaller firm and supply the empirical anchor for the cross-framework demand mapping that follows.

What this means for the SME question. Read together, these findings do more than describe the frontier; they bound the smaller firm's problem. Because even the best-resourced reporters converge on a compact, cross-cutting backbone rather than reporting everything, the SME's task is not to approximate an ever-expanding frontier but to build a short, stable set of disclosures the sector has already revealed to be expected - and that a firm is, on the demand-relevance reading, most likely to be asked for. This is the hard step described at the start of this chapter (Section 5.1); the contribution here is to make it concrete for one sector by naming the topics on which it should begin. That neither disclosure nor declared materiality is expanding in the early CSRD years (Nahrstedt, 2026) only strengthens the case for a proportionate, prioritised starting point over open-ended completeness. *Which* of these topics an individual firm should build first, given the specific demands it faces, is the question the cross-framework mapping takes up next.

Its limits are equally definite and are stated so that the findings are not read beyond what they support. The cohort is a purposive sample of six upper-bound reporters and is not representative of the sector, so no claim about typical or average practice is made. Coverage is binary presence at DR level and speaks to breadth, not to the depth, quality or assurance of disclosure. The mapping from mixed ESRS and GRI source documents onto the ESRS requirement set rests on analyst judgement, for which the member-checking step described in Chapter 3 is a corroboration still outstanding rather than a completed control. And the analysis is a 2024–2025 snapshot of a rapidly moving reporting landscape. Within these bounds, the cohort provides what it was designed to provide: a defensible, document-grounded map of the seafood disclosure frontier.

6. Cross-framework demand mapping: where to build first

This chapter covers two steps rather than one. Step 1 (Chapter 4) fixed the *columns*: the instruments that make up the disclosure ecosystem and how binding each is. Step 2 (Chapter 5) fixed the *rows*: the disclosure topics the sector's leading firms actually report, narrowed to a cohort-evidenced universe.

Step 3, the *overlap*, crosses the two. It scores *who demands what*, topic by topic and instrument by instrument, then reads the resulting matrix for the thesis's central question: given a fragmented wall of overlapping demands, which topics satisfy the most of them if built first. Step 4, *building the tool*, turns that finding into a decision a firm can act on - a short ranked list of topics to build first, and a diagnostic that tailors the list to its own situation. Together the two steps answer the third research question and produce the thesis's central practical contribution.

Figure 9 shows a representative excerpt of the underlying coverage matrix, so the scale and shape of the raw scoring are visible before the chapter turns to how it is built.

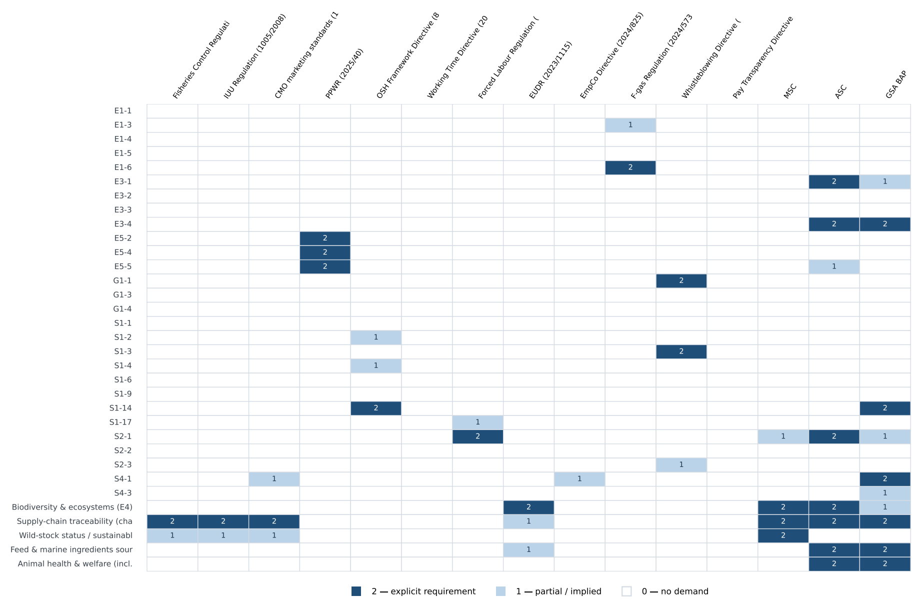

Figure 9: Representative excerpt of the coverage matrix (*Phase3_Coverage_Matrix_v18.xlsx*, sheet "Coverage matrix"): topic rows against scored-instrument columns, coloured by score (0/1/2). Shows all 33 topic rows and the first 15 of the 31 scored instrument columns (Tier 1–2); the full matrix is in the source workbook.

6.1 The convergence logic

The move from a wall of demands to a short build-first list rests on one idea: that the demands, though issued by different actors in different vocabularies, draw on a much smaller set of underlying capabilities, so that a single capability can answer many demands at once. Two established literatures make this precise. The first is the *resource-based view* of the firm and the dynamic-capabilities tradition (Barney, 1991; Teece, 2007), which treat the ability to produce and reuse structured data as an organisational capability that, once built, is redeployable across uses at low marginal cost. The second is *stakeholder-salience* theory (Mitchell, Agle & Wood, 1997): a firm facing many legitimate claimants cannot serve each separately and must triage, and the rational triage is toward the topics on which the most salient claimants converge. Salience in that account is the combination of three attributes - the *power* to impose consequences, the *legitimacy* of the claim, and its *urgency* - and the tiering of Chapter 4 is essentially an empirical reading of the first two: a Tier 1 legal obligation carries power and legitimacy together, a Tier 4 voluntary commitment neither. To these three the SME case adds a fourth constraint that Mitchell's framework treats as given, namely the *resources* available to answer any claim at all. Triage for a resource-constrained firm is therefore not only a question of which claimants matter most, but of how few distinct capabilities the answer can be built from - which is what the convergence measure below is designed to capture.

Convergence is therefore the organising construct. Where a topic is demanded by a regulator, an investor framework and an industry standard alike, capability built for that topic is not spent once but several times over - the compound return that distinguishes a proactive posture from a reactive one (Chapter 2). A topic demanded through only one channel yields a narrower return. The analytical task of this chapter is thus to *measure* convergence across the fragmented landscape and to identify the topics where it is highest - the hotspots at which building once satisfies many.

6.2 Assembling the matrix: rows and columns

The matrix has rows and columns, and neither comes straight from the previous chapters. The rows start from the topics the cohort actually reports (Chapter 5), which are then trimmed to those the sector agrees on and extended with four topics the ESRS has no row for. The columns start from the instruments catalogued in Chapter 4, which are then cut back to those that genuinely ask the processor for something. Both changes are set out here, before any cell is scored, because what goes into the matrix decides what it can show: a row for a topic nobody is asked about, or a column for an instrument that demands nothing, would distort every count taken from it.

Rows: from the cohort frontier to the topic universe. The rows begin from the cohort frontier (Chapter 5) and are built up in three steps:

1. **The requirements the cohort converges on enter directly** - those reported by at least five of the six firms. These are the topics the sector's own practice has revealed to be expected (28 requirement-level rows). The set includes the climate transition plan (E1-1), which reaches the cut only after the direct screening of report bodies described in Section , because no GRI disclosure maps to it.
2. **One topic clears the same bar one level up.** Biodiversity (E4) is

reported by at least five firms when a firm counts as covering the topic if it reports *any* of the topic's requirements, but no individual E4 requirement reaches the five-of-six cut on its own - the best-reported reach four. E4 therefore enters as a topic-level row: the same consensus rule at coarser granularity, not an expert-judgement exception. Consumers and end-users (S4) was admitted the same way in an earlier version of this analysis; after the bridge rebuild two of its requirements, S4-1 and S4-3, clear the cut individually, so S4 is now represented at requirement level and needs no topic-level rescue. With the requirement-level set it forms the cohort-evidenced core (29 rows).

3. **Four seafood-defining topics have no ESRS home at all** and are carried in against the GRI 13 sector standard and EU sector legislation: supply-chain traceability, wild-stock status, feed and marine ingredients, and animal health and welfare (antimicrobial use folded into the last). These are flagged as sector-material rather than cohort-evidenced. Each carries a production-system tag - wild, aqua or both - so a firm drops the rows that do not concern it.

The funnel therefore runs from the 70 topical ESRS disclosure requirements, narrows to the cohort-evidenced core, then *extends* by four. Those four are the one place where the ESRS blueprint is insufficient and the sector standard supplies the rows instead. Stated as a sequence, with each figure used once:

1. **70** topical ESRS disclosure requirements form the starting universe (Chapter 5).
2. **28** of those are reported by at least five of the six firms at requirement level, and enter directly. E1-1 is among them, reached through the body screening of Section rather than through the bridge.
3. **+1** topic-level row, biodiversity (E4), where the topic clears the consensus cut but no single requirement within it does - giving **29** cohort-evidenced rows.
4. **+4** sector-material rows carried in from GRI 13 and EU sector law, for which no ESRS requirement exists: supply-chain traceability, wild-stock status, feed and marine ingredients, and animal health and welfare.
5. **33** rows in total, scored against **31** instrument columns - 30 instruments, with the VSME split into its Basic and Comprehensive modules - giving the **1,023** cells of the matrix.

They are not chosen ad hoc. Each of GRI 13's 26 material topics (Agriculture, Aquaculture and Fishing) was classified against the ESRS as *covered*, a *reporting gap* (an ESRS mapping exists, but cohort reporting on it falls below the $\geq 5/6$ consensus threshold), or a *structural gap* (no ESRS disclosure requirement exists at all).

The four divide evenly between two of those categories. Animal health and welfare and supply-chain traceability are structural gaps: the ESRS has no requirement covering either, so nothing in the standard asks for them at all. Wild-stock status and feed and marine-ingredient sourcing are different. Both are production-specific disclosures nested inside biodiversity (GRI 13.3), which is itself covered and already in the universe. The broad subject is therefore present, but these two specific disclosures within it are not. The full classification is given in Appendix B - GRI 13 to ESRS coverage classification.

Columns: from the full ecosystem to the scored instruments. The columns begin from the instruments catalogued in the ecosystem (Chapter 4). That catalogue is deliberately the whole field, so it includes instruments that place no scorable demand on the processor in the end. The scored set is the subset that does.

Three groups are set aside. The *context* instruments bind Member States rather than the firm. The *excluded* instruments fail the second of the two admission gates screened at catalogue stage - ESG relevance first, then organisation-level obligation - which are defined and applied instrument by instrument in Appendix C - Instrument scope classification. The CSRD/ESRS standard is held out separately as the *baseline* reference against which the rows are defined, rather than scored as one more demand.

That leaves **30** scored instruments. Only one of them, the VSME, is split into two columns, for its Basic and Comprehensive modules, because it is the only instrument with two officially defined and nested reporting tiers - which is why 30 instruments occupy 31 columns. Each scored column inherits its stakeholder source and its obligation locus (direct, indirect or context) from Step 1. Some columns are switched off for a given firm because its profile puts the instrument out of scope, and two kinds of switch apply. The first is the production system: a wild-only firm switches off the aquaculture certifications (ASC, BAP). The F-gas switch is separate from production system and turns on equipment rather than sourcing: it is switched *on* for the modelled firm, since refrigeration is near-universal in seafood processing, and exists only for the minority of sites that have already converted entirely to natural refrigerants such as ammonia or CO_2 and hold no fluorinated gas. The second switch is size: within the SME band several obligations turn on headcount - the Pay Transparency Directive's gender pay-gap reporting applies from 150 employees and the whistleblowing obligation from 50 - so a firm below a threshold switches off the instruments above it. A switched-off instrument is excluded from that firm's demand sums, though a customer may of course still ask for the same content contractually, in which case the demand reaches the firm through the customer channel rather than the legal one. The full instrument-by-instrument classification - scored, baseline, context or excluded, with the reason for each - is provided in Appendix C - Instrument scope classification.

The demand matrix is therefore a grid of topic rows against instrument columns, every cell recording how strongly a given instrument demands a given topic, on the scale set out next.

6.3 Scoring the matrix

The coverage scale. Each cell is coded on a three-level scale defined ex ante:

- **0 - not covered:** the instrument does not address the topic, or its demand maps only to datapoints outside the row universe (the out-of-frame rule below).
- **1 - touched:** the instrument references the topic but imposes no binding requirement on any of its datapoints.
- **2 - required:** the instrument explicitly and bindingly requires the entity to achieve, address, comply with, certify, report on or disclose at least one datapoint belonging to the topic.

Binding *conformity* counts equally with binding *disclosure*: a mandatory recycled-content target (e.g. the PPWR), an accident-prevention duty (e.g. the OSH Framework Directive) or a market-access prohibition on forced-labour products (e.g. the Forced Labour Regulation) scores 2 even where no report is filed, because it bindingly addresses the datapoint substance. The score measures *how strongly* an instrument demands a topic; it says nothing about *how* that demand reaches the firm, which is a different question. A demand that arrives indirectly - CSDDD, which for the sub-threshold processor modelled here binds its large customers and reaches it as a cascaded supplier requirement, though it would bind a larger processor directly - is still scored on its strength: if it bindingly imposes a topic's datapoints it scores 2, exactly as a directly binding rule would. Whether the route is direct or indirect is recorded separately, as the obligation locus, and never raises or lowers the score. Keeping the two apart stops an indirect-but-binding demand from being under-counted merely because it is transmitted rather than imposed.

Datapoint-grounded coding. A score is assigned against the *constituent datapoints* of a requirement, not against the title of its disclosure requirement. Each of the 29 cohort-evidenced rows is decomposed into its official EFRAG datapoints using the ESRS datapoint master (Commission Delegated Regulation (EU) 2023/2772): the 26 DR-level rows resolve to 356 datapoints and the two topic-level rows (E4, S4) to a further 152. An instrument scores 2 where its indicators demand at least one of a requirement's datapoints, 1 where it references the topic but demands none of them, and 0 where it is silent. This replaces the broad question of whether instrument f addresses topic r with the concrete and reproducible one of whether f demands at least one datapoint of r. It also surfaces corrections. Whistleblowing demand, for instance, resolves to the governance row that actually carries the whistleblowing datapoints rather than to an adjacent anti-corruption row. Once such corrections are logged they are findings about the coding process rather than matters of method.

When self-reporting counts as a binding demand. Many instruments are self-reported, which raises a scoring question with real consequences: whether completing a supplier questionnaire places the same demand on a topic as reporting against a defined standard. Treating the two alike would inflate the matrix, so a single rule separates them - and what matters is not *who* reports but whether the *datapoints are fixed in advance*:

- An *open-question self-assessment*, where the firm describes itself against broad prompts with no defined datapoints and no external threshold (e.g. EcoVadis, the Sedex SAQ), shows the topic is relevant to the firm but pins down no specific data. It scores **1**.
- A *defined-datapoint standard*, where the datapoints to report are fixed ahead of time - voluntary (GRI, CDP, TNFD) or mandatory (the ESRS) - imposes the datapoint even though it too is self-reported. It scores **2**.
- An *enforced-conformity* requirement, where the firm must meet a bar and fix failures (a SMETA audit it must pass to keep the buyer, or a binding legal duty), likewise scores **2**.

A last line separates a rating the firm *feeds* from one built *about* it: a rating is scored only where the firm must supply the data itself (EcoVadis, scored 1); purely external ratings assembled from public data (MSCI, Sustainalytics, the Dow Jones Best-in-Class Indices) place no demand on the firm and are not scored at all, since such a rating consumes disclosures rather than asking for them.

Sector-material rows. For the four seafood-specific gap rows (traceability, wild-stock status, feed and marine ingredients, animal health and welfare) no ESRS datapoint exists by construction; these are scored against the criteria of the demanding instruments themselves - MSC Principle 1 stock-status reference points for wild-stock, the GDST key data elements for traceability, and ASC antibiotic and survival criteria for animal health and welfare. A pinned rule prevents double-counting on the wild-stock/E4 boundary: stock-status and legal-sourcing demand (MSC Principle 1, the IUU catch-certificate regime) codes to the wild-stock row, whereas generic biodiversity and ecosystem-impact demand (ESRS E4, GRI 101, TNFD, MSC Principle 2 on bycatch and habitat) codes to E4; a single instrument may legitimately score on both rows through different provisions.

These four rows are not equivalent kinds of measure, and the distinction matters for how their scores should be read. Wild-stock status is an *outcome*: it asks whether the stock a firm buys from is healthy, and answering it requires stock assessment science and reference points the processor does not generate itself. Supply-chain traceability is an *assurance mechanism*: it establishes that a product came from where it is claimed to have come from, and says nothing on its own about whether that source is sustainable. A firm can operate complete traceability on a stock that is severely overfished. The high demand score traceability carries in this matrix therefore records how many instruments require it, not that meeting it evidences a sustainability outcome. The two rows are kept separate for exactly this reason, and the build-first ranking that follows should be read as a map of what the environment *asks for* rather than of what most improves the state of the resource.

Are wild-stock status and traceability actually live practice, not just a GRI category? Neither has an ESRS DR, so neither was captured by the cohort's DR-level coverage coding in Chapter 5 - which raises a fair question: is including them here just following GRI 13's classification, or is there evidence the sector's own leading firms actually treat them as real, active topics? All six cohort firms do, on both counts (Table 2).

Table 2: Sector-material topics in the cohort's own reporting. Each firm discloses both a sustainable wild-sourcing certification level and a traceability system, though neither topic has an ESRS disclosure requirement. Figures as published by each firm; wording follows the source report.

Firm	**Sustainable sourcing, as reported**	**Traceability**
Thai Union (2024)	98.9% of tuna MSC-certified, in MSC assessment, or FIP-covered	Working toward GDST compliance
Bolton (2024)	99.7% of tuna from responsible fishing practices: MSC-certified, in full MSC assessment, or covered by a credible and comprehensive FIP	Working toward GDST compliance
Nomad Foods (2025)	99.9% of fish and seafood certified sustainable or responsibly sourced	Own digital traceability system
Espersen (2025)	99% of fish and seafood sourced in accordance with third-party certification	Own electronic traceability system
Mowi (2025)	100% of harvest volume certified to a GSSI-recognised standard (ASC, BAP or GLOBALG.A.P.)	Own digital traceability system
Profand (2025)	MSC- and ASC-certified sourcing	Active GDST participant

The figures in the middle column are not comparable across firms and are not used comparatively here. Each firm defines its own numerator: some report certified volume alone, others bundle certified volume with product in assessment and with fisheries covered by improvement projects, and some state a figure for one species rather than the whole portfolio. The column records that each firm makes a sourcing claim and on what basis, not how the firms rank against one another; no claim in this thesis rests on the difference between 99.9% and 98.9%.

The two topics fall outside the ESRS structure this thesis uses as its main coordinate system, but they are not a marginal or invented addition: every firm in the cohort reports on both, independently of each other and of GRI 13's classification. This is stated here as corroboration, not as scored evidence - these mentions were not coded against a defined datapoint set the way the 70-DR cohort analysis was, so no coverage percentage is claimed for them.

Does this reach a firm that only processes, not farms or fishes? A pure processor buys already-caught or already-farmed product; it does not manage a stock or run a farm. So how much of the wild-stock, feed and animal-welfare demand above actually lands on it? All of it, but in two different ways, and the matrix tags which is which. *Direct*: the processor itself holds a Chain-of-Custody (CoC) certificate (MSC, ASC, GLOBALG.A.P., BAP, Friend of the Sea) and keeps its own segregation and lot records to prove it - this is an obligation on the processor itself. *Indirect*: to hold that CoC certificate at all, its supply contracts must specify sourcing from a stock, farm or feed mill that meets the scheme's upstream standard - for instance MSC Principle 1 on stock health, or ASC's feed and antibiotic limits. These are illustrative rather than exhaustive: both schemes are full standards, and a supplier is certified against the whole of one, not against a single principle. MSC covers stock status, environmental impact and management system across its three principles, and ASC likewise extends well beyond feed; no buyer accepts conformity with one principle alone. The examples are named here only because they are the clauses through which the upstream demand reaches the *processor*.

This transmission also runs in the other direction, which qualifies the common assumption that a processor is simply a demand-taker. A purchasing policy that specifies certified sourcing exerts real pressure upstream, and a large enough book of such policies can move a fishery: the Argentine hake fishery (*Merluccius hubbsi*) has moved toward MSC assessment in response to buyers declining uncertified product (O. Spring, Group Sustainability Manager, Nomad Foods, personal communication, August 2026). The effect is weaker for a single sub-threshold SME than for a large buyer, but it is not absent, and it is one route by which a small processor's disclosure capability converts into influence rather than only cost.

The processor doesn't manage the upstream operation, but it cannot keep its certificate without sourcing from one that does - so the demand is real, just transmitted rather than performed directly. Each cell is scored on the same 0/1/2 strength scale either way (a core criterion such as MSC P1 stock status or ASC feed-conversion limits scores 2, a lighter good-practice mention 1); direct or indirect only records *how* the demand reaches the processor, and - as stated above - never changes the score itself.

One locus tag needs qualifying. EUDR's due-diligence duty on feed and marine-ingredients sourcing binds the feed manufacturer importing raw soy or palm, not a processor further downstream buying finished feed. For the modelled sub-threshold processor the demand therefore arrives indirectly, through the feed supplier's own compliance. The source workbook records obligation locus once per instrument column rather than per cell, so EUDR carries a single *direct* tag there; the qualification stated here applies to the feed row specifically. Because locus is descriptive and never enters the score, no coverage figure in this chapter is affected.

Out-of-frame demand. Some indicator-level demand maps cleanly onto an ESRS datapoint that lies *outside* the cohort-defined universe: the Pay Transparency Directive's gender pay-gap requirement maps to S1-16, the Working Time Directive to S1-11/S1-15, none of which cleared the $\geq 5/6$ consensus and so have no row. Such demand scores 0 against the in-universe rows and is *not* reassigned to the nearest available row (which would misattribute and inflate it); instead it is recorded in a separate out-of-frame register (instrument $\rightarrow$ demanded DR $\rightarrow$ reason). A framework's contribution to the matrix is thus the *intersection* of its demand with the cohort-evidenced universe, not the union of everything it asks.

This produces a finding in its own right rather than a mere bookkeeping note. **Mandatory labour-standards disclosure and the voluntary reporting universe occupy partly disjoint spaces.** EU labour law requires disclosures - adequate wages, work-life balance, working-time protections - that the sector's own leading reporters largely do not make, because those requirements never clear the consensus threshold that defines the topic universe. The demand is legally binding and the reporting practice has not followed it. For a firm, this means the build-first set derived here does not discharge its labour-law obligations, which have to be met whether or not peers report against them; for policy, it marks a place where mandatory demand and voluntary convergence have not met, and where the assumption that market practice will carry legal requirements does not hold.

Methodology providers versus disclosure askers. Instruments that supply methods but do not themselves pressure an entity to report - e.g. the GHG Protocol or the ISO standards such as ISO 14001 - are held in an enabling layer and excluded from the demand sums, distinct from the disclosure askers (e.g. CDP, SBTi, MSC) that require an entity to submit or certify information. The same treatment applies to the food-safety schemes: BRCGS and IFS require chain of custody and full traceability, and so materially help a processor meet the traceability row, but they are certification programmes a buyer requires rather than disclosure demands in their own right, and are therefore carried as enablers rather than scored columns. Scored cells are documented in a justification log giving a one-line rationale, a citation to the source clause or indicator, and the identifier(s) of the ESRS datapoint(s) the demand maps to - the audit trail that protects the coding against challenge. The log covers 120 of the 188 scored cells. The 68 without an individual entry are concentrated in a small number of instruments whose scoring logic is uniform across topics - the Sedex self-assessment questionnaire accounts for 15 of them, GSA BAP for 8 - where the rationale is given once at instrument level in the coding protocol rather than repeated per cell. The gap is therefore one of documentation granularity rather than of unrecorded judgement, but it is stated here so the audit trail is not claimed to be more complete than it is. The complete coded matrix and its justification log, together with the per-cell coding protocol, are provided in

Appendix D - Cross-framework coverage matrix and justification log (Figure 14). A representative excerpt of the scored matrix itself is shown in Figure 9 at the start of this chapter.

6.4 Hotspot measures and stage assignment

With the matrix populated, the columns are collapsed into the four stakeholder-driver channels (regulator, customer, investor, industry/NGO), and two complementary measures are computed for every topic:

- **Breadth (Σ)** - how much demand a topic attracts in total: add up its scores across every instrument, counting a binding requirement (2) as double a passing mention (1). A high breadth means many instruments ask for the topic, or a few ask for it bindingly. It measures the *volume* of demand.
- **Convergence (0–4)** - the number of the four stakeholder channels in which at least one instrument demands the topic. Because it counts *channels*, not instruments, it is robust by construction to within-channel repetition (five overlapping certifications collapse to a single customer signal) and captures cross-stakeholder consensus rather than sheer volume.

Convergence is the primary signal and breadth the secondary one: convergence is the dedup-robust measure of how widely the fragmented landscape agrees on a topic, while breadth - which overlapping instruments can inflate - discriminates between topics of equal convergence.

Topics are then triaged into three stages on the two measures jointly:

- **Stage 1 - build first:** convergence ≥ 3 *and* breadth ≥ 12. Demanded across at least three of the four channels *and* in the top quartile of demand volume - the no-regret core, where capability built for one channel serves several.
- **Stage 2 - next:** convergence ≥ 3 but breadth below the quartile cut. Broad consensus, less intense demand; build after the core.
- **Stage 3 - monitor:** convergence ≤ 2. Demanded through only one or two channels; required for compliance where in scope, but not yet drawing broad demand.

Both thresholds are anchored in the empirical distribution rather than set arbitrarily (Figure 10). The convergence cut of three is the simple majority of the four channels. The breadth cut of twelve is the *third quartile* of the breadth distribution across the 33 topics, so the Stage-1 volume gate selects the top quartile and recalibrates automatically if the scores change. In the modelled firm no topic below convergence three reaches the breadth cut, so the two criteria identify one coherent Stage-1 set rather than two competing ones.

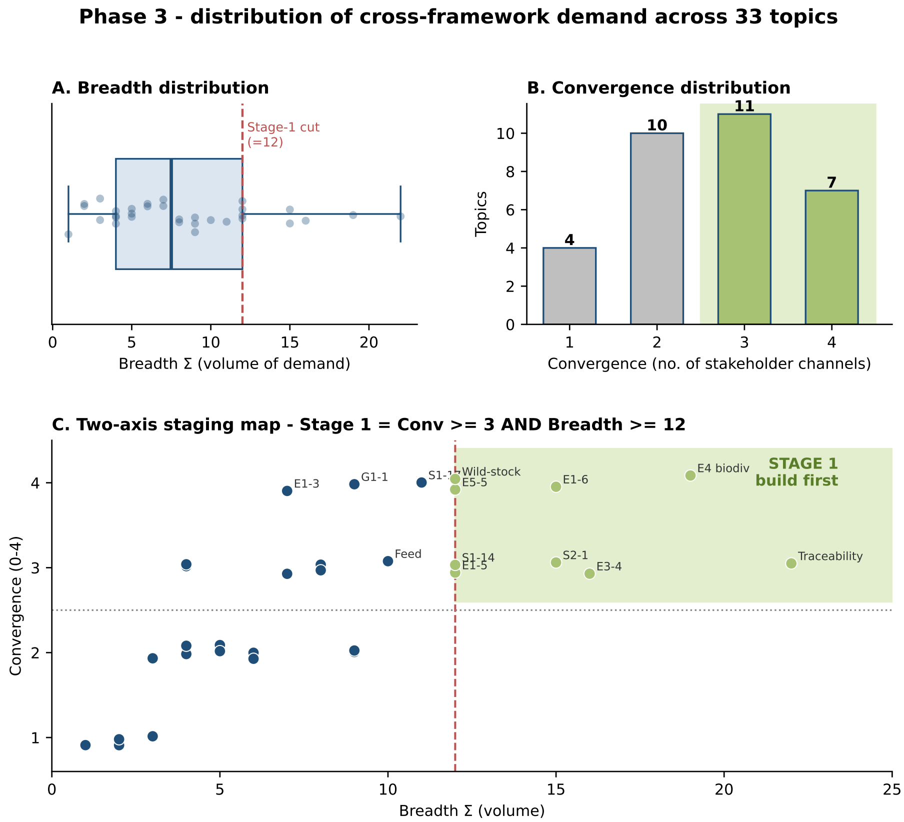

Figure 10: Distribution of cross-framework demand across the 33 topics for the modelled firm. (A) Breadth Σ (volume of demand); the Stage-1 cut is the third quartile, Q3 = 12. (B) Convergence - the number of the four stakeholder channels demanding each topic. (C) The two-axis staging map: Stage 1 (*build first*) requires both broad consensus (convergence ≥ 3) and high volume (breadth ≥ 12); the shaded region is the Stage-1 set. All 33 topics are plotted in panel C, but only the nine Stage-1 topics and the nearest few below the cut are labelled, to keep the panel legible; the remaining points are unlabelled rather than absent. E1-1, the climate transition plan added to the universe by the bridge rebuild, plots at breadth 8 and convergence 3 and so falls below the volume gate.

6.5 The composition of the demand environment

The populated matrix is read first as a description of the demand *environment* the modelled sub-threshold SME (50–250 employees, non-tuna) faces. Its 31 scored columns are distributed unevenly across the four channels (Figure 11): the environment is dominated by regulators (12 columns) and customers (11), which together account for 23 of the 31 columns (74%), while investors are the thinnest channel (3) and industry/NGO initiatives the remainder (5). The necessity tiers fall 12/7/8/4 across Tiers 1–4, so the regulatory legal-licence layer is the single largest block.

The matrix is sparse by construction: of the 1,023 instrument×topic cells, 188 (18%) carry a demand, split almost evenly between *binding* requirements (92 cells scoring 2) and *touched* references (89 cells scoring 1). The demand environment is therefore not a dense web in which every framework touches every topic, but a structured field in which a minority of concentrated, mostly regulator- and customer-borne demands define the landscape.

Source and necessity tier are coded on independent criteria, yet the two axes correlate: the instruments at Tier 4 (strategic differentiation) cluster almost entirely in the industry/NGO channel (SBTi, the UN Global Compact, voluntary coalitions), while the binding licence-to-operate tiers are dominated by the regulator channel. Strategic, proactively-adopted initiatives tend to be NGO- or coalition-run rather than regulator-mandated or capital-demanded. This is more than the truism that voluntary means non-mandatory: it locates *who owns the elective frontier*. For a seafood processor, the route to a proactive posture therefore runs through civil-society standards rather than through the state or the capital markets. The sector's voluntary leadership agenda is set by NGOs and coalitions, not by investors - a point the discussion returns to.

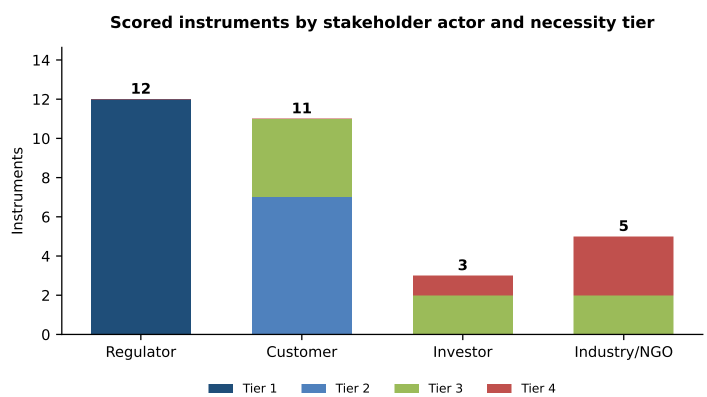

Figure 11: Composition of the scored demand environment: the 31 instrument columns by stakeholder channel and necessity tier. Regulators and customers dominate; investors are the thinnest channel.

6.6 Narrow law, broad cascade

A second pattern concerns not *who* demands but how *broadly* each instrument demands, and it reframes the SME's problem. Individual regulations are narrow: each of the twelve regulatory instruments touches on average only two topics, and none more than three - the Packaging and Packaging Waste Regulation covers waste and packaging, the occupational-health directive covers health and safety, the F-gas Regulation covers emissions. Breadth - and therefore the overlap that creates the SME's burden - comes overwhelmingly from the market and voluntary side: customer instruments demand on average nine topics each (the CSDDD eighteen), and industry and NGO reporting frameworks almost eight (GRI alone twenty-five).

The wall of overlapping demand is therefore not the product of the SME being directly over-regulated. It is the product of a few *broad* horizontal instruments - CSRD, CSDDD, SFDR - that apply to the firm's large customers and financiers and *cascade* down the value chain as supplier questionnaires, together with the voluntary certification and rating landscape. Direct regulation on the processor is targeted and lean; the breadth is transmitted, not imposed. This has a direct consequence for what an SME should do. The broad horizontal instruments that generate the cascade - CSRD and its ESRS - are also, ultimately, the most *comprehensive* statement of what will be asked: a firm fully aligned to the ESRS would, for the general-sustainability topics, already hold almost everything a customer or financier could cascade down to it. Full ESRS alignment is therefore the sensible end-state. But it is a large ask for a resource-constrained firm, and in the interim the demand actually placed on SMEs is being deliberately capped - by the VSME and by the Omnibus value-chain provisions - to a proportionate subset. The pragmatic path this chapter identifies sits between the two: rather than attempt full alignment at once or simply wait for the caps to settle, a firm builds first the *most convergent* ESRS topics - those the demand mapping shows are asked across the most channels - which are exactly the topics that secure its licence to operate and to sell today, while laying ESRS-aligned foundations it can extend, topic by topic, toward the fuller proactive posture. The same structure carries a policy corollary, developed in the discussion: because the burden originates in the downward cascade of large-firm regulation rather than in narrow direct rules, it is the coordination and capping of that cascade - the aim of the VSME and the Omnibus - that would most reduce it.

6.7 The build-first set by production system

Because several seafood instruments and topics are production-system-specific, the diagnostic returns a different Stage-1 (build-first) set depending on whether the firm is wild-capture, aquaculture, or both (Figure 12). For the standard modelled profile the Stage-1 set is nine topics for a mixed-sourcing firm, eight for an aquaculture-only firm, and five for a wild-capture-only firm.

The *overlap* is a universal core of four topics that should be build-first under every production system: energy (E1-5), greenhouse-gas emissions (E1-6), biodiversity and ecosystems (E4), and supply-chain traceability. These are the no-regret priorities for any European seafood processor - demanded broadly, across stakeholder channels, irrespective of business model.

Traceability is also the cheapest of the four to build, which is why it heads the sequence rather than merely appearing in it. Most European processors already hold BRCGS or IFS certification, and both require full traceability and chain of custody as a condition of the certificate (BRCGS, 2022, clause 3.9; IFS Management GmbH, 2023, requirement 4.18). The underlying records therefore largely exist before any sustainability requirement is raised; what is usually missing is not the data but its organisation for external reporting. Meeting the MSC or ASC chain-of-custody requirement on top of an existing BRC or IFS system is correspondingly less burdensome than its position in the demand matrix might suggest. This is the build-once case in its clearest form: the topic with the broadest demand across the ecosystem is also among the least costly to supply.[5]

The *differences* are structured rather than arbitrary. Four further topics - water (E3-4), waste (E5-5), occupational health and safety (S1-14) and value-chain workers (S2-1) - are Stage 1 for mixed and aquaculture firms but slip to Stage 2 for a wild-capture firm. This is a volume effect, not a change in importance. The matrix carries three aquaculture certifications (ASC, BAP, GLOBALG.A.P.) against two wild ones (MSC, Friend of the Sea), so a wild-only firm loses the certification demand that lifts these shared topics over the breadth threshold. Their cross-stakeholder convergence is unchanged, and several retain a regulatory floor (OSH, the Forced Labour Regulation, PPWR).

The counts follow from this. Four core topics plus these four give the eight of an aquaculture-only firm; wild-stock sourcing joins them for a mixed firm, giving nine. A wild-capture firm keeps the four core topics plus wild-stock sourcing, and five is its total. Feed and animal health/antimicrobial use are scored for aquaculture and mixed firms, but reach only Stage 2 and Stage 3 respectively, which is why they do not add to the Stage-1 count. The personalisation therefore refines *which* of a shared backbone a firm should prioritise; it does not produce wholly different agendas for different firms.

[5] This point was raised by a reviewer with sector experience and is reflected here; see also the traceability systems reported by the cohort in Table 2.

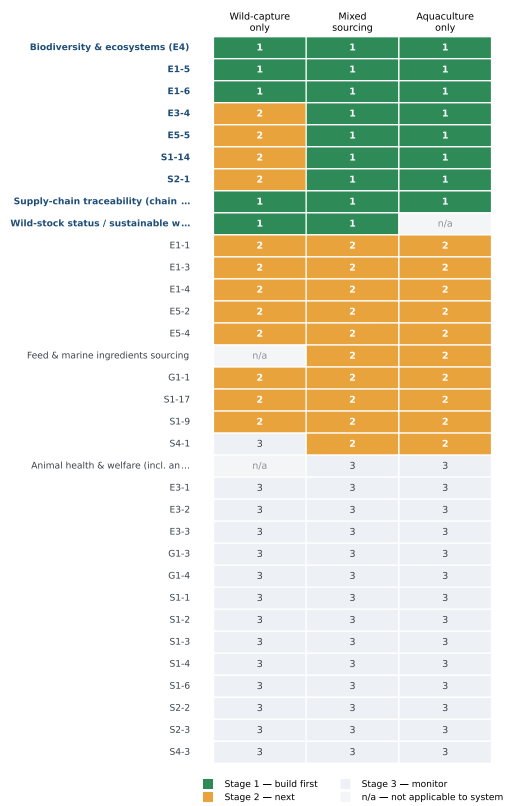

Figure 12: First action points by production system. Each topic's stage (1 = build first, 2 = next, 3 = monitor) under a wild-only, mixed and aquaculture-only profile; "–" marks a topic not applicable to that system. The four topics that are Stage 1 across all systems form the universal core.

6.8 Step 4: from themes to datapoints, and the diagnostic tool

The build-first data layer. The staging returns priority *themes*; a firm still needs to know which *datapoints* to collect for each. A datapoint layer translates every priority theme into the specific data the firm must produce. The ESRS topical requirements decompose into their constituent datapoints (the ESRS datapoint catalogue under Commission Delegated Regulation (EU) 2023/2772): the 26 DR-level Stage rows resolve to 356 official datapoints and the two topic-level rows (E4, S4) to a further 152. For each theme the *proportionate* subset - the realistic starting point for a smaller firm - is identified through the VSME-to-ESRS crosswalk (Appendix E - ESRS to VSME crosswalk), which maps every VSME Basic and Comprehensive module datapoint to the ESRS disclosure requirement it is thematically closest to and flags whether it is realistically self-reportable without external verification. The crosswalk operates at disclosure-requirement level, not at official-datapoint level: no ESRS datapoint is itself tagged as a starting point, but a VSME datapoint mapped onto a theme's requirement shows that a lighter, self-assessable version of that requirement exists. Applied to the Stage-1 set, five of the seven ESRS-anchored topics have solid VSME coverage - energy, GHG emissions, water, waste, and health and safety. Biodiversity is covered only partially, and value-chain workers (S2-1) not at all. This confirms at datapoint resolution the gap already identified in Appendix F - ESRS 2023-to-2026 revision mapping and VSME cross-reference. The four seafood-specific themes have no ESRS requirement and are sourced from GRI 13, so they correctly have no VSME entry either.

The enabling layer. Knowing which datapoints to collect is not the same as knowing how to produce them. Each topic therefore also carries the enabling instruments that generate its data - the GHG Protocol for emissions accounting, GDST for traceability, ISO 14001 for environmental management. These are methods and management systems rather than disclosure demands, which is why they are held outside the demand sums (Section 6.3). Eight of the nine Stage-1 topics carry at least one enabling instrument; biodiversity is the exception, with no established method specific enough to name, which is consistent with its partial VSME coverage and with its being the topic where demand most clearly outruns practice (Section 7.1).

Each *topic* (not datapoint) carries the count of stakeholder demands it serves, taken from the matrix (Section 6.4). This makes the reuse explicit: a single topic typically satisfies several frameworks at once. Because both the ESRS and the VSME are under active Omnibus revision, the layer is a dated snapshot documented with its sources.

The diagnostic tool. The staging logic is operationalised as a working prototype for an individual SME. The tool takes two inputs the firm can supply directly. The first is a short reporting-posture self-assessment of five to eight items: whether it has a formal sustainability strategy, a dedicated function or role, externally or internally initiated reporting, voluntary commitments such as SBTi-aligned targets, and disclosure data that is reusable across requests. The second is its production profile - wild, aquaculture or mixed, and size band. It returns the firm's Stage-1 build-first topics filtered to that profile, and the datapoint layer beneath them. This is the core of the diagnostic: it turns the general staging into a personalised shortlist.

Two extensions are designed but not yet built into the prototype, and are described here as the intended next increment. The first is a *stakeholder-pressure input*: the firm would identify which customers, certifications and investors are actively asking, so the shortlist could be narrowed further to the topics where its *actual* demand and the convergent topics coincide. The second is a fully automated *staged roadmap* sequencing Stage 1 → 2 → 3 against the firm's posture and resourcing. Positioning on the reactive-to-proactive spectrum is left to the self-assessment, on the basis that posture is a behavioural rather than a measurable categorical state that the firm is best placed to judge.

6.9 Step 4 continued: worked case and validation

Worked case: scope and next step. Validating the diagnostic against a real firm's public disclosures is the natural empirical next step after this thesis, and is deliberately scoped here as specified future work rather than attempted under this thesis's time constraints. The design is specified precisely enough to be immediately actionable: a representative European mid-sized seafood-processing SME (50–250 employees, primarily a B2B supplier to large EU retailers) would be profiled using only material the firm has itself published - its sustainability report, certification listings, and customer and product information - so the exercise depends on no privileged access and is fully reproducible by a future researcher or practitioner without a partner firm's cooperation. The diagnostic would be run on that profile, and its output examined for correspondence to the firm's observable position: whether it locates the firm's posture plausibly, and whether the topics it flags as build-first match the demands the firm is visibly subject to. By design the case is singular and *formative rather than confirmatory* - a usability demonstration of the diagnostic, not a test of the underlying empirical claims about the cohort or the cross-framework demand pattern - and no generalisation beyond it would be claimed. Practitioner validation of the diagnosis with the firm itself is a further, separate step, identified here and left to future work regardless of whether the worked case is carried out first.

Figure 13 shows the diagnostic's actual output screen, illustrating the format this validation step would use. The profile shown sits within the scoped parameters - mixed wild-plus-aquaculture sourcing, sub-250 employees - rather than belonging to a named firm. Given a production profile, the tool returns a ranked Stage-1 build-first set and states how much of the demand environment building it answers. A firm therefore sees not only *what* to build first but *why that subset.*

Validation of the mapping and the mechanism. Both the demand mapping

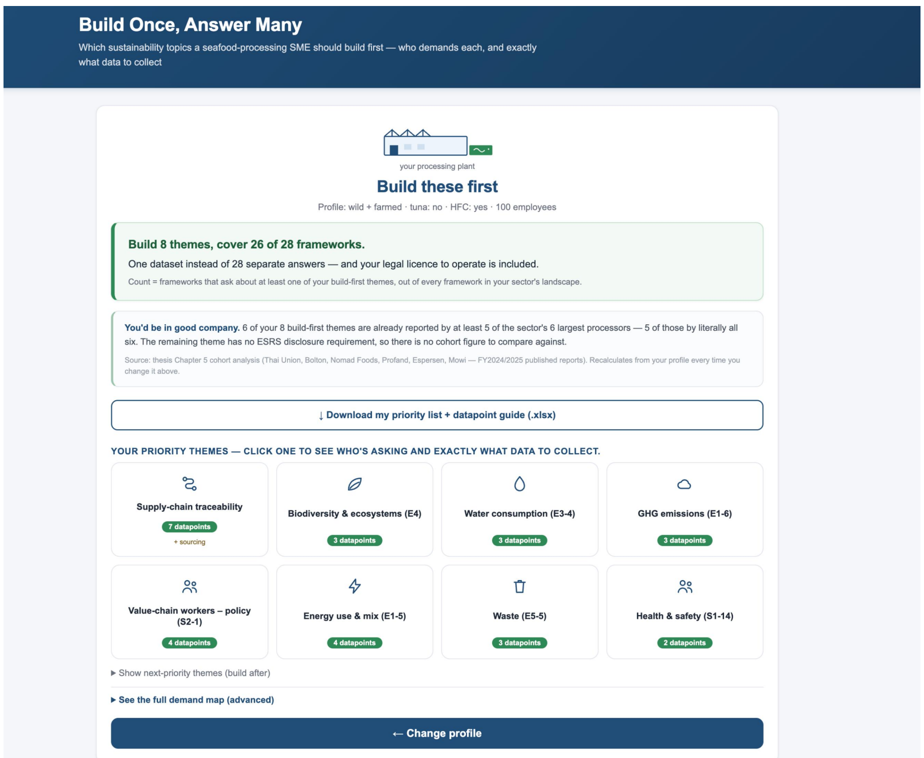

Figure 13: The diagnostic tool's output screen: for a given production profile, it returns the ranked Stage-1 (build-first) themes, the datapoints each requires, and the share of the sector's demand environment they jointly answer. This is the operational form the staging logic of Section 6.4 takes for an individual firm. The mixed profile shown returns nine Stage-1 topics but eight tiles, because the tool presents wild-stock sourcing within supply-chain traceability - the two are evidenced by one set of chain-of-custody data.

and the capability-sharing claim beneath it are substantiated from documents, so that credibility rests on independently inspectable evidence rather than on any single informant.

The *descriptive* claim is that the ecosystem can be organised exhaustively and unambiguously by the necessity, domain and source axes. Its test is the completeness of the classification itself, every instrument placed and every placement reasoned. The classification has also been submitted to the sustainability leads at the cohort firms for member checking (Lincoln & Guba, 1985); that corroboration is outstanding at the time of writing.

The *explanatory* claim is that capabilities are shared across requirements, so that investment compelled by a lower-tier requirement subsidises higher-tier disclosure. This is substantiated through an *input-overlap analysis*: for each shared capability, the inputs the drawing instruments require are read directly from their published specifications. MSC chain-of-custody, the GDST standard and ESRS value-chain disclosure all draw on the same body of lot-level traceability data - shared inputs that are document-verifiable rather than self-reported.

The evidence is therefore *consistent with* the capability-sharing mechanism; it does not prove a causal relationship. Establishing that would take retrospective practitioner testimony on whether capability built for a mandatory requirement reduced the effort of later disclosures. That is the natural next stage, and is left to future work.

The implications for seafood SMEs and for EU disclosure policy are drawn out in the discussion that follows.

7. Discussion and conclusion

7.1 What the four steps establish together

The four steps answer one question, not four unrelated ones. Step 1 catalogued 49 instruments acting on a European seafood-processing SME and found they reach the firm through only four stakeholder channels. Step 2 found the sector's six most advanced reporters cover between 33 and 60 of the 70 screened ESRS disclosure requirements and converge on a 13-requirement core that survives every robustness cut applied to it. Step 3 crossed the two: of 1,023 instrument-by-topic cells only 188 (18%) carry demand at all, and that sparse demand concentrates on a Stage-1 set of nine topics for a mixed-sourcing firm, which between them touch a topic asked about by 26 of the 30 frameworks active in the sector.[6] Step 4 turned that into a diagnostic a firm can apply to its own profile (6.8 Step 4: from themes to datapoints, and the diagnostic tool).

[6] Counting a framework whether it legally binds the firm or is a certification the firm may choose to pursue.

The answer to the organising question follows directly. A resource-constrained seafood-processing SME does not face an undifferentiated wall of demand. It faces a small, identifiable, cross-cutting core, surrounded by a long tail of narrower requirements. What to build first is nameable, and now toolable.

A closer look at where Step 2's and Step 3's two "cores" agree, and where they do not, sharpens this picture further. Of Step 3's four-topic universal Stage-1 core - the topics a firm should build first irrespective of production system - two (energy consumption, E1-5, and gross GHG emissions, E1-6) sit squarely inside Step 2's 13-requirement reported baseline: the sector's mature reporters already treat them as standard practice, and the cross-stakeholder demand structure independently confirms they are exactly where demand concentrates. The other two, however, are not yet part of that reported baseline: biodiversity (E4) only clears Step 2's near-universal threshold through the topic-level union rule rather than as a core requirement outright, and supply-chain traceability has no ESRS home at all and enters Step 3's rows only as a seafood-specific addition (Section 6.2). Conversely, water, governance and workforce topics anchor Step 2's reported baseline without appearing in Step 3's cross-system core. This is not a contradiction between the two steps but a finding in its own right. Reporting practice and cross-stakeholder demand agree that climate accounting is foundational. They diverge in two directions after that. Reporting *lags* demand on biodiversity and traceability, the newer and harder-to-operationalise topics. It runs *ahead* of demand on water, governance and workforce. The likely reason for the second gap is that those topics are anchored by requirements specific to particular sourcing models (Section 6.7), which a cross-system view averages away.

7.2 Reading the findings through the conceptual framework

Chapter 2 set out three lenses for reading multi-stakeholder demand. Each now returns something the findings sharpen, and in one case something the findings complicate.

Stakeholder salience: transmission, not just attributes. Mitchell, Agle and Wood (1997) explain how firms triage competing claims by the power, legitimacy and urgency a claimant holds. Applied directly to an SME, that account predicts the firm attends most to whoever can most immediately damage it. The narrow-law/broad-cascade finding (Section 6.6) suggests the mechanism is more indirect than that. The instruments generating most of the demand a small processor experiences - CSRD, CSDDD, SFDR - are not ones for which the processor is a claimant's target at all; they bind its *customers*, and reach the processor as transmitted questionnaires. The salient party, in the sense the theory intends, is the retailer. But the substance of what is asked is set by a regulator with no direct relationship to the processor whatsoever.

Salience, on this evidence, is not only a property of the claimant a firm faces. It is a property of the *chain* through which an obligation travels: a high-power regulator acting on a high-power customer generates more demand on a small supplier than a regulator binding that supplier directly. This is why Chapter 4 codes instruments to the channel through which they arrive rather than to their nominal issuer, and it is a refinement the seafood case makes visible because the sector's SMEs sit almost entirely below the CSRD threshold while selling almost entirely to firms above it.

Institutional isomorphism: which mechanism? The 13-requirement universal core (Section 5.5) is a convergence result, and DiMaggio and Powell (1983) offer three candidate explanations. Coercive isomorphism fits poorly on its own: the six firms report under different regimes, three under the ESRS and three under the GRI at two different application levels (Section 5.3), so no single mandate compels the overlap. That leaves the mimetic and normative mechanisms, which the evidence cannot fully separate. What it can show is that the core survives every cut applied to it - by threshold, by cohort composition, by EU versus non-EU membership (Section 5.7) - which is what one would expect of a settled professional norm and less of what one would expect from firms imitating a single visible leader.

The finding that would distinguish the two is not available here and is worth naming as such: mimetic pressure predicts a diffusion *sequence*, with later adopters following earlier ones topic by topic, whereas a normative account predicts firms arriving at the same core independently through shared professional and assurance practice. A longer panel than two years would let a future study adjudicate this directly.

The resource-based view, and where it cuts against the argument. The capability-sharing claim rests on the resource-based view and its dynamic-capabilities extension (Barney, 1991; Teece, 2007): structured data, once built, is redeployable across many uses at low marginal cost. The redeployability half holds, and the input-overlap analysis (Section 6.9) documents it.

Barney's other criteria do not, and this should be stated plainly rather than passed over. A capability confers sustained competitive advantage only if it is also rare and costly to imitate. Lot-level traceability data is neither. Every processor can build it, the enabling instruments that produce it are public (Section 6.8), and the sector's leading firms have already built it. On a strict reading of Barney, the build-first core is not a source of advantage at all: it is the price of remaining in the market.

That is not a defeat for the argument; it changes what the argument claims. The return on building the convergent core is not differentiation but *efficiency and access* - the same licence to operate and to sell obtained at materially lower cost than answering each demand separately, which is the compound return Chapter 2 describes and Chapter 6 measures. Advantage, if it exists, sits one level up: in the organisational capability to identify which data to build and to reuse it quickly, which is a dynamic capability in Teece's sense rather than a resource in Barney's. The diagnostic developed here is best understood as an attempt to lower the cost of acquiring precisely that capability for firms that cannot develop it through experience alone.

This is the precise sense in which compliance obligations become strategic advantage, and it is worth stating exactly because the phrase invites a stronger reading than the evidence supports. The obligations themselves confer nothing: every competitor faces them, and every competitor can meet them. What is open to advantage is the *manner* of meeting them - once, deliberately, and in a form that also serves the next demand. The return is realised as cost avoided and access retained rather than as a position rivals cannot occupy, and it accrues to the firm that builds the underlying capability earliest rather than to the one that holds any particular dataset.

7.3 Contribution to seafood-processing SMEs

For seafood-processing SMEs, the contribution is the Stage-1 build-first set itself and the datapoint layer beneath it (Section 6.8), not as an abstract finding but as a concrete starting list: energy and GHG accounting (E1-5, E1-6), water (E3-4), biodiversity (E4), supply-chain traceability, and - for mixed and aquaculture profiles - waste, occupational health and safety, and value-chain-worker policy. Building these first is a *sequencing* argument, not a completeness one: because even the cohort's best-resourced reporters leave roughly a quarter of the screened requirements uncovered (Section 5.4), completeness is the wrong target for a smaller firm, and the diagnostic's contribution is to say which subset to prioritise rather than to promise a shortcut to full compliance.

The diagnostic tool (Section 6.8) turns this from a static finding into a usable instrument: a firm supplies a short self-assessment and its production profile, and receives its own filtered Stage-1 set and datapoint layer rather than the sector-wide one. This matters because the build-first set is not identical across production systems (Section 6.7), and the reason is a genuine feature of the current instrument landscape rather than an artefact of how the tool is built: the catalogue carries three aquaculture certifications (ASC, BAP, GLOBALG.A.P.) against only two for wild-capture (MSC, Friend of the Sea). For a shared topic such as water or waste, a wild-capture-only firm loses the extra breadth those additional aquaculture certifications would have contributed, so the topic's volume score falls below the Stage-1 cut even though its *convergence* - how many of the four stakeholder channels ask about it - is unchanged. A wild-capture-only firm therefore has a narrower Stage-1 set not because the tool treats it differently in principle, but because fewer certifications currently exist for its sourcing model; a generic, one-size-fits-all list would consequently over- or under-state what any individual firm actually needs.

The diagnostic's output format is illustrated in Section 6.9 (Figure 13); a worked case applying it end to end to a specific, publicly reconstructable firm and checking its flagged Stage-1 topics against that firm's actual disclosures is the natural closing demonstration for this section, once that case is identified.

7.4 Academic contribution

The thesis makes four contributions to the literatures reviewed in Chapter 2. First, it supplies what that review found missing (Section 2.1, "the gap"): a systematic mapping of cross-stakeholder disclosure demand at sector level, built on the defined 0/1/2 coverage scale set out in Section 6.3 - not covered, touched, or bindingly required - rather than an impressionistic count of "how many frameworks" apply.

Second, the narrow-law, broad-cascade finding (Section 6.6): direct regulation on the processor is targeted and lean, while breadth is transmitted down the value chain from a handful of horizontal instruments. To the author's knowledge this has not been demonstrated empirically for the sector before, and it reframes the burden question - the wall of demand a small firm experiences is substantially a cascade effect of large-firm regulation (CSRD, CSDDD, SFDR) rather than direct over-regulation of the SME itself. Section 7.2 draws out what this implies for stakeholder-salience theory.

Third, the thesis extends the reactive–proactive and maturity literatures (Baumgartner & Ebner, 2010; Sharma & Vredenburg, 1998; Torugsa, O'Donohue & Hecker, 2013) - which describe the posture and its performance consequences but say little about how the external structure of demand enables or blocks the transition - with a diagnostic-tool operationalisation: a way of making the "build shared capability" prescription concrete for a named sector, rather than leaving it at the level of strategic recommendation.

Fourth, the cohort analysis (Chapter 5) is itself a methodological contribution to the maturity literature: it demonstrates that a sector's disclosure frontier and its universal core can be derived directly from mature firms' own reporting practice, cross-validated for robustness against threshold choice, cohort composition and reporting year (Section 5.7), rather than assumed from theory or asserted by expert judgement alone.

7.5 Implications for EU disclosure policy

The findings speak directly to the live Omnibus simplification debate, and in one respect more concretely than a debate still in progress: as detailed below, the direction this thesis anticipated is now a matter of adopted record rather than proposal. Three points stand out.

First, the narrow-law/broad-cascade finding (Section 6.6) locates the policy lever precisely: because the SME's burden originates overwhelmingly in the downward cascade of large-firm regulation rather than in direct sector-specific rules, it is the *coordination and capping* of that cascade - the explicit aim of the VSME and the Omnibus value-chain provisions - that would reduce SME burden fastest, more than any further simplification of direct obligations on the SME itself.

Second, and more speculatively than the other two, the cohort's contracting disclosure trend (Section 5.7) is worth flagging as a direction to watch rather than a conclusion this thesis can support: coverage fell slightly from 2024 to 2025 for the three cohort firms with both years, in the same direction as the decline in declared materiality documented across 241 large listed EU companies (Nahrstedt, 2026). Three firms over two years cannot establish whether this reflects firms reading the Omnibus's relief as a floor or a ceiling - that would need a far larger panel than this thesis's cohort provides - but the fact that an independent, much larger sample moves the same way is reason enough to flag the question for the re-anchored analysis in Section 7.7 rather than to answer it here.

Third, the sector's own blind spots and the revision's own priorities turn out to point in *opposite* directions, which is a sharper finding than either alone. The requirements that phase in slowest under the Omnibus timeline - anticipated financial effects and pollution disclosure - are exactly the cohort's collective blind spot (Section 5.6); simplification is, for now, deferring precisely what the sector was not yet disclosing anyway. Biodiversity runs the other way. It is the topic the demand structure most clearly flagged as under-reported relative to how widely it is demanded (Section 7.1), and the revision tightens it rather than easing it: the standards adopted on 3 July 2026 make biodiversity transition plans mandatory where they were previously voluntary. Meanwhile the water standard loses roughly 70% of its mandatory datapoints, the deepest cut of any topical area - and water was already part of the cohort's reported baseline (Section 5.5). The revision is therefore easing what mature practice had already absorbed and tightening what this thesis found demand was outpacing practice on. That is consistent with the Chapter 6 demand structure rather than a correction of it.

This alignment is no longer only a matter of timeline projection. On 3 July 2026 the European Commission formally adopted the revised ESRS and a new voluntary reporting standard for smaller companies built on the VSME, cutting mandatory datapoints by over 60% and total datapoints by over 70% (European Commission, 2026) - the reduction this thesis anticipated throughout (Section 1.5) is therefore no longer a proposal under debate but an adopted delegated act, pending only the standard two-to-four-month European Parliament and Council scrutiny period before it applies. The adopted standard also formalises the value-chain cap referenced in the introduction (Section 1.1): CSRD-scoped firms may not require value-chain partners to supply more than the voluntary standard covers. This does not change any finding in this thesis, whose cohort and demand-matrix analysis predate the adoption, but it does mean the policy debate this thesis speaks to has moved, in the ten days before submission, from live negotiation to implementation - and it strengthens rather than weakens the case for the future-work item below: re-anchoring this analysis to the adopted standard once it takes effect.

Fourth, the case this thesis makes for building VSME-aligned capability rests on an assumption worth testing against independent evidence: that customers, banks and investors will actually treat a VSME-based disclosure as sufficient, rather than continuing to ask for more. EFRAG's own market acceptance survey of preparers and users (EFRAG, 2025) gives this some support: it finds developing, positive acceptance of the VSME, with financial institutions and investors expecting its modules to satisfy a substantial share of the requests they currently address directly to SMEs. The finding is preliminary, and self-reported to the standard-setter that designed the VSME, so it should be read as encouraging rather than conclusive. It nonetheless points the same way as the narrow-law/broad-cascade finding above. The demand a resource-constrained firm faces is transmitted through a small number of large channels - customers and financiers - which a proportionate, standardised disclosure can plausibly satisfy. It is not an open-ended list that every actor insists on customising.

7.6 Limitations

Each step's own limitations are stated where the relevant claims are made (Sections 5.*x*, 6.7, 7.*x*); four apply to the thesis as a whole and are drawn together here rather than left scattered.

Small, purposive samples throughout. The cohort (N=6) and the demand matrix's instrument catalogue are both purposive constructions - an upper-bound proxy and a comprehensiveness claim respectively - not statistical samples, so no claim of population-level representativeness is made anywhere in the thesis. The robustness checks in the Section 5.7 section substitute cross-validation for statistical inference; they cannot substitute for it.

Analyst judgement in the coding. The GRI-to-ESRS mapping (Section 5.3), the per-cell demand scoring (Section 6.3) and the necessity/domain classification (Section 4.1) all rest on a single analyst's documented judgement, checked against defined criteria and logged for audit but not independently double-coded. The member-checking review submitted to the cohort firms (Section 4, Validation) is a corroborating step, not a formal inter-rater-reliability test, and at the time of writing it is still outstanding. A second-coder check remains future work (Section 7.7). One firm's review has since returned and is reflected in Section 5.3.

What the framework bridge cannot carry. Cohort coverage is coded from each firm's own framework index, and three of the six firms are indexed under the GRI Standards. Two consequences follow, both set out in Section 5.3. A firm whose index is selective scores below what its report body contains, which is a question of indexing practice. More structurally, 14 of the 70 topical disclosure requirements have no counterpart anywhere in the GRI-ESRS interoperability index, so no GRI-indexed firm can score them at all; those 14 can be reported by at most three of the six firms and can never enter the universal core or the near-universal set. The effect on the findings is bounded and small. None of the 14 could reach six of six even if every GRI-indexed firm reported it in full, because each already lacks at least one ESRS-native reporter, so the universal core of 13 is exact rather than a floor. Only two, E1-1 and S2-5, could in principle reach five of six, placing an upper bound of 28 on the near-universal set of 26. The limitation is real, and it does distort the firm-by-firm coverage comparison, but neither headline result rests on the unmeasured cells.

The sector-material row set is curated, not exhaustive. The four sector-material rows (6.2 Assembling the matrix: rows and columns) were chosen because GRI 13 asks for genuinely different operational data - species, volumes, stock status - that no ESRS or generic GRI disclosure asks for at all. GRI 13 contains other sector-specific disclosures that were judged, but not included, because they refine an already-covered topic rather than open a new one: certified-volume reporting under food safety (13.10), and nationality/migrant-status breakdowns under non-discrimination (13.15), for example, sit inside topics the cohort already reports on. One further candidate follows the same logic already used for feed and animal welfare above and was a closer call: ASC's conversion-free-sourcing criterion, which reaches a processor exactly as feed and welfare demand does, as a certification precondition rather than direct operational reporting. It is left out of the four rows here for scope, not because the logic doesn't apply - extending the sector-material row set on that same certification-precondition basis is a natural next check, not a finding this thesis is confident closes the question.

A 2024–2025 snapshot of a moving target. The ESRS, the VSME and the Omnibus simplification were all under active revision throughout this analysis, and the revised ESRS and a new VSME-based voluntary standard were formally adopted only in the final weeks before submission (3 July 2026; Section 7.5), with the standard two-to-four-month Parliament and Council scrutiny period still to run. The coverage figures, the demand-matrix scores and the year-over-year comparison in the Section 5.7 section are dated to the 2024–2025 window that precedes this adoption and will need re-anchoring once the revised standard actually takes effect. Checked against EFRAG's draft revised standards, that re-anchoring is a matter of translation rather than revision. Every disclosure requirement in the 13-requirement universal core (Section 5.5) and in the four-topic Stage-1 core common to all production systems (Section 6.7) survives in substance; none was deleted. Most simply carry a different DR code. Several standards gained new upfront requirements, chiefly on risk identification and resilience, which shifted everything after them down by one to three positions - energy consumption moves from E1-5 to E1-7, GHG emissions from E1-6 to E1-8. The full old-to-new code mapping, cross-referenced against the VSME, is given in Appendix F - ESRS 2023-to-2026 revision mapping and VSME cross-reference.

One formative, not confirmatory, worked case. The diagnostic's usability is demonstrated on a single public case (Section 6.9); no claim of generalisation beyond that case is made, and practitioner validation with a firm willing to share non-public information is explicitly left to future work.

7.7 Future work

Several extensions are flagged at the point they arise in Chapters 4–6; gathered here, they form a natural research agenda. *Practitioner validation*: running the diagnostic with a firm willing to share internal data beyond its public disclosures, to test the worked case's formative finding against a confirmatory one (Section 6.9). *Reliability*: a second-coder pass on the GRI-ESRS mapping and the demand-matrix scoring, to convert the current documented-judgement audit trail into a measured inter-rater agreement (Section 5.3 and Section 6.3). *Stakeholder-pressure input*: the tool extension, already scoped but not yet built (Section 6.8), that lets a firm identify which of its own customers, certifications and investors are actively asking, narrowing the generic Stage-1 set to the firm's *actual* demand. *A staged roadmap*: automating the Stage 1 → 2 → 3 sequencing against a firm's posture and resourcing, rather than leaving that positioning to the self-assessment alone (Section 6.8). *Re-anchoring to the adopted ESRS revision*: repeating the cohort and demand-matrix analysis once the standards adopted on 3 July 2026 (Section 7.5) clear Parliamentary and Council scrutiny and take effect, both to test whether the contracting-disclosure trend (Section 5.7) persists and to keep the diagnostic's datapoint layer current against the roughly 70%-smaller datapoint set. *Sectoral transfer*: testing whether the build-first core identified here is seafood-specific or reflects a more general cross-sector pattern, in the other resource-constrained manufacturing and agri-food SME contexts the introduction argues the approach should transfer to (Section 1.4). *From diagnosis to implementation*: this thesis stops at telling a firm *which* topics to build first; the natural next step is piloting the framework with actual firms to see how a prioritised topic list translates into the internal processes, ownership and data infrastructure needed to build it, and to learn where firms get stuck turning a build-first list into a working data-collection routine. Beyond validating the diagnostic's outputs (the practitioner-validation item above), this would test the thesis's implicit second claim - that naming the right topics is enough to unblock action - which this thesis has not itself tested.

7.8 Conclusion

This thesis set out to answer where, amid a thicket of overlapping sustainability demands, a resource-constrained European seafood-processing SME should begin. Three linked findings answer it. The disclosure ecosystem acting on such a firm is wide but structured, not undifferentiated (Chapter 4). The sector's own leading reporters already converge on a compact, cross-cutting disclosure core, not an ever-expanding one (Chapter 5). And that empirical core is exactly where cross-stakeholder demand is most concentrated, so that building it first answers the largest share of what the firm is actually asked, operationalised here as a diagnostic an individual firm can apply to its own situation (Chapter 6).

What is now known that was not. Three things. First, the demand a small processor faces is measurably sparse: 18 per cent of the possible instrument-by-topic space carries any demand at all, which is what makes prioritisation coherent rather than arbitrary. Second, the burden originates overwhelmingly in transmission rather than in direct regulation, so the wall a small firm experiences is a cascade from large-firm rules and not over-regulation of small firms themselves. Third, the sector's revealed reporting baseline and its cross-stakeholder demand structure substantially agree, and where they diverge - biodiversity and traceability, where demand runs ahead of practice - they identify precisely the topics a forward-looking firm should build before it is asked.

What remains open. The diagnostic has not been run with a firm that can supply internal data, so the claim that naming the right topics unblocks action remains untested (Section 7.7). The coding rests on a single analyst's documented judgement, with member checking outstanding at the time of writing. And whether the convergent core identified here is a seafood phenomenon or a general feature of resource-constrained manufacturing and agri-food sectors is an empirical question this thesis frames but does not answer: the *method* transfers by construction - any sector with a defined disclosure taxonomy, an identifiable set of leading reporters and a catalogue of instruments can be mapped the same way - but the *particular* build-first set almost certainly does not.

What it means in practice. For a firm deciding what to do on Monday morning, the answer is narrower than the literature's usual advice to become strategic about sustainability. Build energy and greenhouse-gas accounting, water, biodiversity and supply-chain traceability first, in that order, and build them as reusable data rather than as answers to whoever asked last. Those four are demanded across every stakeholder channel and under every production system studied here. The rest can wait, and knowing what can wait is the scarcer piece of information.

Together these findings turn "build a reusable sustainability-data capability" from a strategic aspiration into a named, sequenced and practically actionable starting list - the missing view this thesis set out to supply.

Declaration on the use of generative AI

Generative AI tools were used in the preparation of this thesis, under the author's direction and review throughout. They assisted with language and presentation - drafting, rewriting for clarity and concision, and copy-editing - and with computational work, including scripting the extraction of data from the source workbooks, generating the figures from that data, and checking the numerical claims in the text against the underlying files.

The research design, the coding decisions, the analysis and the interpretation of the results are the author's own, as is the responsibility for any errors that remain.

References

Adams, Carol A., and Subhash Abhayawansa. "Connecting the COVID-19 Pandemic, Environmental, Social and Governance (ESG) Investing and Calls for 'Harmonisation' of Sustainability Reporting." *Critical Perspectives on Accounting* 82 (2022): 102309. https://doi.org/10.1016/j.cpa.2021.102309.

Afolabi, Hammed, Ronita Ram, and Gunnar Rimmel. "Harmonization of Sustainability Reporting Regulation: Analysis of a Contested Arena." *Sustainability* 14, no. 9 (2022): 5517. https://doi.org/10.3390/su14095517.

Ahmad, Khursheed, Maria Grazia Marchesano, Valentina Popolo, Roberto Revetria, and Anastasiia Rozhok. "Development of a European Sustainability Reporting Standards Compliant Sustainability Assessment Framework for Manufacturing Organisations Using Analytic Hierarchy Process." *Sustainability* 17, no. 11 (2025): 4772. https://doi.org/10.3390/su17114772.

Ashurst. "European Commission Consults on Revised ESRS and Sustainability Reporting Standard for Voluntary Use." Accessed May 13, 2026. https://www.ashurst.com/en/insights/european-commission-consults-on-revised-esrs-and-sustainability-reporting-standard-for-voluntary-use/.

Barclay, Kate, and Alice Miller. "The Sustainable Seafood Movement Is a Governance Concert, with the Audience Playing a Key Role." *Sustainability* 10, no. 1 (2018): 180. https://doi.org/10.3390/su10010180.

Barney, Jay. "Firm Resources and Sustained Competitive Advantage." *Journal of Management* 17, no. 1 (1991): 99–120. https://doi.org/10.1177/014920639101700108.

Bataleblu, Ali Asghar, Erwin Rauch, David S. Cochran, and Dominik T. Matt. "Impact of European Sustainability Reporting Standards Guidelines on the Design of Sustainable Factories and Manufacturing Systems." In *Advances in Manufacturing IV*, edited by Adam Hamrol, Marta Grabowska, and Marcin Hinz. Lecture Notes in Mechanical Engineering. Springer Nature Switzerland, 2024. https://doi.org/10.1007/978-3-031-56474-1_18.

Battisti, Martina, and Martin Perry. "Walking the Talk? Environmental Responsibility from the Perspective of Small-Business Owners." *Corporate Social Responsibility and Environmental Management* 18, no. 3 (2011): 172–85. https://doi.org/10.1002/csr.266.

Baumgartner, Rupert J., and Daniela Ebner. "Corporate Sustainability Strategies: Sustainability Profiles and Maturity Levels." *Sustainable Development* 18, no. 2 (2010): 76–89. https://doi.org/10.1002/sd.447.

Bolton Group. *Sustainability Report 2024*. 2025. https://www.bolton.com/sites/default/files/2025-04/bolton-sustainability-report_2024.pdf.

Blank, Christine. "Major UK, US Grocery Chains Step Up Seafood Sustainability Initiatives." *SeafoodSource*, 22 April 2025. https://www.seafoodsource.com/news/foodservice-retail/major-us-uk-grocery-chains-step-up-seafood-sustainability-initiatives.

BRCGS. *Global Standard Food Safety, Issue 9*. London: BRCGS, 2022. https://www.brcgs.com/product-safety-standards/food-safety/.

Bush, Simon, and Peter Oosterveer. *Governing Sustainable Seafood*. Routledge, 2019. https://doi.org/10.4324/9781315780429.

Chan, Ricky Y. K., Jennifer W. M. Lai, and Namwoon Kim. "Strategic Motives and Performance Implications of Proactive versus Reactive Environmental Strategies in Corporate Sustainable Development." *Business Strategy and the Environment* 31, no. 5 (2022): 2127–42. https://doi.org/10.1002/bse.3011.

Cöster, Mathias, Gunnar Dahlin, and Raine Isaksson. "Are They Reporting the Right Thing and Are They Doing It Right? - A Measurement Maturity Grid for Evaluation of Sustainability Reports." *Sustainability* 12, no. 24 (2020): 10393. https://doi.org/10.3390/su122410393.

CSR Tools. "VSME-Nachhaltigkeitsbericht: Effizientes Mapping der ESRS-Themen." 2024. https://csrtools.eu.

DiMaggio, Paul J., and Walter W. Powell. "The Iron Cage Revisited: Institutional Isomorphism and Collective Rationality in Organizational Fields." *American Sociological Review* 48, no. 2 (1983): 147–60. https://doi.org/10.2307/2095101.

Dinh, Tami, Anna Husmann, and Gaia Melloni. "Corporate Sustainability Reporting in Europe: A Scoping Review." *Accounting in Europe* 20, no. 1 (2023): 1–29. https://doi.org/10.1080/17449480.2022.2149345.

EFRAG. *Voluntary Sustainability Reporting Standard for Non-Listed Micro-, Small- and Medium-Sized Undertakings (VSME Standard).* 2024. https://www.efrag.org/sites/default/files/sites/webpublishing/SiteAssets/VSME%20Standard.pdf.

EFRAG. *VSME Market Acceptance Survey Report.* December 2025. https://www.efrag.org/sites/default/files/media/document/2025-12/VSME%20Market%20Acceptance%20Survey%20Report.pdf.

European Commission. *Commission Delegated Regulation (EU) 2023/2772 of 31 July 2023 Supplementing Directive 2013/34/EU of the European Parliament and of the Council as Regards Sustainability Reporting Standards.* OJ L, 22 December 2023. https://eur-lex.europa.eu/eli/reg_del/2023/2772/oj.

European Commission. *Commission Recommendation of 30 July 2025 on Voluntary Sustainability Reporting Standards for Non-Listed Micro-, Small- and Medium-Sized Undertakings*, C(2025) 4984 final. 30 July 2025. https://eur-lex.europa.eu/eli/reco/2025/1710/oj.

European Commission. "Commission Adopts Revised Sustainability Reporting Standards." 3 July 2026. https://finance.ec.europa.eu/news/commission-adopts-revised-sustainability-reporting-standards-2026-07-03_en.

The European House – Ambrosetti (TEHA Group), with Erion. *Non-Negotiable Priorities for Businesses and the Future of the Sustainable Transition* (2026 Strategic Report). Presented at the Erion Forum, Rome, 17 June 2026. https://www.ambrosetti.eu/en/news/sustainable-transition-non-negotiable-priorities-for-the-future-of-businesses/.

Fazli, Haleem, Sami Farooq, Cheng Yang, and Brian Vejrum Wæhrens. "Proactive and Reactive Approaches towards Sustainable Practices in Manufacturing Companies: Emerging Economies Perspective." *Sustainability* 15, no. 17 (2023): 12796. https://doi.org/10.3390/su151712796.

"Feedback on Sustainability Reporting Standards: Additional Explanatory Information

Regarding the Value Chain Cap - Finance." Accessed May 13, 2026. https://finance.ec.europa.eu/news/feedback-sustainability-reporting-standards-additional-explanatory-information-regarding-value-chain-2026-05-06_en.

Fragidis, Garyfallos, and Triantafyllos Papafloratos. "An Information Architecture for the European Sustainability Reporting Standards (ESRS)." *Sustainability* 17, no. 17 (2025): 7675. https://doi.org/10.3390/su17177675.

Himanen, Satu. *Master's Programme in Circular Economy, Master´s Thesis.* 2025.

Global Sustainability Standards Board (GSSB). *GRI 13: Agriculture, Aquaculture and Fishing Sectors 2022.* Global Reporting Initiative, 2022. https://www.globalreporting.org/standards/sector-program/sector-standards/.

GRI and EFRAG. *GRI–ESRS Interoperability Index.* November 2024. https://www.globalreporting.org/media/qzmoeixv/esrs-gri-interoperability-index-november-2024.pdf.

Hummel, Katrin, and Dominik Jobst. "An Overview of Corporate Sustainability Reporting Legislation in the European Union." *Accounting in Europe* 21, no. 3 (2024): 320–55. https://doi.org/10.1080/17449480.2024.2312145.

IFS Management GmbH. *IFS Food: Standard for Assessing Product and Process Compliance in Relation to Food Safety and Quality, Version 8.* Berlin: IFS Management GmbH, 2023. https://www.ifs-certification.com/en/standards/food.

Jabareen, Yosef. "Building a Conceptual Framework: Philosophy, Definitions, and Procedure." *International Journal of Qualitative Methods* 8, no. 4 (2009): 49–62. https://doi.org/10.1177/160940690900800406.

Lincoln, Yvonna S., and Egon G. Guba. *Naturalistic Inquiry.* Beverly Hills, CA: Sage Publications, 1985.

Lozano, Rodrigo. "Are Companies Planning Their Organisational Changes for Corporate Sustainability? An Analysis of Three Case Studies on Resistance to Change and Their Strategies to Overcome It." *Corporate Social Responsibility and Environmental Management* 20, no. 5 (2013): 275–95. https://doi.org/10.1002/csr.1290.

Mitchell, Ronald K., Bradley R. Agle, and Donna J. Wood. "Toward a Theory of Stakeholder Identification and Salience: Defining the Principle of Who and What Really Counts." *Academy of Management Review* 22, no. 4 (1997): 853–86. https://doi.org/10.2307/259247.

Nahrstedt, Jan. "Double Materiality in Practice: Evidence from Two Years of CSRD Reporting." SSRN working paper, University of Hamburg, 6 May 2026. https://ssrn.com/abstract=6724699.

Nomad Foods. *Sustainability Report 2025.* 2026. https://sustainability.nomadfoods.com/.

Nielsen, Christian. "ESG Reporting and Metrics: From Double Materiality to Key Performance Indicators." *Sustainability* 15, no. 24 (2023): 16844. https://doi.org/10.3390/su152416844.

Philips. *Beyond Auditing: Philips' Approach to Supplier Sustainability.* 2025. https://www.philips.com/c-dam/corporate/en_AA/about/about-us/esg/downloads/b eyond-auditing-philips-approach-to-supplier-sustainability.pdf.

Russo, Michael V., and Paul A. Fouts. "A Resource-Based Perspective on Corporate Environmental Performance and Profitability." *Academy of Management Journal* 40, no. 3 (1997): 534–59. https://doi.org/10.5465/257052.

Sharma, Sanjay, and Harrie Vredenburg. "Proactive Corporate Environmental Strategy and the Development of Competitively Valuable Organizational Capabilities." *Strategic Management Journal* 19, no. 8 (1998): 729–53.

STECF (Scientific, Technical and Economic Committee for Fisheries). *Economic Report on the Fish Processing Industry (STECF 25-15).* Edited by L. Malvarosa, J. Virtanen, and J. Guillen. Luxembourg: Publications Office of the European Union, 2025. https://op.europa.eu/en/publication-detail/-/publication/5b285dab-e076-11f0-8439-01aa75ed71a1/language-en.

Suchman, Mark C. "Managing Legitimacy: Strategic and Institutional Approaches." *The Academy of Management Review* 20, no. 3 (1995): 571–610. https://doi.org/10.2307/258788.

Tanase, Auras. "Supplier ESG Assessment: A Quick Guide." Veridion, January 9, 2025. https://veridion.com/insights/articles/supplier-esg-assessment.

Teece, David J. "Explicating Dynamic Capabilities: The Nature and Microfoundations of (Sustainable) Enterprise Performance." *Strategic Management Journal* 28, no. 13 (2007): 1319–50. https://doi.org/10.1002/smj.640.

Thai Union Group PCL. *Sustainability Report 2025.* 2025. https://www.thaiunion.com/uploads/Sustainability_Report_2025_en.pdf.

Torugsa, Nuttaneeya Ann, Wayne O'Donohue, and Rob Hecker. "Proactive CSR: An Empirical Analysis of the Role of Its Economic, Social and Environmental Dimensions on the Association between Capabilities and Performance." *Journal of Business Ethics* 115, no. 2 (2013): 383–402. https://doi.org/10.1007/s10551-012-1405-4.

Tugliani, Teresa, Marianna Guareschi, and Filippo Arfini. "Evaluating the Applicability of the European Voluntary Sustainability Reporting Standards for Agrifood Small- and Medium-Sized Enterprises." *Italian Review of Agricultural Economics* 81, no. 1 (2026): 39–55. https://oajournals.fupress.net/index.php/rea/article/view/16734.

US Foods Holding Corp. *Seafood Policy.* Effective 25 March 2025. https://www.usfoods.com/content/dam/usf/pdf/Policies/Responsibly_Sourced_Seafood_Policy.pdf.

World Wildlife Fund. "What Are Fishery Improvement Projects?" Accessed June 4, 2026. https://www.worldwildlife.org/our-work/oceans/improving-fisheries/what-are-fishery-improvement-projects-and-how-do-they-work/.

Yin, Robert K. *Case Study Research and Applications: Design and Methods.* 6th ed. Thousand Oaks, CA: SAGE, 2018.

Primary sources - cohort sustainability reports

A. Espersen A/S. *Sustainability Report 2025.* 2026.

A. Espersen A/S. *Sustainability Report 2024.* 2025.

Bolton Group. *Sustainability Report 2024*. 2025.

Mowi ASA. *Annual Report 2025*. 2026. https://mowi.com/wp-content/uploads/2025/05/Mowi-Annual-Report-2025.pdf.

Nomad Foods Limited. *Sustainability Report 2025*. 2026.

Nomad Foods Limited. *Sustainability Report 2024*. 2025.

Profand Fishing Holding S.L.U. *ESG Annual Report 2025*. 2026. https://profand.com/en/esg-annual-report-2025/.

Profand Fishing Holding S.L.U. *ESG Annual Report 2024*. 2025.

Thai Union Group PCL. *Sustainability Report 2024*. 2025. https://www.thaiunion.com/uploads/sd-report-2024-en.pdf.

Appendices

Supplementary material supporting the chapters above.

Appendix A - Ecosystem Glossary

A.1 - Foundation and enabling instruments

Two bands of instruments sit outside the 49-instrument demand landscape catalogued in Appendix A.2 and are recorded here instead. **Tier 0** is the food-safety and food-law floor every EU seafood processor clears before sustainability strategy begins: in scope of the firm, out of scope of the sustainability-disclosure analysis. **Layer 2** supplies the accounting methods, management systems and data infrastructure a firm builds capability on; these are excluded from the demand sums because they do not themselves request a disclosure (Section 6.3).

Snapshot June 2026. EU sustainability law was under active simplification at the snapshot date; verify against EUR-Lex before citing.

Tier 0 - Foundation (food integrity)

Instrument	Objective and role
General Food Law (Reg 178/2002)	General food-safety principles, the one-step-back / one-step-forward traceability duty, and RASFF. The baseline food-safety and traceability regime.
Hygiene of Foodstuffs (Reg 852/2004)	General hygiene rules for food businesses.
Specific Hygiene Rules, Animal Origin (Reg 853/2004)	Hygiene rules for products of animal origin, including fishery products.
Food Information to Consumers (Reg 1169/2011)	Mandatory labelling and consumer information (ingredients, allergens, provenance) - distinct from the sustainability claims EmpCo regulates.
Food Contact Materials (Reg 1935/2004)	Safety of materials in contact with food; a packaging-material safety floor, distinct from PPWR's environmental requirements.
Microbiological Criteria (Reg 2073/2005)	Microbiological safety limits for foodstuffs.
Maximum Contaminant Levels (Reg 1881/2006)	Maximum permitted contaminant levels, relevant to seafood heavy-metal screening.
Pesticide Residues (Reg 396/2005)	Maximum residue levels in food and feed, relevant chiefly to aquaculture feed inputs.
REACH (Reg 1907/2006)	Registration and authorisation of chemical substances; governs processing aids and cleaning agents rather than disclosure.
Official Controls Regulation (Reg 2017/625)	Framework for official food-safety controls and enforcement across the EU.
IFS Food	Certifies food-safety and quality-management systems for processors supplying mainly continental-European retail. Effectively non-negotiable for retail listing; commercially it functions much like a Tier-2 market licence.

Instrument	Objective and role
BRCGS	Food-safety certification widely required by UK and international retailers; a near-substitute for IFS Food, buyer choice largely geographic.
FSSC 22000	ISO 22000 plus sector-specific prerequisite programmes; the third common GFSI-benchmarked option, interchangeable with IFS and BRCGS in most buyer specifications.
GFSI	Benchmarks food-safety schemes for mutual recognition. The reason IFS, BRCGS and FSSC behave as near-substitutes, and why they are not counted as three independent demands.

Layer 2 - Enablement / methodology layer

Instrument	Objective and role
GHG Protocol	The dominant corporate GHG accounting methodology, defining Scopes 1–3 and boundary-setting rules. Underlies every climate disclosure (ESRS E1, CDP, SBTi) without demanding reporting itself - a pure capability enabler.
ISO 14001	Environmental management system on a plan-do-check-act basis. Process backbone for environmental data.
ISO 45001	Occupational health and safety management system; underpins own-workforce H&S (S1) data, the management-system counterpart to the OSH Directive's legal duty.
ISO 50001	Energy management system including energy-performance monitoring; supports the auditability of energy-consumption data (ESRS E1).
ISO 14064	GHG quantification, reporting and third-party verification. Complements the GHG Protocol; commonly referenced in verification of ESRS E1 and CDP disclosures.
LCA methodologies	Quantify environmental impacts across a product's life cycle. Underpin environmental claims under EmpCo and product-footprint requests from EcoVadis and retail buyers.
PEF / PEFCR	The EU's Product Environmental Footprint method and category rules; the technical basis for substantiating EmpCo claims.
ISAE 3000	Assurance standard for non-financial information; sits behind CSRD's mandatory limited-assurance requirement and voluntary VSME assurance.
AA1000	Stakeholder-based assurance standard; an alternative to ISAE 3000's audit-style approach.
Supplier portals / data-exchange platforms	Digital infrastructure through which buyers collect standardised supplier data. The practical channel through which CSRD, CDP and EcoVadis value-chain requests reach the processor.

Selected literature and evidence base

Source	**What it supports**
Gutiérrez, N. L. et al. (2012), *PLoS ONE* 7(8): e43765	MSC-certified stocks trend healthier; associative, not causal.
van Putten, I. et al. (2020), *PLoS ONE* 15(5): e0233237	Reviews certification outcomes and FIP accountability concerns.
Sampson, G. S. et al. (2015), *Science* 348(6234): 504–506	Widely cited on FIPs and market access for developing-country fisheries.
EFRAG (2024), GRI–ESRS Interoperability Index; EFRAG–TNFD correspondence mapping	Official interoperability mappings underpinning the framework-connection logic.
FAO (2009), *Guidelines for the Ecolabelling of Fish and Fishery Products from Marine Capture Fisheries*, Rev. 1	Primary normative reference behind GSSI recognition criteria.

A.2 - Instrument catalogue (the 49-instrument demand landscape)

What this covers	Every instrument in the disclosure landscape (Chapter 4), classified on the necessity–domain map and grouped by necessity tier.
Status column	Whether the instrument is *scored* in the demand matrix, held out as *baseline*, or marked *context* (binds Member States, not the processor) or *excluded*.
Duplicates the workbook	Domain, Source, Locus and Status - see *Necessity_Domain_Matrix_v8.xlsx*, sheet "Classification", for the same fields in spreadsheet form.
Not in the workbook	The **Reference / founded** column below (legal citation and official source URL for all 49 instruments) - kept here in full for that reason.

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
Tier 1 - Legal licence to operate						
Fisheries Control Regulation (Reg 2023/2842)	Seafood	Regulator	Direct	Scored	Regulation (EU) 2023/2842, amending Reg (EC) 1224/2009 eur-lex.europa.eu - Reg (EU) 2023/2842	Traceability and catch-documentation obligations bind operators handling fishery products directly.
IUU Regulation (Reg 1005/2008)	Seafood	Regulator	Direct	Scored	Council Regulation (EC) No 1005/2008 - applies from 2010 eur-lex.europa.eu - Reg (EC) 1005/2008	Catch certificate must be held by the importer/processor; direct condition of placing product on the EU market.
CMO marketing standards (Reg 1379/2013)	Seafood	Regulator	Direct	Scored	Regulation (EU) No 1379/2013 eur-lex.europa.eu - Reg (EU) 1379/2013	Marketing standards and mandatory consumer information bind the processor at point of sale.
Common Fisheries Policy (Reg 1380/2013)	Seafood	Regulator	Context	Context	Council Directive (EU) 2017/159 eur-lex.europa.eu - Dir (EU) 2017/159	Frames the regime; binds Member States. Processor-facing obligations flow through FCR/IUU, not CFP itself.
Marine Strategy Framework Directive (Dir 2008/56/EC)	Seafood	Regulator	Context	Context	-	Binds Member States on marine environmental status; no direct processor obligation.
RFMOs (Regional Fisheries Management Orgs)	Seafood	Regulator	Context	Context	Council Directive (EU) 2017/159 eur-lex.europa.eu - Dir (EU) 2017/159	Set stock-level catch rules via Member State implementation; reach the processor through IUU documentation.
EMFAF (Reg 2021/1139)	Seafood	Regulator	Context	Context	-	A funding instrument the processor may access; imposes no compliance or disclosure obligation.
Packaging & Packaging Waste Regulation / PPWR (Reg 2025/40)	General	Regulator	Direct	Scored	Regulation (EU) 2025/40 - in force 11 Feb 2025 eur-lex.europa.eu - Reg (EU) 2025/40	Processor places packaged product on the market; recyclability, recycled-content and labelling obligations bind it directly.

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
OSH Framework Directive (Dir 89/391/EEC)	General	Regulator	Direct	Scored	Directive 89/391/EEC eur-lex.europa.eu - Dir 89/391/EEC	Workplace health & safety duties bind the processor as employer.
Working Time Directive (Dir 2003/88/EC)	General	Regulator	Direct	Scored	Directive 2003/88/EC eur-lex.europa.eu - Dir 2003/88/EC	Working-time limits bind the processor as employer.
Forced Labour Regulation (Reg 2024/3015)	General	Regulator	Direct	Scored	Regulation (EU) 2024/3015 - published Dec 2024 eur-lex.europa.eu - Reg (EU) 2024/3015	Prohibits placing products made with forced labour on the EU market; binds the processor directly.
EU Deforestation Regulation / EUDR (Reg 2023/1115)	General	Regulator	Direct	Scored	Regulation (EU) 2020/852 eur-lex.europa.eu - Regulation (EU) 2020/852	Due-diligence obligation on relevant commodities (e.g. packaging, feed inputs) placed on the market.
Empowering Consumers Directive / EmpCo (Dir 2024/825)	General	Regulator	Direct	Scored	Directive (EU) 2024/825 - in force 26 Mar 2024 eur-lex.europa.eu - Dir (EU) 2024/825	Bans generic/unsubstantiated environmental claims; binds the processor's product communication.
Water Framework Directive (Dir 2000/60/EC)	General	Regulator	Context	Context	-	Binds Member States; reaches the processor via national discharge permits, not directly.
Urban Waste Water Treatment Directive (Dir 2024/3019)	General	Regulator	Context	Context	-	Binds Member States/agglomerations; processor wastewater regulated via national permits.
Waste Framework Directive / EPR (Dir 2008/98/EC)	General	Regulator	Context	Context	-	EPR obligations for packaging now consolidated under PPWR for the processor.

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
F-gas Regulation (Reg 2024/573)	General	Regulator	Direct	Opt-in	Regulation (EU) 2024/573 - in force March 2024 (repealing Reg 517/2014) eur-lex.europa.eu - Reg (EU) 2024/573	Operating refrigeration/freezing equipment triggers direct duties (leak checks $\geq$5 t CO2e, 5-yr records, reporting via F-gas portal Art 26, phase-down/quota). No SME exemption - obligations attach to equipment/operator, not company size.
Whistleblowing Directive (Dir 2019/1937)	General	Regulator	Direct	Opt-in	Directive (EU) 2019/1937 - in force for 50–249-employee entities since 17 Dec 2023 eur-lex.europa.eu - Dir (EU) 2019/1937	Legal entities with 50+ employees must establish internal whistleblowing reporting channels, procedures and reporter protection (in force for 50-249 since 17 Dec 2023). Binds the processor directly as employer.
Pay Transparency Directive (Dir 2023/970)	General	Regulator	Direct	Opt-in	Directive (EU) 2023/970 - transposition deadline 7 June 2026 eur-lex.europa.eu - Dir (EU) 2023/970	Mandatory gender pay-gap reporting. Transposition deadline 7 June 2026. 150-249 employees report every 3 yrs on 2026 data (first report 2027); 250+ annually; 100-149 from 2031. Binds the processor directly above the headcount trigger.
Tier 2 - Market licence to operate						
MSC (Marine Stewardship Council)	Seafood	Customer	Direct	Scored	Founded 1997 msc.org	Customer-demanded certification; voluntary, but a direct obligation on the processor once adopted.
ASC (Aquaculture Stewardship Council)	Seafood	Customer	Direct	Scored	Founded 2010 asc-aqua.org	Customer-demanded aquaculture certification held directly by the processor.
Best Aquaculture Practices (BAP)	Seafood	Customer	Direct	Scored	From 2002 onwards bapcertification.org	Customer-demanded aquaculture certification.
Dolphin Safe	Seafood	Customer	Direct	Excluded	-	Dolphin-safe tuna labelling scheme; largely superseded and obsolete for an EU processor. Retained for ecosystem completeness.
GLOBALG.A.P. Aquaculture	Seafood	Customer	Direct	Scored	1997 (EUREPGAP) / aquaculture module 2007 globalgap.org	Customer-demanded aquaculture production-standard certification.

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
Friend of the Sea	Seafood	Customer	Direct	Scored	Founded 2008 friendofthesea.org	Customer-demanded sustainable-seafood certification.
Fisheries labour conditions (Council Dir 2017/159)	Seafood	Customer	Indirect	Excluded	Council Directive (EU) 2017/159 eur-lex.europa.eu - Dir (EU) 2017/159	Binds vessel owners, not the processor. Reaches the processor as value-chain content via buyer SMETA audits and ESRS S2.
Sedex / SMETA	General	Customer	Direct	Scored	Founded 2004 sedex.com	Customer-demanded social-audit membership held directly by the processor.
EcoVadis	General	Customer	Direct	Scored	Founded 2007 ecovadis.com	Customer-demanded sustainability rating the processor completes directly.
RSPO [conditional]	General	Customer	Indirect (scope-conditional)	Excluded	Founded 2004 rspo.org	Commodity certification for palm oil. Reaches a seafood processor via upstream feed, not direct certification, UNLESS it physically handles palm as a processing input - only then is the processor the certified entity (Direct). Demoted from a first-class entry: the deforestation-commodity exposure (palm AND soy) is already carried by EUDR at Tier 1. Soy equivalents (RTRS / ProTerra) likewise excluded unless directly handled, as are the timber and fibre schemes (FSC / PEFC), whose deforestation exposure is carried by EUDR and whose packaging exposure by the PPWR. RSPO is listed rather than these because palm is the commodity most commonly present in a seafood processor's own inputs, via oils and coatings.

Tier 3 - Competitive expectation

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
VSME (EFRAG voluntary SME standard)	General	Customer	Direct	Scored	Delivered to the Commission Dec 2024; Commission Recommendation adopted 30 July 2025 ec.europa.eu - VSME Recommendation (30 July 2025)	The SME's own proportionate sustainability-reporting standard, adopted directly.
CSRD / ESRS (Dir 2022/2464 + Del. Reg 2023/2772)	General	Customer	Indirect	Baseline	Directive (EU) 2022/2464 - in force January 2023 eur-lex.europa.eu - Directive (EU) 2022/2464	Most 50-250 employee processors fall below mandatory scope; pulled in as value-chain partners of large reporting customers.
CSDDD (Dir 2024/1760)	General	Customer	Indirect	Scored	Directive (EU) 2024/1760 eur-lex.europa.eu - Directive (EU) 2024/1760	Applies to large undertakings; the SME is affected as an upstream supplier subject to due-diligence requests.
CDP	General	Customer	Indirect	Scored	Founded 2000 cdp.net	Completed at the request of buyers or investors.
FIPs / FisheryProgress	Seafood	Industry/NGO	Direct	Excluded	Platform from ~2017 fisheryprogress.org	Processor sourcing commitment to a fishery improvement project.
GDST (Global Dialogue on Seafood Traceability)	Seafood	Industry/NGO	Direct	Scored	Convened 2017; GDST 1.0 standard, 2020 thegdst.org	Interoperable seafood-traceability data standard adopted by the processor.
GSSI (Global Sustainable Seafood Initiative)	Seafood	Industry/NGO	Indirect	Excluded	Founded 2015 ourgssi.org	Benchmarks certification schemes; shapes which certs buyers accept rather than binding the processor directly.
GRI Standards (incl. GRI 13)	General	Industry/NGO	Direct	Scored	GRI 13 'Agriculture, Aquaculture and Fishing Sectors', 2022 globalreporting.org	Voluntary reporting standard the processor adopts directly; GRI 13 covers agriculture, aquaculture and fishing.

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
EU Taxonomy (Reg 2020/852)	General	Investor	Indirect	Excluded	Regulation (EU) 2020/852 eur-lex.europa.eu - Regulation (EU) 2020/852	Reaches the SME via customers' and financiers' taxonomy-alignment data requests.
SFDR (Reg 2019/2088)	General	Investor	Indirect	Scored	Regulation (EU) 2019/2088 - applies from 2021 eur-lex.europa.eu - Regulation (EU) 2019/2088	Binds financial market participants; reaches the processor only via investor/lender data requests.
IFRS S1/S2 (ISSB - incorporating SASB and TCFD)	General	Investor	Indirect	Scored	Created 2015; recommendations 2017 fsb-tcfd.org	Investor-oriented disclosure standard requested by capital providers; consolidates the former SASB and TCFD frameworks.
Tier 4 - Strategic differentiation						
ISSF (Intl Seafood Sustainability Foundation)	Seafood	Industry/NGO	Direct	Opt-in	Founded 2000 unglobalcompact.org	Voluntary industry coalition (tuna-focused).
SeaBOS (Seafood Business for Ocean Stewardship)	Seafood	Industry/NGO	Direct	Excluded	Founded 2016 seabos.org	CEO-led keystone-company coalition; aspirational for an SME.
SBTi (Science Based Targets initiative)	General	Industry/NGO	Direct	Scored	Founded 2015 sciencebasedtargets.org	Processor sets validated emissions-reduction targets directly.
UN Global Compact	General	Industry/NGO	Direct	Scored	Founded 2000 unglobalcompact.org	Voluntary corporate-responsibility commitment.
UN Sustainable Development Goals	General	Industry/NGO	Direct	Excluded	Founded 2000 unglobalcompact.org	Reference framework for goal alignment and communication.
Move to Minus 15	General	Industry/NGO	Direct	Excluded	Initiated 2023 movetominus15.com	Industry coalition on frozen-supply-chain temperature.
TNFD (Taskforce on Nature-related Financial Disclosures)	General	Investor	Direct	Scored	Recommendations finalised 2023 tnfd.global	Voluntary nature-related disclosure framework the processor adopts directly.
Coller FAIRR Protein Producer Index	General	Investor	Indirect	Excluded	Index since 2018 fairr.org	An investor benchmark the processor is rated by, not a member of.

Instrument	Domain	Source	Locus	Status	Reference / founded	Role in the ecosystem
Dow Jones Best-in-Class Indices (S&P)	General	Investor	Indirect	Excluded	-	An investor index the processor is rated by, not a member of. Renamed from the Dow Jones Sustainability Indices, February 2025.

Appendix B - GRI 13 to ESRS coverage classification

Scope note	GRI 13 names four applicability categories - crop, animal, aquaculture, fishing (GSSB, 2022, p. 6) - not processing. Its *Topic Standard disclosures* (climate, biodiversity, water, etc.) apply to any organization regardless, and are what the classification below rests on; its *Additional sector disclosures* (catch volumes, farmed species, stock assessments) are restricted to primary producers and do not apply to a processor directly.
Categories & counts	Of 26 material topics: **Covered** (13, ESRS DR mapping clears the cohort's $\geq 5/6$ threshold, carried into the 28-row universe); **Reporting gap** (8, a mapping exists but cohort reporting falls below $\geq 5/6$, not carried); **Structural gap** (5, no ESRS DR exists for the topic at all).
Sector-material rows (2 origins)	2 come from Structural-gap topics with no ESRS home at all - animal health & welfare (13.11), supply-chain traceability (13.23). 2 come from sector-specific sub-disclosures (GRI 13.3.10, 13.3.11) nested inside Biodiversity (13.3), itself Covered. The remaining 3 Structural-gap topics (soil health, food security, anti-competitive behaviour) are judged not sector-defining for seafood and are not carried.
A known edge case	Natural ecosystem conversion (13.4) clears Covered status on one generic disclosure only; its substantive content (conversion-free sourcing %, hectares converted) sits in Additional sector disclosures a processor doesn't generate. Reporting burden here is closer to Structural-gap than "Covered" suggests - same issue flagged for ASC's conversion-free criterion in Section 7.6.
Full classification (26 rows)	*GRI_13_Topics_FINAL.xlsx*, sheet "Classification"

Appendix C - Instrument scope classification

The two admission gates	*Gate 1 - ESG relevance*: bears on an environmental, social or governance matter (excludes fiscal, customs, data-protection law). *Gate 2 - organisation-level obligation*: places a binding disclosure, due-diligence, certification or market-access requirement on the processor itself, not merely a Member State or another actor. Gate 1 is applied before an instrument is even added to the catalogue, so every exclusion below is a Gate-2 outcome.
Outcome	Passing both gates: *scored* (30 instruments, 4 further gated by firm profile as *opt-in*). CSRD/ESRS: passes both but held out as *baseline*. The remaining 18: *context* (7) or *excluded* (11) - regrouped by reason below.

Instrument	**Gate-2 outcome**	**Reason**
Common Fisheries Policy (Reg 1380/2013)	Context	Binds Member States on quota-setting; reaches the processor only through the Fisheries Control and IUU Regulations.
Marine Strategy Framework Directive (Dir 2008/56/EC)	Context	Binds Member States on marine environmental status; no processor-facing obligation.
RFMOs (Regional Fisheries Management Organisations)	Context	Set stock-level catch rules via Member State implementation; reach the processor only through IUU catch documentation.
EMFAF (Reg 2021/1139)	Context	A Member-State-administered funding instrument the processor may access; imposes no compliance or disclosure duty.
Water Framework Directive (Dir 2000/60/EC)	Context	Binds Member States; reaches the processor via national discharge permits, not directly.
Urban Waste Water Treatment Directive (Dir 2024/3019)	Context	Binds Member States/agglomerations; processor wastewater regulated via national permits.
Waste Framework Directive / EPR (Dir 2008/98/EC)	Context	Member-State-transposed EPR duties; now operationally consolidated for the processor under PPWR.

Instrument	Gate-2 outcome	Reason
Dolphin Safe	Excluded - superseded	Largely obsolete labelling scheme for an EU processor; retained in the catalogue for completeness only.
Fisheries labour conditions (Council Dir 2017/159)	Excluded - wrong bound party	Binds vessel owners, not the processor; the labour content it covers reaches the processor anyway via buyer SMETA audits and ESRS S2, so scoring it separately would double-count.
RSPO (conditional)	Excluded - redundant exposure	Certifies palm-oil sourcing; for a seafood processor this is a feed-input exposure already carried by EUDR at Tier 1, so it is not scored as a second instrument on the same exposure unless the processor directly handles palm as an input. The soy schemes (RTRS, ProTerra) and the timber and fibre schemes (FSC, PEFC) are excluded on the same reasoning; RSPO is catalogued rather than these because palm is the commodity most often present in a processor's own inputs, through oils and coatings.
FIPs / FisheryProgress	Excluded - no own obligation	FisheryProgress is a registry and progress-tracking platform for Fishery Improvement Projects rather than a standard: participating fisheries self-report against their own workplans, with no independent audit against a defined bar, which is the basis of the "FIP forever" criticism levelled at long-running projects. It therefore places no scorable disclosure obligation on the processor; the topics it touches are already scored via MSC, ASC, GDST and the IUU Regulation.
GSSI (Global Sustainable Seafood Initiative)	Excluded - rates schemes, not the processor	Benchmarks certification schemes and shapes which certs buyers accept; it does not itself demand anything of the processor.
EU Taxonomy (Reg 2020/852)	Excluded - no eligible activity	The Taxonomy's technical screening criteria define no eligible economic activity for NACE C10.20 (seafood processing), so it imposes no disclosure trigger on this activity; reaches the processor only indirectly via customer/financier data requests.

Instrument	Gate-2 outcome	Reason
SeaBOS (Seafood Business for Ocean Stewardship)	Excluded - closed membership	A CEO-led keystone-company coalition with no admission path for an SME; aspirational rather than an obligation it could ever hold.
UN Sustainable Development Goals	Excluded - no reporting obligation	An aspirational reference framework for goal alignment; imposes no corporate reporting requirement or datapoint.
Move to Minus 15	Excluded - no reporting obligation	An industry decarbonisation campaign (cold-chain temperature); no reporting requirement attaches to participation.
Coller FAIRR Protein Producer Index	Excluded - rated, not a member	An investor benchmark the processor is rated by, not one it joins or reports into.
Dow Jones Best-in-Class Indices (S&P)	Excluded - rated, not a member	An investor index the processor is rated by, not one it joins or reports into.

Firm-profile opt-in switches. Four scored instruments are further gated by the profile of the individual firm being modelled, and are switched off (excluded from that firm's demand sums, though they remain scored columns in the matrix itself) when the condition below is not met.

Instrument	Switch dimension	Condition to be switched on
ASC / GSA BAP	Production system	Firm produces or sources farmed (aquaculture) product; a wild-only firm switches both off.
F-gas Regulation (2024/573)	Equipment profile	Firm operates fluorinated-refrigerant refrigeration or freezing equipment; a firm with none switches it off.
Pay Transparency Directive (2023/970)	Headcount	Applies from 150 employees (gender pay-gap reporting); below that threshold the instrument is switched off.
Whistleblowing Directive (2019/1937)	Headcount	Applies from 50 employees; below that threshold the instrument is switched off.
ISSF (Intl Seafood Sustainability Foundation)	Species profile	Firm sources tuna; a non-tuna firm switches it off.

The full per-instrument reference - domain, stakeholder source, obligation locus, legal citation and official source URL, for all 49 catalogued instruments including those in the tables above - is given in Appendix A - Ecosystem Glossary.

Appendix D - Cross-framework coverage matrix and justification log

The complete coded instrument $\times$ topic matrix and the per-cell justification log (rationale, source clause, and mapped ESRS datapoint), which covers 120 of the 188 scored cells (Chapter 6), are held in *Phase3_Coverage_Matrix_v18.xlsx* (sheets “Coverage matrix” and “Justifications log”). Figure 14 below sets out the decision-tree coding protocol applied to every cell; a representative excerpt of the matrix itself is shown in Chapter 6 (Figure 9), and the full matrix is available in the source workbook.

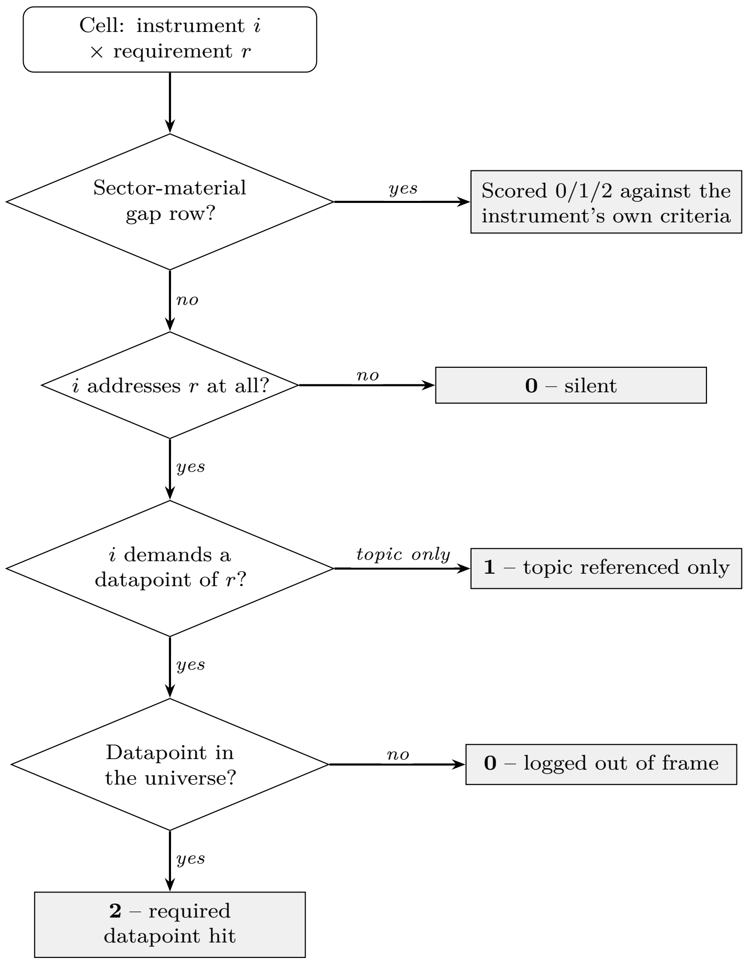

Figure 14: Per-cell coding protocol. Each instrument × requirement cell is scored against the requirement's constituent ESRS datapoints; gap rows, which have no ESRS datapoints by construction, are scored against the demanding instrument's own criteria; an instrument silent on the requirement scores 0, and demand mapping only to out-of-universe datapoints also scores 0 and is logged separately.

Appendix E - ESRS to VSME crosswalk

What this maps	Every datapoint in the VSME Basic (B1–B11) and Comprehensive (C1–C9) modules to its thematically closest ESRS disclosure requirement, flagging whether it's realistic for an SME to self-report as a low-effort "start-here" item.
Sources & limitation	No official EFRAG VSME-to-ESRS crosswalk existed at time of writing. Combines: topic-level mapping from CSR Tools (2024); the author's own DR-level extension of that mapping (checked against Commission Delegated Regulation (EU) 2023/2772); and, for a minority of rows, direct author mapping. **Operates at DR level, not datapoint level** - a match shows shared topic, not equivalent scope (Appendix F - ESRS 2023-to-2026 revision mapping and VSME cross-reference gives DR-to-datapoint depth separately). A July 2026 check of EFRAG's Knowledge Hub and VSME XBRL taxonomy found no official crosswalk that supersedes this one.
Cross-check vs. build-first set	Of the 7 ESRS-anchored Stage-1 build-first topics (6.7 The build-first set by production system), 5 have direct VSME coverage (energy, GHG, water, waste, health & safety). Biodiversity (E4) has coverage only via one site-count datapoint. **Value-chain workers (S2-1) has no VSME datapoint at all** - confirming, at datapoint level, the gap already flagged at topic level in Appendix F - ESRS 2023-to-2026 revision mapping and VSME cross-reference.
Full crosswalk (all datapoints)	*VSME_Basic_and_Comprehensive.xlsx*, sheet "Crosswalk"

Appendix F - ESRS 2023-to-2026 revision mapping and VSME cross-reference

The European Commission adopted a revised ESRS and a new VSME-based voluntary reporting standard on 3 July 2026 (Section 7.5), while this thesis was built on the 2023 ESRS Set 1 structure throughout. This appendix checks every disclosure requirement in the thesis's 13-requirement universal core (Section 5.5) and four-topic Stage-1 universal core (Section 6.7) against EFRAG's November 2025 draft revised standards - the technical basis for the adopted delegated act - for the six topical standards this thesis draws on (E1, E3, E4, G1, S1, S2), and cross-references each against the VSME voluntary standard's Basic and

Comprehensive modules.

Headline finding. Every one of these disclosure requirements survives the revision in substance; none was deleted. Most carry a new DR code, because two or three new requirements were inserted earlier in several standards (chiefly on risk identification and resilience for climate, and on actions/targets for business conduct), shifting everything after them down by one to three positions. One requirement - prevention and detection of corruption and bribery, old G1-3, part of the thesis's universal core - is no longer a separately numbered DR: its substance is now distributed across the policies (G1-1) and actions (G1-2) requirements instead.

VSME cross-reference. The clearest asymmetry is that the VSME voluntary standard has no dedicated module for value-chain workers, unlike the mandatory ESRS's ESRS S2: it surfaces only inside a general practices-and-policies template and a single yes/no incidents question. Climate targets and detailed water/biodiversity metrics are similarly absent from the VSME's Basic module, appearing only if a firm reports at Comprehensive level. Both observations are consistent with this thesis's broader finding (Section 6.6) that disclosure burden cascades down the value chain from large-firm obligations rather than being symmetric with what a smaller firm's own operations generate.

The complete requirement-by-requirement mapping - old code, old title, new code, new title, type of change, and VSME module cross-reference, each row sourced to the specific EFRAG draft standard it was verified against - is held in *ESRS_VSME_revision_mapping.xlsx* (sheets "DR_mapping" and "VSME_crossref"; full source list in the "Sources" sheet). This mapping reflects EFRAG's November 2025 drafts and the Commission's 3 July 2026 adoption; it should be re-checked against the final Official Journal text once the Parliament and Council scrutiny period concludes.

Appendix G - GRI/ESRS cohort coverage analysis and admission rules

This appendix documents the disclosure-requirement-level analysis underlying Chapter 5's cohort funnel (Section 5.5) and gives the admission rule by which each of the 70 disclosure requirements (DRs) in the ESRS universe was classified.

Admission tiers. Two evidentiary tiers are applied at the same threshold (≥ 5 of the 6 cohort firms) but at different levels of aggregation. *Tier 1* admits a DR directly where at least five firms report it: 28 DRs qualify, matching the "reported by at least five" band in Figure 5. Seven of the 70 DRs have no route through the GRI-ESRS bridge and so cannot be reported by more than three of the six firms; six of those seven are reported by no firm at all (Section 5.3). *Tier 2* admits a *topic* where signal is dispersed across its sub-DRs such that no single sub-DR clears the bar alone but at least five firms report *some* DR within the topic; two topics qualify this way - Biodiversity (E4) and Consumer health & food safety (S4). A third, qualitatively different *Tier 3* records four sector-material topics that the cohort-consensus test cannot surface at all because ESRS has no corresponding DR for them: supply-chain traceability, wild-stock status, feed and marine-ingredient sourcing, and animal health and welfare (with antimicrobial use folded into the last). These are anchored instead in the sector-specific GRI 13 standard and the relevant sector law (IUU Regulation, IHR/GDST, MSC/ASC chain of custody), and are included as an explicit structural gap rather than a cohort finding.

GRI-reported firms as a cross-check. Five of the six cohort firms' year-editions publish a GRI content index rather than a native ESRS index (Nomad Foods 2024 and 2025; Profand 2024 and 2025; Thai Union). Their disclosures were independently mapped to ESRS through the official EFRAG/GRI interoperability crosswalk (GRI and EFRAG, *GRI–ESRS Interoperability Index*, November 2024) rather than read directly, giving a second, differently-sourced route into the same coverage matrix. The two routes converge on the same output schema and are combined without distinction in the coverage counts above.

Where each part of the analysis is held. The underlying material is split across two workbooks, two notebooks, and one methodology note, each covering a specific stage:

- *ESRS_KPI_Mapping_V10.xlsx* - the per-company ESRS mapping database. Sheet `ESRS_MASTER` holds the full 70-DR universe; sheets `Bolton_Mapping`, `Mowi_Mapping`, `Espersen_Mapping` / `Espersen2025_Mapping`, `Nomad_Mapping` / `Nomad2025_Mapping`, `Profand_Mapping` / `Profand2025_Mapping` and `ThaiUnion_Mapping` hold each firm-year's reported DRs; `Mapping_ALLStacked_TU_N25` stacks all six current firm-years into the single table the coverage analysis reads. (`Nomad_Mapping` and `Profand_Mapping` used to contain live external-workbook-link formulas pointing back to *GRI_ESRS_v8.xlsx*; these were converted to static values on 2026-08-01, verified cell-by-cell against the original, so the file is now safe to open or resave normally.)
- *ESRS_Coverage_Analytics_v7.ipynb* - reads `Mapping_ALLStacked_TU_N25`, builds the coverage matrix, and applies the Tier 1/Tier 2 admission rule above.
- *ESRS_Coverage_Results.xlsx* - the notebook's output, consolidated: sheets `Coverage_Matrix` (70 DRs $\times$ 6 companies), `Topic_Universe` (the 29 Tier 1/2 rows - 28 at requirement level plus biodiversity E4 at topic level - and the 4 hand-added Tier 3 rows), `BACKBONE` (the datapoint-level shortlist) and `Excluded_DR`, plus a `Summary`, `Coverage_Rate`, per-standard heatmap and `DR_Frequency` breakdown computed from them.
- *GRI_ESRS_v8.xlsx* - the GRI-to-ESRS bridge (renamed from *v5* once the file was located and re-verified, 2026-07-30). Sheet `Interop_Index` is the transcribed EFRAG/GRI crosswalk; sheets ending `_GRI_Index`

(one per GRI-reporting firm-year) are each firm's own published content index; sheets ending `_Topical_Mapping` are the per-firm result of joining the two (full row-level detail: GRI standard/number, description, location in report, mapped ESRS topic/standard/DR, paragraph reference, notes, coverage). A further-condensed layer of sheets, all following the *Firm*`_Mapping` pattern as of 2026-08-01 (previously inconsistent - some were named `Mapping_`*Firm*), holds just Company/Topic/DR/Datapoint/Page/Comments; this is the format `ESRS_KPI_Mapping_V10.xlsx`'s own per-company sheets used to pull from via external-workbook-link formulas before those were converted to static values on the same date. This file did not previously have a README sheet; one has now been added as the first sheet.

- *GRI_ESRS_Mapping_v4.ipynb* - performs the join (Lookup_Key match, topical-DR filter, one-row-per-datapoint expansion) for every GRI-reporting firm-year in its `COMPANIES` list. Its own header previously referenced the bridge file as *v5* and named a separate output file (`GRI_ESRS_Topical_Mapping_AllCompanies.xlsx`) that, in practice, was merged directly into the bridge workbook instead of staying separate - both now corrected in the notebook's own documentation.
- *Mapping_Methodology.docx* - **dropped per your 2026-07-30 instruction ("we don't include mapping_methodology")**; the join/filter/expand/condense procedure is instead documented directly in the two notebooks' own headers.

Both notebooks are reproducible end to end from the two workbooks above; the only manual step is transcribing a newly-added firm's own published index into a new `_GRI_Index` or `_Mapping` sheet before re-running.

Appendix H - Master file index (all phases)

This appendix consolidates, in one place, every workbook and notebook underlying the three empirical phases (Chapter 4 ecosystem scan, Chapter 5 cohort scan, Chapter 6 cross-mapping and tool), plus the diagnostic tool itself. All workbooks below carry a README sheet as their first sheet as of the 2026-07-30 verification pass, except where noted.

Phase	File	Type	Contents	Status
Phase 1 - Chapter 4, the disclosure ecosystem				
Phase 1	*Necessity_Domain_Matrix_v8.xlsx*	Excel	The 49-instrument necessity/domain catalogue underlying Appendix A.2, plus a Tier-0 food-integrity foundation band, a Layer-2 enabler list, and a count-reconciliation sheet tying 49 catalogued instruments down to the 30 Phase 3-scored columns.	Verified 2026-07-30; numbers match Ch5/Ch8 prose exactly (49→30, 26 always-scored + 4 opt-in). README already first sheet.
Phase 1	*necessity_domain_matrix.pptx / .pdf*	Slides/PDF	Visual rendering of the necessity/domain matrix (Figure 2).	Visual companion only, not a data source - see the Excel above for the underlying rows.
Phase 2 - Chapter 5, the cohort frontier				
Phase 2	*ESRS_KPI_Mapping_V10.xlsx*	Excel	Per-company ESRS mapping database (see Appendix G for full sheet guide).	Corrected (Profand 2025 fix). **Regenerated 2026-08-04** from the rebuilt bridge: all five GRI-firm mapping sheets and all three stacked sheets refreshed. External-workbook-link formulas in `Nomad_Mapping` / `Profand_Mapping` converted to static values 2026-08-01 (surgically, cell-by-cell verified against the original with zero value changes elsewhere) - now safe to open or resave normally.

Phase	File	Type	Contents	Status
Phase 2	*GRI_ESRS_v8.xlsx*	Excel	GRI-to-ESRS bridge for the 5 GRI-reporting firm-years (see Appendix G).	Verified 2026-07-30; README added. Naming inconsistency fixed 2026-08-01. **Rebuilt 2026-08-04:** `Interop_Index` regenerated in full from the GRI-EFRAG interoperability index (previous version was under-populated); old bridge retained as `Interop_Index_v6_ARCHIVE`, changes logged in `Bridge_CHANGES`, and the seven ESRS requirements with no GRI counterpart screened directly in the reports (`Unlinked_DR_Screening`).
Phase 2	*GRI_ESRS_Mapping_v4.ipynb*	Notebook	Builds the bridge workbook's `_Topical_Mapping` sheets.	Doc header corrected 2026-07-30 (was referencing v5).
Phase 2	*ESRS_Coverage_Analytics_v7.ipynb*	Notebook	Builds the coverage matrix and applies the Tier 1/2 admission rule.	Doc header corrected 2026-07-30 (was referencing an old V5 database and a stale 5-company preset); config cell itself was already correctly set. Renamed from `...v7 (1).ipynb` to drop the duplicate-download suffix.
Phase 2	*ESRS_Coverage_Results.xlsx*	Excel	The notebook's output, consolidated with summary/heatmap/frequency analytics (see Appendix G).	Regenerated 2026-08-01 with the Profand-2025 fix and the animal-health/AMU topic merge.
Phase 2	Six firms' sustainability reports	External (web)	Primary source documents.	Linked via URL in references.tex, not held locally.

Phase	File	Type	Contents	Status
Phase 3 - Chapter 6, cross-mapping and the tool				
Phase 3	*GRI_13_Topics_FINAL.xlsx*	Excel	GRI 13 topic-to-ESRS reconciliation underlying Appendix B.	Verified 2026-07-30; re-checked against the rebuilt matrix 2026-08-08. Numbers match Appendix B/G exactly: a 29-row cohort-evidenced universe (28 requirement-level rows at the ≥5-of-6 cut, plus biodiversity E4 at topic level) and 4 sector-material rows, giving 33. README moved to first sheet.
Phase 3	*Phase3_Coverage_Matrix_v18.xlsx*	Excel	The full instrument×topic demand matrix underlying Appendix D (`Coverage matrix` + `Justifications log` sheets).	README already first sheet. **Rebuilt 2026-08-04** from v17: G1-5 removed and the S4 topic-level row replaced by S4-1 and S4-3 following the bridge rebuild, E1-1 added following the body screening; 33 rows × 31 instruments. All 30 unchanged rows verified to recompute identically to v17. Thirteen new scored cells logged in `Justifications log`.
Phase 3	*VSME_Basic_and_Comprehensive.xlsx*	Excel	VSME-to-ESRS crosswalk underlying Appendix E.	Verified 2026-07-30; matches Appendix E's description exactly. Added a workbook-level orientation block to its legend sheet and moved it to first position.

Phase	File	Type	Contents	Status
Phase 3	*SME_Datapoint_Guide.xlsx*	Excel	Master datapoint→enabling-instrument mapping behind the diagnostic tool (Section 6.8).	`Sources & methodology` renamed `READ ME` and moved to first sheet 2026-08-01, consistent with the rest of the package.
Phase 3	Diagnostic tool	Web app	The live Seafood SME Priority Navigator.	https://seafood-sme-priority-navigator.netlify.app/ - confirmed live 2026-08-07.

Appendix I - Year-over-year reporting in the cohort

Three of the six cohort firms - Nomad Foods, Profand and Espersen - published a structured report in both 2024 and 2025, allowing a like-for-like comparison. Reported coverage was flat or fell very slightly: Nomad Foods from 53 to 52 requirements, Profand unchanged at 43, and Espersen from 36 to 34, driven by a handful of requirements dropped and none added (Figure 15).

These are changes in what the indexes carry, and not necessarily changes in what the firms disclose. Nomad Foods' single decline illustrates the distinction. E1-9 was reached in 2024 through GRI 201-2, indexed to the firm's TCFD statement; the 2025 index no longer carries that disclosure, yet the TCFD statement itself is still published and linked from the 2025 report. The firm confirms that its own position on the requirement did not change between the two years. The measurement limitation set out in Section therefore applies to the year-over-year reading at least as strongly as to the levels, and arguably more so: a change in coverage can be produced by a change in indexing practice alone.

Two further cautions apply. Only firms publishing in both years can be examined, so the comparison covers half the cohort and is silent on the other half; the three firms not examined are not thereby shown to be stable. And with three firms over two years, no weight can be placed on the result beyond its direction. It is reported because that direction runs against the common assumption of ever-expanding disclosure, and because a comparable movement appears at scale. The first large-scale longitudinal study of double-materiality outcomes under the CSRD, covering 241 large listed EU companies, finds materiality rates declining moderately between FY2024 and FY2025, concentrated in the environmental standards other than climate (Nahrstedt, 2026). That study measures which topics firms judge *material* rather than how much they disclose, so it is a parallel indicator and not a direct comparison; the two nonetheless move the same way. The discussion returns to this under the Omnibus simplification.

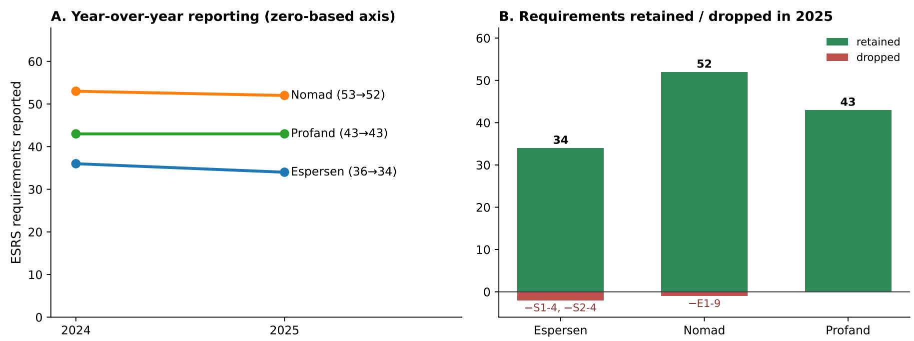

Figure 15: Year-over-year reporting for the three firms with both a 2024 and a 2025 report. Panel A: total requirements reported, on a zero-based axis, so that the size of the change is not visually exaggerated. Panel B: requirements retained and dropped in 2025; no firm added a requirement. Change is dominated by retention, with a small net reduction confined to three requirements across the three firms.